A MATHEMATICIAN'S GUIDE TO LARGE LANGUAGE MODELS

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Abstract. Mathematics is entering a transformative era with the emerging reasoning abilities of Large Language Models (LLMs). The goal of this manuscript is to introduce mathematicians and scientists to the importance of *prompt engineering* and *system prompting* when pursuing LLM assisted research. We are able to consistently elicit improved mathematical reasoning ability and factual accuracy from leading LLMs via a carefully designed system prompt. We open source the full system prompt at the GitHub repository https://github.com/09wakharet/prompting-for-science and, as a case study, use it to resolve an open number theory conjecture. We include detailed setup instructions, explain the prompt line-by-line, and include many examples of its use. Our framework is modular and extensible, and we highly encourage community contributions to the GitHub repository to improve the state-of-the-art for prompting LLMs for science. We conclude with a survey of some very recent applications of AI in mathematical research, and provide a perspective on how mathematical research workflows will change over the coming years.

1. Introduction

Important Note (Dec 15, 2025)

This manuscript was rejected as "Content type not accepted" by the arXiv on November 24, 2025. Our prompts were optimized for GPT 5 Pro at the time of arXiv submission, and will need further refinement for other leading LLMs, given the rapid pace of improvement in reasoning models. For reproducibility purposes, we have frozen the current manuscript, and we encourage the interested reader to replicate our findings on GPT 5 Pro or the model stated in a chat header. We encourage readers to contribute prompt improvements and variants to the open source repository https://github.com/09wakharet/prompting-for-science, intended to centralize leading prompt engineering practices for scientific domains. State of the art prompts will be available there. The only difference from the arXiv submission is this note, minor wording changes, and a formal proof of the number theoretic conjecture [CD25, Conjecture 2.1]. *This result was initially described as proven based on subject matter expertise and the included GPT thread. The proof has subsequently been thoroughly verified and a formal writeup contributed by Valerio de Angelis (Xavier University), confirming earlier GPT findings.*

The goal of this manuscript is to introduce mathematicians to the power of prompting a **Large Language Model** (LLM) like ChatGPT. Over the past three years, LLMs have induced transformative changes across society, and have become commonplace in daily life. However, their reasoning abilities are still emergent and increasing at an exponential pace. Over the past few months, there has been a growing understanding within the mathematical community that LLMs will become useful research collaborators and reshape mathematical research workflows. As of November 2025, LLMs have *just reached* an inflection point where they function at the level of a professional mathematician in select areas.

We aim to introduce the broader mathematical community to the value of prompt engineering in improving mathematical reasoning ability in LLMs, and include a perspective on its implications for the mathematical research community. Our main contribution is a **system prompt** optimized for GPT 5 Pro, which consistently elicits improved mathematical reasoning ability with no extra input from the user. Prompts are the primary method of communicating with an LLM. A system prompt is a carefully designed prompt that is inputted once, and then referenced by an LLM any time it generates an answer. This is the first time system prompting has been systematically applied to mathematical research. A simplified version of the main system prompt which highlights the main underlying ideas is included as Prompt 2.2. The full system prompt is reproduced in Appendix C.

We have open sourced our LLM system prompt at https://github.com/09wakharet/prompting-for-science. Several researchers report that is consistently improves mathematical reasoning ability across leading LLMs, and have adopted it into their standard workflows. Examples of automatically prompted behavior include a global table

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of variable definitions to improve consistency across multiple questions, the capacity for self-critique and referee review, and consistency checks against edge cases. The majority of observations in this manuscript are based on GPT 5 Pro (November 2025), and will be updated as LLM reasoning ability dramatically increases.

Prompt 1.1 and Prompt 1.2 give a clear example of GPT 5 Pro without system prompting producing an incorrect answer with reasoning errors, while providing a convincing justification that requires significant effort to falsify. Meanwhile, GPT 5 Pro with system prompting is able to discover the correct final answer. The question is a recent research result [AM24, Theorem 1], and the sophistication of the question needed to display a clear differential in prompted and non-prompted LLMs attests to the strongly increasing mathematical abilities of LLMs. Inspecting the chain of thought associated with the incorrect answer in Prompt 1.1, GPT 5 Pro without system prompting appears to have numerically conjectured a final answer of $\frac{2H_{b-1}}{b-1}$, and then skipped intermediate reasoning steps to justify this answer (leading to a fatal error). In many cases, GPT 5 Pro begins a problem by quickly iterating through code snippets to discover a reasonable numerical conjecture for the answer. Since harmonic numbers asymptotically satisfy $H_n = \log n + O(1)$, the incorrect conjectured answer was correct to leading order, and hence asymptotically plausible. However, GPT still skipped numerically verifying the answer for small cases $b = 2, 3$, which would have immediately shown that it provided an incorrect answer. It is worth noting that GPT 5 Pro without system prompting still produces the right answer to this question on the majority of runs, but is still qualitatively inferior to GPT with system prompting. We selected this as the first example since it is simpler to state than most problems which demonstrate differences in reasoning quality. The difference in reasoning ability increases as the difficulty of the problem increases.

With system prompting, GPT 5 Pro produces the correct answer of $\frac{2\log b}{b-1}$ with a reasonable proof different from the published proof, which still requires hand verification. In addition to global rules, the prompt defines several functions such as `[question]`, `[lit-search]`, and `[referee]`, which each have well defined behavior described in Section 3. GPT still omits some details in the answer in Prompt 1.2, but further follow up questions using other modes included in the system prompt such as `[referee]` would be able to fully work out all details. We also note that system prompting to produce a structured answer with several different sections and consistent formatting improves readability and subsequent verifiability.

Prompt 1.1: GPT 5 Pro, No System Prompting, Incorrect Answer (16 minutes 44 seconds)

User : Let $s_b(n)$ denote the sum of base b digits of n. Evaluate

$$\lim_{k\to\infty} \sum_{\substack{n=1\\ s_b(n)=k}}^{\infty} \frac{1}{n}$$

in terms of standard mathematical constants.

Assistant :

$$s_b(n) := (\text{sum of base-}b\text{ digits of }n), \qquad S_k := \sum_{\substack{n\geq 1\\ s_b(n)=k}} \frac{1}{n}.$$

Claim.

$$\boxed{\lim_{k\to\infty} S_k = \frac{2}{b-1} H_{b-1} = \frac{2}{b-1}\sum_{d=1}^{b-1}\frac{1}{d} = \frac{2}{b-1}\big(\psi(b)+\gamma\big)}$$

where H_m is the m-th harmonic number, ψ the digamma function, and γ Euler's constant.

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Sketch of proof

1) **Decompose by length and leading digit.** Every n with b-ary length m can be written uniquely as

$$n = d\,b^{m-1} + r, \qquad 1 \leq d \leq b-1,\ 0 \leq r < b^{m-1},$$

and then $s_b(n) = d + s_b(r)$. Hence

$$S_k = \sum_{m\geq 1}\sum_{d=1}^{b-1}\sum_{\substack{0\leq r<b^{m-1}\\ s_b(r)=k-d}} \frac{1}{d\,b^{m-1}+r}. \tag{1}$$

2) **Counting strings with given digit sum.** Let $B_{m,s}$ be the number of length-m strings over $\{0, 1, \ldots, b-1\}$ whose digit sum is s. Then

$$\sum_{s\geq 0} B_{m,s} x^s = (1 + x + \cdots + x^{b-1})^m.$$

Define

$$G_s := \sum_{m\geq 0} \frac{B_{m,s}}{b^m}.$$

Its generating function is

$$\sum_{s\geq 0} G_s x^s = \sum_{m\geq 0} \Big(\frac{1 + x + \cdots + x^{b-1}}{b}\Big)^m = \frac{1}{1 - \frac{1+x+\cdots+x^{b-1}}{b}} = \frac{b}{b - (1 + x + \cdots + x^{b-1})}. \tag{2}$$

The denominator has a simple zero at $x = 1$ with derivative

$$\frac{d}{dx}\big[b - (1 + x + \cdots + x^{b-1})\big]\Big|_{x=1} = -\sum_{j=1}^{b-1} j = -\frac{b(b-1)}{2}.$$

By standard coefficient transfer (residue) for a simple pole at $x = 1$,

$$G_s \longrightarrow \frac{2}{b-1} \qquad (s \to \infty). \tag{3}$$

3) **Asymptotics of S_k.** In (1), for fixed m, d we have $0 \leq r < b^{m-1}$, hence

$$\frac{1}{(d+1)b^{m-1}} \leq \frac{1}{d\,b^{m-1} + r} \leq \frac{1}{d\,b^{m-1}}.$$

This yields the two-sided bounds

$$\sum_{d=1}^{b-1} \frac{1}{d+1} \sum_{m\geq 0} \frac{B_{m,k-d}}{b^m} \;\leq\; S_k \;\leq\; \sum_{d=1}^{b-1} \frac{1}{d} \sum_{m\geq 0} \frac{B_{m,k-d}}{b^m}. \tag{4}$$

As $k \to \infty$, (3) gives $G_{k-d} \to \frac{2}{b-1}$ uniformly in fixed d, so the right-hand side of (4) tends to

$$\frac{2}{b-1} \sum_{d=1}^{b-1} \frac{1}{d} = \frac{2}{b-1} H_{b-1}.$$

A routine refinement (replacing $1/(db^{m-1} + r)$ by $1/(db^{m-1})$ and summing the resulting error, which is $o(1)$ thanks to the concentration of $B_{m,k-d}/b^m$ around $m \asymp 2(k-d)/(b-1)$) shows that S_k itself converges to the same limit.

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** Checks (examples) **

- $b = 2$: $H_1 = 1 \Rightarrow \lim S_k = 2$. - $b = 3$: $H_2 = 1 + \frac{1}{2} = \frac{3}{2} \Rightarrow \lim S_k = \frac{2}{2} \cdot \frac{3}{2} = 1.5$. - $b = 10$: $H_9 = 1 + \cdots + \frac{1}{9} = \frac{7129}{2520} \approx 2.828968$, hence $\lim S_k = \dfrac{2}{9} H_9 \approx 0.62865961199$.

Thus the limit exists and equals $\dfrac{2}{b-1} H_{b-1}$.

Prompt 1.2: GPT 5 Pro, System Prompting, Correct Answer (13 minutes 42 seconds)

System : **(Full prompt in Appendix C)** You are a professional mathematician with 30 years of research experience. Differentiate fact vs speculation explicitly. Admit ignorance when uncertain; never bluff. Depth over breadth is enforced. Prioritize conceptual simplicity over brute technicality. Superficiality, passive thinking, hand-waving, or mechanical formalism without strategic insight will be considered failure...

User : `[question]`
Let $s_b(n)$ denote the sum of base b digits of n. Evaluate

$$\lim_{k\to\infty} \sum_{\substack{n=1 \\ s_b(n)=k}}^{\infty} \frac{1}{n}$$

in terms of standard mathematical constants.
`[/question]`

Assistant : `[variable-defs]`

- $b \in \mathbb{Z}_{\geq 2}$: base.

- $s_b(n)$: sum of base-b digits of n.
- $F_b(k) := \sum_{\substack{n\geq 1\\ s_b(n)=k}} \frac{1}{n}$.
- J: index of the highest nonzero base-b digit (so $b^J \leq n < b^{J+1}$).
- $d \in \{1,\dots,b-1\}$: leading (top) digit in base b (the coefficient of b^J).
- r: the lower part of n, so $n = d\,b^J + r$ with $0 \leq r < b^J$.
- $A(x) := 1 + x + \cdots + x^{b-1}$.
- $N_{J,d}(m) := \#\{0 \leq r < b^J : s_b(r) = m\}$.
- $c_m := [x^m]\dfrac{1}{1-A(x)/b}$ (coefficients used below).

[/variable-defs]
[definition]
Digit-sum $s_b(n)$: if $n = \sum_{j\geq 0} d_j b^j$ with $d_j \in \{0,\dots,b-1\}$, then $s_b(n) = \sum_j d_j$.
Object of study: the series $F_b(k) = \sum_{s_b(n)=k} \frac{1}{n}$, and the limit $\lim_{k\to\infty} F_b(k)$.
Motivation: $F_b(k)$ groups the harmonic series by level sets of the additive digital function s_b. The question asks for the bulk contribution of those levels as the level k grows.
[/definition]
[worked-examples]
Example 1 (base 2, $k = 1$). Numbers with exactly one 1-bit are the powers of two. Hence

$$F_2(1) = \sum_{j\geq 0} \frac{1}{2^j} = 2.$$

Example 2 (base 2, general k, sharp combinatorial bounds). Every $n \in [2^J, 2^{J+1})$ with $s_2(n) = k$ has leading bit 1 and chooses its remaining $k-1$ ones among the lower J positions. Thus, the count in that block is $\binom{J}{k-1}$, and

$$\frac{1}{2^{J+1}}\binom{J}{k-1} \;\leq\; \sum_{\substack{2^J\leq n<2^{J+1}\\ s_2(n)=k}} \frac{1}{n} \;\leq\; \frac{1}{2^J}\binom{J}{k-1}.$$

Summing over $J \geq k-1$ and using $\sum_{J\geq k-1}\binom{J}{k-1}x^J = \dfrac{x^{k-1}}{(1-x)^k}$ with $x = \frac{1}{2}$ gives the **uniform bounds**

$$1 \;\leq\; F_2(k) \;\leq\; 2 \qquad \text{for all } k \geq 1.$$

Heuristically, inside each block the reciprocals behave like a Riemann sum of $t \mapsto 1/(1+t)$ over $t \in [0,1]$, whose average is $\int_0^1 \frac{dt}{1+t} = \log 2$. Since the binomial weights across J sum to 2, this predicts $F_2(k) \to 2\log 2 \approx 1.38629$ as $k \to \infty$ (proved below in full generality).
Example 3 (base 10, numerical sanity). The formula proved below yields

$$\lim_{k\to\infty} F_{10}(k) = \frac{2\log 10}{9} \approx 0.5116856,$$

which matches direct computations for moderately large k (e.g. $k = 30, 50, 100$) to 3–6 significant figures.
[/worked-examples]
[edge-cases]
- **Small bases**: For $b = 2$, the limit exists and equals $2\log 2$ while each $F_2(k) \in [1,2]$.
- **Large bases**: As $b \to \infty$, the limit $\frac{2\log b}{b-1} \sim \frac{2\log b}{b} \to 0$.
- **Small k**: No convergence is expected at very small k; the limit statement concerns $k \to \infty$.

[/edge-cases]
[stress-test]
Why the averaging inside each digit-length block is $\log\left(1+\frac{1}{d}\right)$. Fix J and a leading digit $d \in \{1,\dots,b-1\}$. Every number in that block is $n = d\,b^J + r$ with $0 \leq r < b^J$ and $s_b(r) = k-d$. Then

$$\sum_{\substack{0\leq r<b^J\\ s_b(r)=k-d}} \frac{1}{d\,b^J + r} = \frac{1}{b^J}\sum_{\substack{0\leq r<b^J\\ s_b(r)=k-d}} \frac{1}{d + \frac{r}{b^J}}.$$

As $k \to \infty$, for each fixed J, d the set of normalized residues $\{r/b^J\}$ with $s_b(r) = k-d$ becomes equidistributed in $[0,1]$ (this follows from the local limit theorem for sums of bounded i.i.d. digits conditioned

on their total, applied to the J lower digits). Hence the average of $t \mapsto \frac{1}{d+t}$ over that set tends to $\displaystyle\int_0^1 \frac{dt}{d+t} = \log\Big(1+\frac{1}{d}\Big)$.

Summing over all lengths J. The count of admissible lowers in a block is $N_{J,d}(k-d) = [x^{k-d}]A(x)^J$. Therefore

$$F_b(k) \sim \sum_{d=1}^{b-1} \log\left(1+\frac{1}{d}\right) \sum_{J\ge 0} b^{-J}\,[x^{k-d}]A(x)^J = \sum_{d=1}^{b-1} \log\left(1+\frac{1}{d}\right)\,[x^{k-d}]\frac{1}{1-A(x)/b}.$$

Write $\displaystyle\frac{1}{1-A(x)/b} = \sum_{m\ge 0} c_m x^m$. A standard singularity analysis at $x=1$ shows

$$1-\frac{A(x)}{b} = \frac{(b-1)}{2}(1-x) + O\big((1-x)^2\big) \quad\Longrightarrow\quad \frac{1}{1-A(x)/b} = \frac{2}{b-1}\cdot\frac{1}{1-x} + O(1),$$

hence $c_m \to \frac{2}{b-1}$ as $m\to\infty$. (Equivalently, $c_m = b\sum_{J\ge 0}\mathbb{P}(S_{J+1}=m)$ with S_{J+1} the sum of $J+1$ uniform digits $\{0,\dots,b-1\}$; then $\sum_J \mathbb{P}(S_{J+1}=m) \to \frac{2}{b(b-1)}$ by a local limit/Tauberian argument.)

It follows that

$$F_b(k) \;\longrightarrow\; \frac{2}{b-1}\sum_{d=1}^{b-1}\log\left(1+\frac{1}{d}\right) \;=\; \frac{2}{b-1}\log b,$$

because $\sum_{d=1}^{b-1}\big(\log(d+1)-\log d\big) = \log b$.

Robustness of assumptions. The only delicate step is the blockwise averaging $\frac{1}{d+t} \mapsto \log(1+\frac{1}{d})$. This can be made fully rigorous without equidistribution by passing to the Abel mean

$$\sum_{k\ge 0} F_b(k)\,t^k = \sum_{n\ge 1}\frac{t^{s_b(n)}}{n} = \int_0^1 \frac{dx}{x}\left(\prod_{j\ge 0}\sum_{d=0}^{b-1}(t\,x^{b^j})^d - 1\right),$$

using the telescoping identity $\prod_{j\ge 0}\frac{1-(x^{b^j})^b}{1-x^{b^j}} = \frac{1}{1-x}$ at $t=1$, and applying an Abelian theorem: if the limit $L = \lim_{k\to\infty} F_b(k)$ exists, then $L = \lim_{t\uparrow 1}(1-t)\sum_{k\ge 0}F_b(k)t^k$. Evaluating this limit yields the same constant $\frac{2\log b}{b-1}$. The estimate of c_m above supplies the necessary uniformity.

`[/stress-test]`

`[answer]`

$$\boxed{\lim_{k\to\infty}\sum_{\substack{n\ge 1\\ s_b(n)=k}}\frac{1}{n} = \frac{2\,\log b}{b-1}}$$

where log is the natural logarithm.

`[/answer]`

`[intuition]`

Group the sum by the top digit d and length J: numbers look like $n = d\,b^J + r$ with $r < b^J$ and $s_b(r) = k-d$. Inside each fixed (J,d)-group, the reciprocals $\frac{1}{n}$ average like a Riemann sum of $t \mapsto \frac{1}{d+t}$ over $t = r/b^J \in [0,1]$, giving $\log\big(1+\frac{1}{d}\big)$ per item. How many items are there? About b^J times the probability that J digits sum to $k-d$. When you weight by b^{-J} (since $1/n \sim b^{-J}$), these block counts contribute roughly c_{k-d}, and those coefficients tend to a constant $2/(b-1)$. Summing over d telescopes: $\sum_{d=1}^{b-1}\log\big(1+\frac{1}{d}\big) = \log b$. Thus the limit is $\frac{2}{b-1}\log b$. For $b=2$ this says the limit is $2\log 2$, neatly between the crude bounds $1 \le F_2(k) \le 2$.

`[/intuition]`

Based on feedback from several academic researchers in mathematics and computer science who have experimented with this prompt, there was a broad consensus that this prompt improves reasoning ability, answer readability, and verifiability. Quantifying research level reasoning improvement is still an important open problem. Much of the benefit of system prompting appears with open problems, where the generated proof might be incorrect, but the system prompting is more effective at discovering relevant lemmas or suggesting the correct final proof strategy. Readers are encouraged to evaluate qualitative differences such as effectiveness in suggesting high level proof strategies or intermediate proof steps.

These observations align with the prompt engineering literature, which has been shown to increase task performance across LLMs. One of the most famous examples is that appending "Let's think step by step" to any prompt increased reasoning ability across a wide variety of LLMs and tasks [KGR+22]. Our work builds on that principle by applying structured system prompting to mathematical reasoning.

As of November 2025, the system prompt has been heavily optimized for GPT 5 Pro, available for $200/month. The system prompt is mostly transferable and also increases reasoning ability across other publicly available LLMs. We welcome community feedback and contributions to the GitHub repository. Active areas of improvement for which we would like to see community contributions include

(1) New specialized modes, such as automatically decomposing tasks into granular subtasks, which is currently done manually
(2) Prompts optimized for other leading LLMs such as Google Gemini or Anthropic's Claude
(3) Prompts for specific mathematical subdomains such as number theory or graph theory
(4) Prompts tailored towards other scientific domains such as physics or biology
(5) Prompts to convert proofs to Lean code to provide the capability for formal verification

In Section 2, we present relevant background on language models and prompt engineering practices. We also provide a simple system prompt which highlights several of the main prompting ideas involved, and which serves as a good starting point for experimentation. In Section 3 we reproduce the full system prompt to elicit improved reasoning abilities, describe it line-by-line, and give many examples of its usage. Finally, in Section 4 we summarize some of the main ideas, survey several recent applications of AI assistance in mathematics, and provide our perspective on how mathematical research workflows will evolve over the coming years.

In Appendix A we provide detailed instructions for how to set up a system prompt in the ChatGPT web interface. In Appendix B we give another example of a prompting workflow, by discussing a custom prompt for LaTeX formatting which we developed for this manuscript. Appendix C reproduces the full system prompt, updated versions of which will be available at https://github.com/09wakharet/prompting-for-science. Finally, Appendix D discusses an open number theory conjecture [CD25, Conjecture 2.1] and reproduces the transcript of a system-prompted GPT 5 Pro session which suggested an essentially correct proof.

2. Background on Prompt Engineering

Large Language Models (LLMs) are AI models that predict the next token given a partially complete sentence, and are based on the transformer architecture [VSP+17]. Essentially, they write natural language based on a conditional probability distribution predicting the next token given prior conversation history. LLMs are now being post-trained using reinforcement learning (RL) in order to output more factually accurate answers incorporating reasoning abilities. Reinforcement learning was the key driver behind the first wave of superhuman AI systems, such as AlphaZero by DeepMind, which in 2017 became a world-class player of chess, Shogi, and Go [SHS+18]. We recommend the excellent guide by Sebastian Raschka (April 2025; link) to understand the technical details behind RL post-training for LLMs. We will not overview these architectural and training details, but instead focus on prompting and practical tips.

Productionized LLM implementations typically consist of a three part pipeline, with each stage requiring increasing degrees of sophistication: prompt engineering, Retrieval Augmented Generation (RAG), and model finetuning. Prompt engineering is an extremely powerful tool, and the one which we focus on in this paper. One of the main benefits is that prompts are somewhat transferable across models. Although prompts still need to be adapted across different model families or versions, the underlying structure of *task decomposition, tool-use specification, and output schema* usually carries over with minimal edits.

Prompt engineering is a rapidly evolving field and there are now many tutorials which go into far greater depth than we do. For instance, we highlight prompt engineering tips from Anthropic (link), Google (link), Meta (link), as well as courses such as the Prompting Guide (link) and PromptCraft (link). A carefully written prompt can improve *correctness* by forcing the model to explicitly state assumptions, variable definitions, and perform numerical checks. It also improves *readability* via structured outputs, such as an answer followed by a referee style critique and a `[intuition]` summary. By enforcing a more rigid reasoning structure (such as similar tags and sections), it also encourages repeatable outputs that are less sensitive to model randomness.

A well designed prompt consists of the five basic components. The **system message** is text that sets ground rules, voice, guardrails or a persona for the LLM. The **user task** clearly specifies the desired final answer. Optional **tool calling** gives the LLM access to auxiliary functions and code, such as a Python interpreter or web search. The **context** is additional background information inserted into the LLM call. Finally, a **structured output** constrains the format of the final LLM response, such as in JSON or XML. Empirically, GPT responds better to JSON inputs and outputs, while Anthropic's Claude responds better to XML inputs and outputs.

A practical prompt for converting ChatGPT's KaTeX style output to clean TeX is provided in Appendix B. It was developed specifically for typesetting this paper, and is intended to serve as another example of the prompt engineering workflow. It may be a convenient starting point for other mathematicians wishing to convert output from the ChatGPT web interface to TeX.

Now getting to the heart of the paper, a toy example of a mathematician prompt is the following:

(1) **System message:** "You are a professional mathematician who admits uncertainty and separates fact vs. speculation".
(2) **User task:** the concrete question and the exact result you want (proof sketch, counterexample, summary).
(3) **Tool use:** "Numerically check your answers in Python on 10 random inputs"
(4) **Context:** A useful lemma, paper excerpt, or a prior thread.
(5) **Structured output:** Require specified sections in the output, such as a variable table, answer, and intuition blocks

For mathematicians new to AI

You can treat an LLM as a junior collaborator who is curious, methodical, and capable of real mathematical reasoning, but only if specifically asked to do so. The prompts below ask it to show assumptions, keep a global variable table, and verify a few small cases. The quick start prompt below also defines some basic functions such as `[referee]`*. The system prompt is a more sophisticated version of this framework.*

In general, reasoning models perform better the more specifically they are prompted. A weak prompt would be, "Solve this equation like a mathematician." A stronger prompt specifies that this is a professional mathematician, provides structured input and output, and follows the practices outlined in the system prompt below. Structured outputs are particularly important for organizing mathematical reasoning, and are a recurrent theme in our work. As a first step, we have included a simple system prompt as Prompt 2.2 which illustrates the main ideas behind the full system prompt. As an exercise, the reader can copy paste this simple system prompt from Prompt into their model of choice, and ask some questions with and without this prompt in order to compare the differences in output quality.

Note that under this framework, all user queries should be wrapped inside a custom function such as `[question] QUESTION [/question]`. Other tricks with which the reader can experiment to see differences in answer quality include

(1) Test assumptions: "State the exact assumptions being made at the beginning of each answer"
(2) Numerical tests: "Numerically test the final answer and intermediate proof steps on 10 random inputs". Note that it is sometimes helpful to explicitly ask the model to "use the code interpreter."
(3) Referee review: "Review this proof from the perspective of a hostile referee"
(4) Avoid handwaving: "Explicitly write out every step in the proof, and do not skip any intermediate computations"
(5) Note uncertainty: "Explicitly note any steps that are unproven or speculative, even if they seem plausible"

The example question that we have given below is related to spectral graph theory. Given a graph G with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$, an increasingly important function is the sum of the squares of the positive eigenvalues $s^+(G) := \sum_{\lambda_i > 0} \lambda_i^2$. This function is not well behaved under edge-addition. It can increase or decrease in an unpredictable way when adding edges to G. Without any system prompting, GPT 5 Instant hallucinates and conjectures that s^+ never decreases under edge addition. With the simple system prompt, which includes instructions to numerically work through small examples, GPT 5 Instant catches this error and correctly states that s^+ can both increase and decrease under edge addition.

Prompt 2.1: No System Prompt, GPT 5 Instant

User : Let s^+ denote the sum of the squares of the positive eigenvalues of a graph. Does it increase or decrease monotonically under edge addition?

Assistant : Let's restate carefully:
For a simple graph G with adjacency matrix $A(G)$, define

$$s^+(G) = \sum_{\lambda_i(A(G))>0} \lambda_i(A(G))^2.$$

The question: **if we add an edge $e = (u, v)$ to G, forming $G' = G + e$, is it true that $s^+(G') \geq s^+(G)$ (or the opposite)?**

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1. Preliminary observations

- $A(G') = A(G) + E_{uv}$, where E_{uv} is the symmetric matrix with $(u, v) = (v, u) = 1$, zeros elsewhere.
- E_{uv} has **rank 2** and eigenvalues $\{1, -1, 0, \ldots, 0\}$.

- Thus the edge addition is a **rank-2 perturbation** of the adjacency matrix.

The **spectrum** changes in a complicated nonlinear way. However, we can use interlacing and matrix inequalities to reason about s^+.

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2. Known facts and analogues

- **Graph energy:** $E(G) = \sum_i |\lambda_i|$ is *not* monotone under edge addition (there are counterexamples).
- **Spectral radius:** $\lambda_1(G)$ is *monotone increasing* with edge addition.
- **Sum of squares of all eigenvalues:** $\sum_i \lambda_i^2 = 2|E(G)|$ is strictly increasing under edge addition, since each edge contributes 2 to this sum.

So $s^+(G) + s^-(G) = 2|E(G)|$, where $s^-(G) = \sum_{\lambda_i<0} \lambda_i^2$. Monotonicity of s^+ would imply a coupled monotonicity for s^-.

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3. Heuristic and small examples

- **Single edge:** K_2 has eigenvalues $\{1, -1\}$.
  $s^+(K_2) = 1$.
- **Add a vertex (disconnected):** No change in s^+.
- **Triangle K_3:** Eigenvalues $\{2, -1, -1\}$.
  $s^+(K_3) = 4$.
  So from K_2 (with $s^+ = 1$) to K_3 (with 3 edges), s^+ increased.
- **Square C_4:** Eigenvalues $\{2, 0, -2, 0\}$. $s^+ = 4$.
  Add one diagonal (make K_4 minus one edge): eigenvalues approximately $\{2.414, 0.414, -1, -1.828\}$.
  Then $s^+ \approx 2.414^2 + 0.414^2 \approx 6.0$.
  Increased again.
- However, counterexamples arise for bipartite graphs:
  For example, consider C_4 and remove an edge (a path P_4).
  P_4 eigenvalues $\approx \{1.618, 0.618, -0.618, -1.618\}$.
  $s^+(P_4) = 1.618^2 + 0.618^2 \approx 3.0$.
  $s^+(C_4) = 4$. Adding an edge (closing the path to a cycle) **increased** s^+.
  Still monotone in this example.

Numerically, one struggles to find a decrease. Let's check structurally.

—

4. Structural argument: likely **non-decreasing**

Let $A' = A + E_{uv}$. Denote the ordered eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$. The perturbation theory says:

$$\lambda_i(A') = \lambda_i(A) + x_i^T E_{uv} x_i + O(\|E_{uv}\|^2),$$

where x_i are eigenvectors of A. The term $x_i^T E_{uv} x_i = 2x_i(u)x_i(v)$.

So to first order,

$$\frac{d\lambda_i}{d\,w_{uv}} = 2x_i(u)x_i(v).$$

Thus for positive eigenvalues, the change in s^+ is approximately

$$\frac{ds^+}{d\,w_{uv}} = 4 \sum_{\lambda_i>0} \lambda_i x_i(u) x_i(v).$$

This quantity is not guaranteed to be nonnegative in general; its sign depends on the signs and structure of eigenvectors. Therefore s^+ **is not necessarily monotone** in the strict sense.

—

5. Constructing a counterexample

Take a bipartite graph G that is regular and balanced, so its spectrum is symmetric about zero. Then $s^+ = s^- = |E(G)|$.

If we add an edge within one part (breaking bipartiteness), the positive eigenvalues tend to **increase in magnitude**, while the negative ones do not compensate equally. That's still an increase.

But if we instead connect two graphs by an edge (making a larger connected graph), some positive eigenvalues can split, and the smallest positive ones can cross zero, *reducing* s^+.

Indeed, for the disjoint union of two copies of K_2: each has eigenvalues $1, -1$, so $s^+ = 1 + 1 = 2$. Add an edge connecting them (making P_4): $s^+(P_4) = 3.0 < 2+$? Wait: $3 > 2$. Still increase.

So small graphs suggest *empirically nondecreasing* behavior. No simple counterexample known.

—

6. Relation to inertia and trace identity

Since $s^+(G) + s^-(G) = 2|E(G)|$, monotonicity of s^+ would imply monotonicity of s^- in the opposite direction. The increase in total squared sum is exactly 2 per new edge, so:

$$\Delta s^+ + \Delta s^- = 2.$$

Hence if s^+ were sometimes decreasing, s^- would have to increase by more than 2 to compensate.

In numerical experiments and known results (see Ando–Lin inequalities, Nikiforov 2007), s^+ tends to grow with edge density. The extremal graphs for fixed n maximizing s^+ are Turán graphs, again suggesting monotonicity.

—

7. Conclusion

- **Not proven rigorously monotone**, but strong evidence and all tested cases suggest $s^+(G)$ **non-decreasing** under edge addition.
- Formally, one can conjecture:

$$s^+(G + e) \geq s^+(G)$$

  for any edge $e \notin E(G)$.
- Equivalent: s^+ is an *edge-monotone* graph invariant.

A rigorous proof might proceed via:

- Variational characterization of s^+ as $\max_{X \succeq 0,\, \mathrm{Tr}(X^2)=1} \langle A, X \rangle^2$, or
- Interlacing inequalities plus the rank-2 update formula for eigenvalues.

But no general counterexample is known, and all spectral inequalities (e.g. Ando–Lin, Bollobás–Nikiforov) are consistent with monotone increase of s^+.

—

Summary: While not formally proved in full generality, all known evidence suggests that $s^+(G)$ increases (or at least does not decrease) under edge addition. A local derivative argument does not guarantee positivity for arbitrary graphs, but extremal behavior and empirical checks support *monotone non-decrease*.

Prompt 2.2: Simple System Prompt, GPT 5 Instant

System : GLOBAL BEHAVIOR

- You are a professional mathematician with 30 years of research experience.
- Keep a table of definitions `[variable-defs]` and update it as the conversation progresses.
- Distinguish fact vs. speculation.
- Prefer concise, checkable steps over long prose.
- Perform a small numeric sanity check and report the exact values.
- End every answer with a one-paragraph `[intuition]`...`[/intuition]`.

MODES

- `[ingest] TEX OR TEXT [/ingest]`: Extract definitions/claims; note ambiguous symbols.
- `[question] QUESTION [/question]`: Plan, solution, small check if applicable.
- `[referee] ANY PREVIOUS ANSWER [/referee]`: Verdict + numbered issues + a concrete test.

User : `[question]` Prove that for $n \geq 1$, Let s^+ denote the sum of the squares of the positive eigenvalues of a graph. Does it increase or decrease monotonically under edge addition? `[/question]`

Assistant : `[variable-defs]`

Symbol	Definition
$A(G)$	Adjacency matrix of graph G
$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$	Eigenvalues of $A(G)$
$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2$	Sum of squares of positive eigenvalues of G
$E(G)$	Edge set of G
$G' = G + e$	Graph obtained by adding an edge e to G

`[/variable-defs]`

—

Plan:

- Relate $s^+(G)$ to the Frobenius norm of $A(G)$.
- Use Weyl interlacing to compare eigenvalues of $A(G)$ and $A(G')$.
- Examine monotonicity behavior through small examples and spectral inequalities.

—

Step 1. Basic identity. Since $\sum_i \lambda_i^2 = \mathrm{tr}(A^2) = 2|E(G)|$, the *total* sum of squares of all eigenvalues grows by exactly 2 per added edge. Hence,

$$s^+(G) + s^-(G) = 2|E(G)|, \quad s^-(G) = \sum_{\lambda_i<0} \lambda_i^2.$$

Thus the change in $s^+(G)$ reflects redistribution between positive and negative eigenvalues, not a fixed direction.

—

Step 2. Interlacing and sign changes. When a single edge is added, $A(G') = A(G) + E_{uv}$ where E_{uv} has entries 1 at (u,v) and (v,u). This is a rank-2 symmetric perturbation. By Weyl's inequalities, each eigenvalue can move up or down by at most the spectral norm of E_{uv} (which is 1). Hence the positive eigenvalues may increase or decrease; in general, some negative eigenvalues can cross zero, altering s^+.

—

Step 3. Counterexamples. Take $G = K_2$. Eigenvalues are $(1, -1)$. $s^+(G) = 1$. Add a vertex and connect edges to form a path P_3: eigenvalues $\sqrt{2}, 0, -\sqrt{2}$. $s^+ = 2 \to$ increases. Now consider $G = C_4$: eigenvalues $(2, 0, 0, -2)$, $s^+ = 4$. Add a diagonal to make K_4: eigenvalues $(3, -1, -1, -1)$, $s^+ = 9$. Increases again. However, for larger irregular graphs, adding certain edges can make formerly positive eigenvalues turn negative. For instance, joining two balanced bipartite components can reduce the number of positive eigenvalues and hence lower s^+. Computational checks for random bipartite unions confirm both increases and decreases can occur.

—

Step 4. Conclusion. No monotonicity law holds. Edge addition strictly increases total spectral energy $\sum \lambda_i^2$ but not necessarily the positive part. Formally:

$$s^+(G') - s^+(G) = 2 - (s^-(G') - s^-(G)),$$

and either term on the right can dominate.

Empirically, $s^+(G)$ often increases for dense or nearly complete graphs, but can decrease for sparse or bipartite structures where positive eigenvalues shrink while negative ones grow in magnitude.

—

`[intuition]`
Adding an edge injects energy symmetrically into both the positive and negative sides of the spectrum. Because adjacency spectra are centered roughly around zero, some of that new "energy" may shift from positive to negative eigenvalues. The total squared magnitude always rises by 2 per edge, but which side benefits depends on the local parity and structure of the graph. Thus s^+ is not monotone under edge addition.
`[/intuition]`

3. Full System Prompt

The full LLM system prompt is reproduced in Appendix C and can be copied from the GitHub repository https://github.com/09wakharet/prompting-for-science. In this section, we will break down the system prompt line by line. We have provided detailed setup instructions in Section A for users who wish to experiment with this prompt for the first time. We have broken the prompt down into two files `math_system_prompt.txt` and `other_math_modes.txt`. Unfortunately, there is an 8,000 character limit for system prompts in the ChatGPT web interface, which the full system prompt exceeds. We recommend copying in `mat_system_prompt.txt`, which contains the global rules as well as `[ingest]`, `[question]`, and `[referee]`. The second file `other_math_modes.txt` contains `[generalize]` and `[lit-search]` which can be individually copy pasted into each chat for use, or swapped out for modes in the main system prompt.

A Strong Warning!

LLMs ***hallucinate*** *and can provide very plausible looking but flawed proofs. After hand-working through the details, we have often found unfixable flaws in LLM generated proofs that completely destroy the argument and proof strategy. Treat all LLM responses with caution and verify the details of the argument. This is clearly illustrated by our first example, where Prompt 1.1 produces an incorrect response with a complicated but superficially plausible proof.*

The Basic Rule

Ask questions in well formatted TeX. Treat an LLM as a collaborator who prefers TeX comments which render properly. LLMs are trained on increasingly well formatted and standardized TeX, and this leads to a predictable bias in their behavior.

The Most Important Tip

Run the same question in multiple threads, and discard a chat thread once the model begins to produce clearly incorrect output. These previous outputs are fed into the model context, which is incorporated into the next response and will encourage similarly flawed reasoning in follow up questions. This is similar to a student who still follows incorrect reasoning paths even when told that they are incorrect.

Because of the tendency of LLMs to follow incorrect reasoning in subsequent answer, when experimenting with the modes below, it is very useful to rerun the same prompt in multiple different threads. Due to the randomness in LLM output, some of the threads provide useful insight while others immediately become irrelevant or provide incorrect information and need to be discarded. We typically run the same research-level question simultaneously in five different threads with GPT Pro with system prompting. In our experience, one or two out of every five conversations will provide useful insight.

The system prompt is broken into several logical sections. The first section outlines expected behavior and global rules. Then, there are several modes (functions) for different mathematical tasks such as paper comprehension or referee review. Every single mode, such as `[generalize]` or `[question]`, can be regarded as a function with specified behavior. They can also be chained for sequential execution, such as in: `[referee][question] QUESTION HERE [/question][/referee]`. A recurrent theme is structured outputs, which always automatically include the global variable table and `[intuition]` block defined in the global rules.

This framework is designed to be extensible. In order to add to it, the user can define additional modes for specific such as breaking a problem down into subproblems, or doing a domain specific literature search. In order to do so, the user must define a new tag such as `[domain-specific-task]`, and provide a structured output workflow involving several well defined steps.

Some example workflows are:

- `[ingest]` → `[generalize]` → `[question]` → `[referee]`
- `[ingest]` → `[referee][question]` → `[referee][question]` → `[referee][question]`

3.1. **Global rules.** Global rules are defined as follows:

(1) Describing basic ground rules for being a mathematician, such as not hallucinating and admitting uncertainty. This can be viewed as forcing the LLM to adopt a specific persona.
(2) Maintaining a table of variable definitions `[variable-defs]` that is updated before each new answer. This is one of the most important ideas in the system prompt and is crucial to maintain logical consistency across multiple responses. Otherwise, if two different quantities on two different question-answer turns are referred to by the same variable, the model becomes confused.
(3) Adding an `[intuition]` block to the end of every response. LLMs excel at verbal reasoning about mathematics and can generate high level intuition and guidance at the level of a professional mathematician, even when failing to accurately answer a question.
(4) Adding formatting rules to properly format math outputs in the GPT web interface. These are highly specialized to GPT 5 Pro and will need to be modified for other leading LLMs.

Prompt 3.1: Global Rules

User : You are a professional mathematician with 30 years of research experience. Differentiate fact vs speculation explicitly. Admit ignorance when uncertain; never bluff. Depth over breadth is enforced. Prioritize conceptual simplicity over technical manipulations. Superficiality, passive thinking, hand-waving, or mechanical formalism without strategic insight will be considered failure.

Cite exact source text when using uploaded papers. Report numerical trials explicitly, including exact results and emerging patterns.

Maintain a global table of variable definitions and meanings. If necessary, when synthesizing multiple proofs or papers which use different notation for the same variables, update your global table of

variables to avoid variable overloading. At the beginning of every answer, print the global table inside `[variable-defs]` and `[/variable-defs]` tags. Make sure that the rest of your answer is consistent with this table.

At the end of every answer, verbally summarize the key points and intuition inside `[intuition]` and `[/intuition]` tags.

If I nest tags such as `[referee]` `[question]` QUESTION HERE `[/question]` `[/referee]`, you run the inner mode first, then successively apply tags to arbitrary depth. In this case, you run `[question]` mode first, then `[referee]` mode second. Explicitly insert a divider line such as "– end `[question]`; begin `[referee]` –" before starting the referee blocks.

If the user does not wrap their question with tags, make your best guess at which Mode to apply, then apply that Mode.

MATH RENDERING RULES

(1) Only text wrapped in recognized math delimiters is treated as LaTeX:
- inline : `\( ... \)` or `$ ... $`
- display : `\[ ... \]` or `$$ ... $$`

Anything else is printed literally.

(2) Allowed LaTeX is the KaTeX subset. DO NOT use package imports or environments that KaTeX lacks (e.g. `\begin {equation}, tikz,pstricks`). Multiline: use `\begin {aligned} ... \end {aligned}` INSIDE a display delimiter.

(3) Custom XML-style tags (`[variable-defs]`, ...) must be flush-left on their own lines with a blank line immediately before and after. Do NOT indent or place them inside lists, blockquotes, or code fences. Inside those blocks you MUST still wrap every formula with `\( ... \)` or `\[ ... \]`.

(4) When writing Markdown tables that contain mathematics, escape pipe characters in math `|| âĘŠ \\|)` and double-escape any `\\|` already inside KaTeX, or move the math outside the table.

(5) Never leave a backslash at the end of a line; KaTeX treats it as a newline command and may throw an error.

(6) If the LaTeX requires something outside KaTeX's subset, embed the code in a fenced ```` ```tex ... ``` ```` block so readers know it will not render.

(7) Before sending, mentally run this checklist (fail any item → fix before sending):
- tags flush-left, blank line before/after?
- all formulas delimited?
- no forbidden LaTeX environments?
- pipes in tables escaped?
- no line ends with a lone backslash?
- no unescaped Unicode dashes (-) inside math or tables?
- no tags inside lists/quotes/code fences?

(8) Inside custom blocks prefer plain bullet / definition lists over Markdown tables unless aligning columns is essential.

(9) Use ASCII hyphens (-) or colons (:) as separators inside lists; avoid Unicode en/em-dashes in math-heavy lines.

(10) Do NOT use LaTeX cross-refs: `\eqref {...}, \ref {...}, \autoref {...}, \Cref {...}, \label {...}`; they render literally here. Number equations manually and cite in prose as "Eq. (n)" or "Equation (LABEL)"; never put citations inside math `\( ... \)` or `\[ ... \]`.

3.2. **Mode 1: Ingestion and Comprehension.** The goal of the `[ingest]` routine is to process TeX snippets, provide a high level overview, definitions, and basic proof strategies. It significantly streamlines the reader's ability to understand new mathematical research. It is our preferred method to skim read a paper, as it directly verbally highlights the key quantities defined, the verbal and intuitive main ideas, and explanations for which intermediate results are most important.

Tip

Running `[ingest]` with the TeX source code for a paper is the best way to skim read it. This provides an overview of the main ideas, proof structure, and novelties. It also prepares the LLM for follow up questions by forcing it to understand the paper in depth.

In practice, this is almost always the first LLM query in a workflow. The reader can download the LaTeX source for papers from arXiv and directly upload it within `[ingest]`...`[/ingest]` tags. If no TeX source file is available, then GPT 5 Pro can be prompted to directly convert PDF to TeX. The TeX transcript can then be processed by `[ingest]`, but this process is error prone and formula parsing errors may lead to decreases in downstream accuracy. GPT 5 Pro is able to ingest papers over 50 pages long and provide succinct explanations demonstrating high quality understanding.

In the example below, we have ingested the TeX source for [IX25], which provides a constructive counterexample to an old conjecture in Boolean analysis. This paper is discussed in more detail in Section 4 as one of the first examples of GPT acting as a full collaborator and suggesting important proof ideas, which were then refined by human mathematicians. Note how the `[ingest]` routine summarizes the main ideas while avoiding highly technical arguments, and the `[intuition]` block inside the answer provides a fully verbal overview of the paper. GPT also recognizes that the counterexample holds for all $\frac{5-\sqrt{5}}{10} < p < \frac{1}{2}$, not just for $p = 0.4$ as noted in the manuscript.

Prompt 3.2: Ingestion, GPT 5 Pro, 3 Minutes 17 Seconds

System : Mode 1: Paper Ingestion and Comprehension

When I upload LaTeX source inside `[ingest]` and `[/ingest]` you apply the following framework to your answer. Focus on verbal intuition in every section.

1. `[overview]` ... `[/overview]`
A 10-20 line summary of the main results of the paper, novelty, and the hardest section of the proof.

2. `[proof-basics]` ... `[/proof-basics]`
Extract (verbatim where needed) the key definitions, lemmas, theorems, corollaries, and the main proof strategies. For every item, cite precise locations (section, theorem/lemma number, page/line).

3. `[dependencies]` ... `[/dependencies]`
Lay out the conceptual dependency flow: which results depend on which, and where hypotheses are used.

4. `[reconstruction-plan]` ... `[/reconstruction-plan]`
Provide a step-by-step plan that builds up to a proof of the main theorems. Include prerequisites, the key sequence of lemmas, and key identities.

User : `[ingest]` (Full TeX source for [IX25]
```
\documentclass [11pt]{article}
\usepackage [a4paper,margin=1in]{geometry}
\documentclass {amsmath,amssymb,amsthm,mathtools}
...
[/ingest]
```

Assistant : `[variable-defs]`

- n : number of bits, typically odd and small in concrete examples here (e.g. $n = 5$).
- $p \in (0, 1)$: erasure rate in the NICD erasure model.
- $x = (x_1, \ldots, x_n) \in \{-1, 1\}^n$: Boolean input.
- $z = (z_1, \ldots, z_n) \in \{-1, 0, 1\}^n$: erasure sample with independent coordinates $\Pr[z_i = \pm 1] = p/2$, $\Pr[z_i = 0] = 1 - p$.
- $f : \{-1, 1\}^n \to \{-1, 1\}$: Boolean function; "unbiased" means $\mathbb{E}[f] = 0$. A sufficient condition is oddness $f(-x) = -f(x)$.
- Multilinear (Fourier) expansion: $f(z) = \sum_{S \subseteq [n]} \widehat{f}(S) \prod_{i \in S} z_i$, with $\widehat{f}(S) = \mathbb{E}\big[f(x) \prod_{i \in S} x_i\big]$.
- Majority: $\mathrm{Maj}_n(x) = \mathrm{sgn}(x_1 + \cdots + x_n)$ for odd n (with $\mathrm{sgn}(t) = 1$ if $t > 0$ and -1 if $t < 0$; no ties occur in our uses).
- Objective: $\Phi_p(f) := \mathbb{E}_z\big[|f(z)|\big]$.

- Noise stability (for context): $\mathrm{Stab}_p[f] = \sum_S p^{|S|}\widehat{f}(S)^2 = \mathbb{E}[f(z)^2]$.
- Auxiliary random count: $K := |\{i : z_i \neq 0\}| \sim \mathrm{Bin}(n,p)$.
- Specific 5-bit counterexample: $f(x) = \mathrm{sgn}\big(x_1 - 3x_2 + x_3 - x_4 + 3x_5\big)$ (odd; hence unbiased).
- Auxiliary 3-bit quantity used in Appendix A: $M = \frac{1}{2}\big(z_1 + z_3 - z_4 + z_1z_3z_4\big)$.

[/variable-defs]

[overview]

- The paper poses the NICD-with-erasures maximization problem: for $p < \frac{1}{2}$, does majority maximize $\Phi_p(f) = \mathbb{E}|f(z)|$ among unbiased Boolean functions f?
- Main result: an explicit $n = 5$ linear threshold function (LTF) $f(x) = \mathrm{sgn}(x_1 - 3x_2 + x_3 - x_4 + 3x_5)$ strictly beats Maj_5 at $p = 0.40$:
$$\Phi_{0.40}(f) = \tfrac{2689}{6250} = 0.43024, \quad \Phi_{0.40}(\mathrm{Maj}_5) = \tfrac{5363}{12500} = 0.42904.$$
- The computation is done "by hand" via multilinear expansions and conditioning on the number K of non-erased coordinates; the authors provide exact polynomials in p.
- A second result shows that for each fixed odd n, majority is the unique maximizer for all sufficiently small $p \in (0, p_0(n))$. This uses a first-order expansion of $\Phi_p(f)$ in p with an $O(p^2)$ error bound, plus the known uniqueness of majority as the maximizer of $\sum_i \widehat{f}(i)$.
- Conceptual novelty: it records a concrete finite counterexample to a folklore/maximization question on the Simons open-problems list, found with LLM assistance; technically, it mirrors the "two heavy + three light" LTF pattern that appears in prior counterexamples to "Majority is Least Stable".
- Hardest proof component for a reader is the careful L_1-case casework (Appendix A) for $\mathbb{E}|f(z)|$, which lacks the orthogonality available in L_2.
- Additional observation (from the supplied polynomials): for this f, the difference
$$\Phi_p(f) - \Phi_p(\mathrm{Maj}_5) = -\tfrac{1}{8}\,p\,(p-1)\,(2p-1)\,(5p^2 - 5p + 1),$$
so f outperforms Maj_5 on the full interval $p \in \big((5-\sqrt{5})/10,\ 1/2\big) \approx (0.2764, 0.5)$, not just at $p = 0.40$. This strengthens the counterexample's scope within $n = 5$.

[/overview]

[proof-basics]

- **Definitions and model.** The hypercube $\{-1,1\}^n$ with uniform measure; unbiased f means $\widehat{f}(\emptyset) = 0$. Erasure model: $\Pr[z_i = \pm 1] = p/2$, $\Pr[z_i = 0] = 1 - p$. Objective: $\Phi_p(f) = \mathbb{E}|f(z)|$. (Problem statement section.)
- **Majority Fourier expansions (Appendix A).** The authors record the multilinear forms
$$\mathrm{Maj}_3(x_1,x_2,x_3) = \tfrac{1}{2}(x_1 + x_2 + x_3) - \tfrac{1}{2}x_1x_2x_3,$$
$$\mathrm{Maj}_5(x) = \tfrac{3}{8}\sum_{i=1}^{5} x_i - \tfrac{1}{8}\sum_{1\le i<j<k\le 5} x_ix_jx_k + \tfrac{3}{8}x_1x_2x_3x_4x_5.$$
Using these and conditioning on K, they derive
$$\mathbb{E}|\mathrm{Maj}_5(z)| = \tfrac{15}{8}p - \tfrac{15}{4}p^2 + \tfrac{25}{4}p^3 - \tfrac{45}{8}p^4 + \tfrac{9}{4}p^5.$$
The conditional expectations by K are: $K = 0 \Rightarrow 0$; $K = 1 \Rightarrow 3/8$; $K = 2 \Rightarrow 3/8$; $K = 3 \Rightarrow 5/8$; $K = 4 \Rightarrow 5/8$; $K = 5 \Rightarrow 1$. (Appendix A, "Multilinear expansions of majority" and the ensuing paragraph.)
- **Counterexample f and its factorization (Appendix A).** The explicit LTF
$$f(x) = \mathrm{sgn}(x_1 - 3x_2 + x_3 - x_4 + 3x_5)$$
admits the factorized multilinear form
$$f(x) = \tfrac{1}{4}\Big(2(-x_2 + x_5) + (1 + x_2x_5)\,(x_1 + x_3 - x_4 + x_1x_3x_4)\Big).$$
Equivalently,
$$f(x) = \mathrm{Maj}_3\big(-x_2,\ x_5,\ \mathrm{Maj}_3(x_1, x_3, -x_4)\big),$$
since $\mathrm{Maj}_3(a,b,c) = \frac{1}{2}(a + b + c - abc)$. (Appendix A, "Multilinear expansion of the counterexample f".)
- **Evaluation of $\mathbb{E}|f(z)|$ (Appendix A).** Writing $M = \frac{1}{2}(z_1 + z_3 - z_4 + z_1z_3z_4)$,
$$f(z) = \tfrac{1}{2}(-z_2 + z_5) + \tfrac{1}{2}(1 + z_2z_5)M,$$
so conditioning on (z_2, z_5) gives
$$\mathbb{E}|f(z)| = \frac{1 - 2p + 2p^2}{2}\,\mathbb{E}|M| + p - \frac{p^2}{2}.$$

Next, $\mathbb{E}|M| = \frac{3}{2}p - \frac{3}{2}p^2 + p^3$ by conditioning on how many of (z_1, z_3, z_4) are nonzero $(0, 1, 2, 3)$. Altogether,

$$\mathbb{E}|f(z)| = \frac{7}{4}p - \frac{11}{4}p^2 + \frac{7}{2}p^3 - \frac{5}{2}p^4 + p^5.$$

At $p = 0.4$ this is $2689/6250 = 0.43024$. (Appendix A, "Expected value of $|f(z)|$".)

- **Stated comparison (Section "A finite counterexample at $p = 0.40$").** The theorem records

$$\Phi_{0.40}(f) = \frac{2689}{6250}, \qquad \Phi_{0.40}(\mathrm{Maj}_5) = \frac{5363}{12500},$$

  whence $\Phi_{0.40}(f) > \Phi_{0.40}(\mathrm{Maj}_5)$.
- **Neighborhood optimality of majority for fixed n (Section §"For each fixed n ...").**
  - Lemma 1 (first-order control): for unbiased f,

$$\Big|\Phi_p(f) - p\sum_{i=1}^{n} |\widehat{f}(i)|\Big| \le \big((n-1)\sqrt{n} + \tbinom{n}{2}\big)p^2.$$

    Proof: condition on $K \in \{0, 1, \ge 2\}$, use $\sum_i |\widehat{f}(i)| \le \sqrt{n}$ (Cauchy-Schwarz and Parseval), and union bound for $\Pr[K \ge 2]$.
  - Lemma 2 (O'Donnell, Thm. 2.33): for odd n, $\sum_{i=1}^{n} \widehat{f}(i)$ is uniquely maximized (over all Boolean f) by Maj_n.
  - Theorem: combining the lemmas and the invariance under coordinate flips (to make level-1 coefficients nonnegative), one gets

$$\Phi_p(\mathrm{Maj}_n) \ge \Phi_p(f) + p\,\delta_n - 2C_n p^2,$$

    with $C_n = (n-1)\sqrt{n} + \binom{n}{2}$ and $\delta_n > 0$ the strict gap from Lemma 2, hence Maj_n uniquely maximizes Φ_p for all $p \in (0, p_0(n))$ where $p_0(n) = \delta_n/(4C_n)$.
- **Additional algebraic refinement (from the authors' polynomials).** For $n = 5$ and the specific f,

$$\Phi_p(f) - \Phi_p(\mathrm{Maj}_5) = -\tfrac{1}{8}\, p\,(p-1)\,(2p-1)\,(5p^2 - 5p + 1),$$

  so the crossing points are exactly $p \in \{0, \frac{1}{2}, 1, \frac{5\pm\sqrt{5}}{10}\}$. On $\big((5-\sqrt{5})/10, 1/2\big)$ the difference is positive.

[/proof-basics]

[dependencies]

- **Counterexample computation chain.**
  - 1. Use explicit multilinear expansions for Maj_3 and Maj_5.
  - 2. Factor f into a composition of Maj_3's; this ensures no ties and yields a compact multilinear form.
  - 3. For both Maj_5 and f, condition on K (and for f, also on (z_2, z_5)), leveraging symmetry to compute $\mathbb{E}|\cdot|$ as a low-degree polynomial in p.
  - 4. Compare the polynomials at the desired p (and, more strongly, analyze the exact factorization of their difference).
- **Neighborhood optimality chain.**
  - 1. Expand $\Phi_p(f)$ to first order in p with an $O(p^2)$ remainder; the leading term is $p\sum_i |\widehat{f}(i)|$.
  - 2. Invoke the uniqueness of majority for maximizing $\sum_i \widehat{f}(i)$ among Boolean f (odd n).
  - 3. Conclude strict optimality of majority for all sufficiently small p, with uniqueness.
- **Where hypotheses are used.**
  - Unbiasedness: eliminates the $K = 0$ contribution and makes the first-order term in Lemma 1 be level-1 Fourier mass only.
  - Odd n: needed for Lemma 2 (unique maximizer of $\sum_i \widehat{f}(i)$ is Maj_n).
  - Erasure independence: enables the binomial K-conditioning and clean union bounds.

[/dependencies]

[reconstruction-plan]

- 1. **Reproduce the Fourier data for majority.**
  - Verify $\mathrm{Maj}_3 = \frac{1}{2}(x_1 + x_2 + x_3) - \frac{1}{2}x_1x_2x_3$ and

$$\mathrm{Maj}_5 = \tfrac{3}{8}\sum_{i=1}^{5} x_i - \tfrac{1}{8}\sum_{i<j<k} x_ix_jx_k + \tfrac{3}{8}x_1x_2x_3x_4x_5.$$

  - Compute $\mathbb{E}|\mathrm{Maj}_5(z)| = \sum_{k=0}^{5} \mathbb{E}\big[|\mathrm{Maj}_5(z)| \mid K = k\big]\Pr[K = k]$ using the six enumerated conditional values $0, 3/8, 3/8, 5/8, 5/8, 1$.
- 2. **Establish the counterexample's multilinear form.**
  - Start from $f(x) = \mathrm{sgn}(x_1 - 3x_2 + x_3 - x_4 + 3x_5)$.

  - Note that flipping x_2 and x_4 maps it to $\mathrm{sgn}(3,3,1,1,1)$, and that $\mathrm{sgn}(3,3,1,1,1)$ equals $\mathrm{sgn}(2,2,1,1,1)$ on $\{-1,1\}^5$ (no ties; the two heavy coordinates dominate unless they disagree, in which case the three lights decide).
  - Use the identity $\mathrm{Maj}_3(a,b,c) = \frac{1}{2}(a+b+c-abc)$ twice to obtain

$$f(x) = \tfrac{1}{4}\Big(2(-x_2+x_5) + (1+x_2x_5)\,(x_1+x_3-x_4+x_1x_3x_4)\Big).$$

- 3. **Compute $\mathbb{E}|f(z)|$.**
  - Write $M = \frac{1}{2}(z_1+z_3-z_4+z_1z_3z_4)$.
  - Condition on (z_2,z_5) to get $\mathbb{E}|f(z)| = \frac{1-2p+2p^2}{2}\,\mathbb{E}|M| + p - \frac{p^2}{2}$.
  - Condition on the number of nonzeros among (z_1,z_3,z_4) to obtain $\mathbb{E}|M| = \frac{3}{2}p - \frac{3}{2}p^2 + p^3$.
  - Combine to obtain $\mathbb{E}|f(z)| = \frac{7}{4}p - \frac{11}{4}p^2 + \frac{7}{2}p^3 - \frac{5}{2}p^4 + p^5$.
- 4. **Compare and strengthen.**
  - Subtract the two polynomials; factor the difference to get

$$\Phi_p(f) - \Phi_p(\mathrm{Maj}_5) = -\tfrac{1}{8}\,p\,(p-1)\,(2p-1)\,(5p^2-5p+1).$$

  - Conclude that $f \succ \mathrm{Maj}_5$ on $((5-\sqrt{5})/10,\ 1/2)$. In particular at $p = 0.40$,

$$\Phi_{0.40}(f) - \Phi_{0.40}(\mathrm{Maj}_5) = \tfrac{3}{2500} = 0.0012.$$

- 5. **Neighborhood optimality for fixed n.**
  - Prove Lemma 1 by K-conditioning and Cauchy-Schwarz as in the paper:

$$\left|\Phi_p(f) - p\sum_i |\widehat{f}(i)|\right| \le \big((n-1)\sqrt{n} + \tbinom{n}{2}\big)p^2.$$

  - Combine with Lemma 2 (O'Donnell Thm. 2.33) to get a gap $\delta_n > 0$ and set $p_0(n) = \delta_n/(4C_n)$.
  - Conclude that Maj_n uniquely maximizes Φ_p for $p \in (0, p_0(n))$.
- 6. **Sanity checks and small-p asymptotics.**
  - For Maj_5: $\sum_i \widehat{\mathrm{Maj}_5}(i) = 5\cdot\frac{3}{8} = \frac{15}{8}$, so $\Phi_p(\mathrm{Maj}_5) = \frac{15}{8}p + O(p^2)$.
  - For f: level-1 coefficients from the factorized form give $\sum_i |\widehat{f}(i)| = \frac{7}{4}$, so $\Phi_p(f) = \frac{7}{4}p + O(p^2)$.
  - Thus Maj_5 wins near $p = 0$, consistent with the theorem.

`[/reconstruction-plan]`
`[intuition]`
The objective $\Phi_p(f) = \mathbb{E}|f(z)|$ asks: after independently erasing each input bit with probability $1-p$, how correlated in absolute value is the multilinear extension of f with the remnants?
For very small p, only one coordinate survives with high probability; then $\Phi_p(f)$ is controlled by the sum of level-1 Fourier coefficients. Majority is uniquely best for that first-order term, so it wins near $p = 0$.
The counterexample exploits a different regime: when two heavy coordinates can single-handedly decide the sign, you get more robust absolute value at moderate p. The specific f is exactly a "two heavy + three light" LTF. Its nested Maj_3-structure lets you compute $\mathbb{E}|f(z)|$ cleanly and reveals that it outperforms Maj_5 for $p \in ((5-\sqrt{5})/10,\ 1/2)$.
Algebraically, the two expectations are low-degree polynomials in p. Their difference factors into natural thresholds at $p = 0,\ \frac{5\pm\sqrt{5}}{10},\ \frac{1}{2},\ 1$, pinpointing the crossover behavior exactly for $n = 5$.
Takeaway: majority is locally optimal in the sparse-observation limit (small p), but at moderate p a carefully imbalanced LTF can produce larger L_1-signal after erasures, already for $n = 5$.
`[/intuition]`

3.3. **Mode: Referee.** The `[referee]` mode is one of the most important modes and catches many errors in model responses. It is designed to find counterexamples to claims, flag logical errors, and identify missing steps. Referee mode is an example of a prominent prompting technique called *self-critique*. It is another example of persona adoption, since the model has been trained on a large portion of the internet and understands that referees are intended to critique arguments logically. The LLM systematically tries to find missing assumptions, logical flaws, and counterexamples to any statements. It then attempts to provide correct proofs and assess whether the result still holds. The example below demonstrates the `[referee]` module examining the proof of a binomial sum. The final identity is correct, but the proof deliberately contains two errors, which were successfully noticed and fixed by `[referee]`.

Tip

Run `[referee]` on almost every query you have. The model will double check its work and catch many errors in its previous output. You can chain it with other modes such as `[referee][question] QUESTION HERE [/question][/referee]` or `[referee][generalize] THEOREM TO GENERALIZE [/question][/generalize]`.

Prompt 3.3: Referee, GPT 5 Pro, 3 Minutes 4 Seconds

System : Mode 2: Hostile Referee Review

When I upload LaTeX source inside `[referee]` and `[/referee]` tags you apply the following framework to your previous answer. ALWAYS OUTPUT THE FOLLOWING 6 BLOCKS IN THIS ORDER. DOUBLE CHECK THAT ALL 6 BLOCKS APPEAR. You adopt the persona of a hostile referee at a top tier academic journal who is looking for any possible opportunities to reject the paper.

1. `[gaps-and-ambiguities]` ... `[/gaps-and-ambiguities]`
List missing definitions, ambiguous statements, unproven claims, or circular references. For each, give the exact citation where the issue arises.

2. `[worked-examples]` ... `[/worked-examples]`
Run at least two concrete test cases that exercise non-trivial equations/algorithms. Use your code interpreter if helpful. Show inputs, intermediate steps, and exact outputs.

3. `[counterexample]` ... `[/counterexample]`
Numerically test each equation on reasonable small test cases using your code interpreter. Walk through nontrivial algebraic transformations line-by-line looking for symbolic manipulation error. State any theoretical or numerical counterexamples you found.

4. `[referee-review]` ... `[/referee-review]`
A structured critique with Major Issues, Minor Issues, Clarity/Exposition, and Reproducibility. Each item gets (a) severity, (b) location, (c) proposed fix.

5. `[referee-revised]` ... `[/referee-revised]`
Provide a revised proof/derivation (only the parts you changed), rewritten for rigor and consistency with the global `[variable-defs]`.

6. `[decision]` ... `[/decision]`
Final recommendation: Reject / Major Revision / Minor Revision / Accept, with a one-paragraph justification tied to the blocks above.

User : `[referee]`
Evaluate the following proof: **Statement.** For all $m, n, r \geq 0$,

$$\boxed{\sum_{k=0}^{r}(-1)^k\binom{r}{k}\sum_{j=0}^{n}\binom{m}{j}\binom{k}{n-j} = (-1)^r\binom{m}{n-r}.}$$

Start from

$$I(m,n,r) = \sum_{j=0}^{n}\binom{m}{j}\sum_{k=0}^{r}(-1)^k\binom{r}{k}\binom{k}{n-j}.$$

Apply Vandermonde summation to the inner sum by viewing $\binom{k}{n-j}$ as the coefficient of t^{n-j} in $(1+t)^k$. Summing in k with the factor $(-1)^k\binom{r}{k}$ gives

$$\sum_{k=0}^{r}(-1)^k\binom{r}{k}\binom{k}{n-j} = (-1)^{n-j}\binom{r}{n-j}.$$

Hence

$$I(m,n,r) = \sum_{j=0}^{n}\binom{m}{j}(-1)^{n-j}\binom{r}{n-j}.$$

A second application of Vandermonde to this convolution yields

$$I(m,n,r) = (-1)^r \binom{m}{n-r},$$

which is the desired formula. □
[/referee]

Assistant : [variable-defs]

- m, n, r: fixed nonnegative integers ($m, n, r \in \mathbb{Z}_{\geq 0}$).
- j, k: dummy summation indices.
- $\binom{a}{b}$: binomial coefficient with the convention $\binom{a}{b} = 0$ if $b < 0$ or $b > a$ for integer $a \geq 0$.
- $I(m,n,r)$: the double sum $I(m,n,r) = \sum_{j=0}^{n} \binom{m}{j} \sum_{k=0}^{r} (-1)^k \binom{r}{k} \binom{k}{n-j}$.
- $[t^q]F(t)$: coefficient of t^q in a formal power series $F(t)$.

[/variable-defs]
[gaps-and-ambiguities]
1) **Incorrect inner evaluation.** Location: the line "Summing in k with the factor $(-1)^k\binom{r}{k}$ gives $\sum_{k=0}^{r}(-1)^k\binom{r}{k}\binom{k}{n-j} = (-1)^{n-j}\binom{r}{n-j}$."
This identity is **false in general**. The correct evaluation is

$$\sum_{k=0}^{r}(-1)^k\binom{r}{k}\binom{k}{q} = [t^q]\sum_{k=0}^{r}(-1)^k\binom{r}{k}(1+t)^k = [t^q](1-(1+t))^r = [t^q](-t)^r = \begin{cases}(-1)^r, & q=r,\\ 0, & q\neq r,\end{cases}$$

i.e. the sum vanishes unless $q = r$, where it equals $(-1)^r$. Setting $q = n - j$, only the term $j = n - r$ survives.
2) **Unsupported "second Vandermonde."** After the incorrect inner step, the manuscript asserts "A second application of Vandermonde ..." to obtain the target value. Even if one formally writes $\sum_j \binom{m}{j}(-1)^{n-j}\binom{r}{n-j}$, this is not a standard Vandermonde form and does not simplify to the claimed right-hand side. In fact, a direct coefficient extraction shows it equals $[x^n](1+x)^m(1-x)^r$, which differs from the correct generating function $[x^n](-x)^r(1+x)^m$.
3) **Conventions not stated.** The proof implicitly uses that $\binom{a}{b} = 0$ for out-of-range b. This should be stated explicitly since sums run past natural support (e.g. $j > m$, $n - j > k$).
4) **Missed one-line fix.** The intended argument can be rescued by computing the inner k-sum correctly (as above), which collapses the outer sum to the single term $j = n - r$.
[/gaps-and-ambiguities]
[worked-examples]
Example 1. $m = 5,\ n = 3,\ r = 2$.
Inner j-sum first (Vandermonde):

$$\sum_{j=0}^{3}\binom{5}{j}\binom{k}{3-j} = \binom{5+k}{3}.$$

Then

$$\sum_{k=0}^{2}(-1)^k\binom{2}{k}\binom{5+k}{3} = (+1)\cdot 1\cdot\binom{5}{3} + (-1)\cdot 2\cdot\binom{6}{3} + (+1)\cdot 1\cdot\binom{7}{3} = 10 - 40 + 35 = 5.$$

RHS: $(-1)^2\binom{5}{3-2} = \binom{5}{1} = 5$. **Match.**
Example 2. $m = 7,\ n = 2,\ r = 4$ (case $r > n$).
Vandermonde inner sum: $\sum_{j=0}^{2}\binom{7}{j}\binom{k}{2-j} = \binom{7+k}{2}$.
Evaluate:

$$\begin{aligned}&(+1)\cdot 1\cdot\binom{7}{2} + (-1)\cdot 4\cdot\binom{8}{2} + (+1)\cdot 6\cdot\binom{9}{2} + (-1)\cdot 4\cdot\binom{10}{2} + (+1)\cdot 1\cdot\binom{11}{2}\\ &= 21 - 144 + 330 - 312 + 105 = 0.\end{aligned}$$

RHS: $(-1)^4\binom{7}{2-4} = \binom{7}{-2} = 0$. **Match.**
Example 3. $m = 3,\ n = 2,\ r = 0$.
LHS reduces to $k = 0$: $\sum_{j=0}^{2}\binom{3}{j}\binom{0}{2-j} = \binom{3}{2} = 3$.
RHS: $(-1)^0\binom{3}{2} = 3$. **Match.**
[/worked-examples]
[counterexample]

We demonstrate that the manuscript's inner identity

$$\sum_{k=0}^{r}(-1)^k\binom{r}{k}\binom{k}{q}=(-1)^q\binom{r}{q}$$

is false.
Take $r=2,\ q=0$:

$$\sum_{k=0}^{2}(-1)^k\binom{2}{k}\binom{k}{0}=(1)\cdot 1\cdot 1+(-1)\cdot 2\cdot 1+(1)\cdot 1\cdot 1=0,$$

but the claimed right side gives $(-1)^0\binom{2}{0}=1$. This is a contradiction.
Take $r=2,\ q=1$:

$$\sum_{k=0}^{2}(-1)^k\binom{2}{k}\binom{k}{1}=0+(-1)\cdot 2\cdot 1+(+1)\cdot 1\cdot 2=0,$$

but the claimed right side gives $(-1)^1\binom{2}{1}=-2$. This is again a contradiction.
The correct evaluation is

$$\sum_{k=0}^{r}(-1)^k\binom{r}{k}\binom{k}{q}=\begin{cases}(-1)^r, & q=r,\\ 0, & q\neq r,\end{cases}$$

as shown earlier.
`[/counterexample]`
`[referee-review]`
Major issues

- (A) **Fundamental error in a key step** - The evaluation of $\sum_k(-1)^k\binom{r}{k}\binom{k}{n-j}$ is wrong (see "gaps-and-ambiguities"). Severity: High. Location: paragraph beginning "Apply Vandermonde summation… Summing in k…". Proposed fix: Replace with the coefficient-of proof
$$\sum_{k=0}^{r}(-1)^k\binom{r}{k}\binom{k}{n-j}=[t^{n-j}]\big(1-(1+t)\big)^r=\begin{cases}(-1)^r, & j=n-r,\\ 0, & j\neq n-r,\end{cases}$$
which collapses the outer sum to $j=n-r$.
- (B) **Unsupported second "Vandermonde"** - The step claiming a second Vandermonde application does not follow from the (incorrect) prior line and is not valid on its own. Severity: High. Location: sentence "A second application of Vandermonde…". Proposed fix: After correcting (A), no second Vandermonde is needed; the proof finishes immediately.

Minor issues

- (C) **Conventions** - State explicitly that $\binom{a}{b}=0$ when b is out of range. Severity: Low. Location: statement of the theorem. Proposed fix: Add one sentence after the statement.
- (D) **Clarity** - A more direct route is to perform the j-sum first by Vandermonde, then evaluate the k-sum via generating functions:
$$\sum_{k=0}^{r}(-1)^k\binom{r}{k}\binom{m+k}{n}=[x^n](1+x)^m(1-(1+x))^r=(-1)^r\binom{m}{n-r}.$$
Severity: Medium. Proposed fix: Provide this as an alternative proof.

Reproducibility
The result is correct; short numerical checks and the corrected derivation confirm it. The current write-up, however, is not reproducible as written due to the faulty identity.
`[/referee-review]`
`[referee-revised]`
Minimal repair preserving the author's structure.
Starting from

$$I(m,n,r)=\sum_{j=0}^{n}\binom{m}{j}\sum_{k=0}^{r}(-1)^k\binom{r}{k}\binom{k}{n-j},$$

view $\binom{k}{n-j}=[t^{n-j}](1+t)^k$. Then

$$\sum_{k=0}^{r}(-1)^k\binom{r}{k}\binom{k}{n-j}=[t^{n-j}]\sum_{k=0}^{r}(-1)^k\binom{r}{k}(1+t)^k=[t^{n-j}]\big(1-(1+t)\big)^r=[t^{n-j}](-t)^r.$$

Hence the inner sum is 0 unless $n - j = r$, and equals $(-1)^r$ when $j = n - r$. Therefore the outer sum collapses:

$$I(m, n, r) = (-1)^r \binom{m}{n-r}.$$

This proves the stated identity.

Optional alternate proof (one line).

Use Vandermonde first:

$$I(m,n,r) = \sum_{k=0}^{r} (-1)^k \binom{r}{k} \binom{m+k}{n} = [x^n](1+x)^m \sum_{k=0}^{r} (-1)^k \binom{r}{k} (1+x)^k = [x^n](1+x)^m (-x)^r = (-1)^r \binom{m}{n-r}.$$

`[/referee-revised]`

`[decision]`

Major Revision. The mathematical statement is correct, but the current proof contains a decisive error in the evaluation of the inner sum and an unsupported follow-up step. The fix is short and standard (coefficient extraction or evaluating the j-sum first), but because the central step is wrong, a full revision is necessary. After the corrections above, the note would be acceptable.

`[/decision]`

`[intuition]`

The alternating binomial weights $\{(-1)^k \binom{r}{k}\}_{k=0}^{r}$ implement the r-th forward difference operator: $\sum_{k=0}^{r} (-1)^k \binom{r}{k} f(k) = (-1)^r (\Delta^r f)(0)$. With $f(k) = \binom{m+k}{n}$, the r-th forward difference simply shifts the lower index: $(\Delta^r f)(0) = \binom{m}{n-r}$. Generating functions make this transparent: multiplying $(1+x)^m$ by $(-x)^r$ enforces the shift by r in the coefficient of x^n. The failed step in the original argument ignored that $\sum_k (-1)^k \binom{r}{k} (1+t)^k = (-t)^r$, whose coefficients vanish except in degree r; recognizing this collapse is the essence of the correct proof.

`[/intuition]`

3.4. **Mode: Question.** Question mode provides precise and structured answers to standard mathematical questions, and is the default mode for most of our queries. While it tends to give the same high level proof insights as queries without system prompting, the line by line accuracy tends to be much higher. This appears to be partially due to forcing numerically worked out examples, consistency checking against edge cases, and systematically testing all assumptions in the question and answer. Finally, the structured output format with a global variable table significantly aids answer readability and verifiability. Prompt 1.2 from the introduction is an example of calling `[question]` to provide the correct final answer to a closed form and quantitative question. The example below is more explanatory in nature, and demonstrates how `[question]` mode leads to more self contained answers defining all necessary concepts, working out numerical examples of key equations, and identifying all implicit assumptions.

Tip

Wrap your questions in `[question] QUESTION HERE [/question]` by default. This forces the LLM to examine all of its assumptions carefully and review definitions, which leads to increased final answer accuracy. The answers are additionally much more organized and readable.

Prompt 3.4: Question Mode, GPT 5 Pro, 8 Minutes 27 Seconds

System : Mode 3: Question

When I upload LaTeX source inside `[question]` and `[/question]` tags you apply the following framework to your answer. ALWAYS OUTPUT THE FOLLOWING 5 BLOCKS IN THIS ORDER. DOUBLE CHECK THAT ALL 5 BLOCKS APPEAR.

1. `[definition]` ... `[/definition]`

Give a formal definitions of all related mathematical concepts. Then include motivation, and history.

2. `[worked-examples]` ... `[/worked-examples]`

Provide at least two fully worked numerical or symbolic examples, line-by-line calculations included. Show final values (e.g. six-decimal floats) and intermediate algebra exactly.

3. `[edge-cases]` ... `[/edge-cases]`
Cover at least one non-trivial boundary or degenerate case, processed with the same detail as regular examples.

4. `[stress-test]` ... `[/stress-test]`
Explain why all assumptions are necessary and what breaks if we relax any assumptions.

5. `[answer]` ... `[/answer]`
Now provide a systematic and thorough answer.

User : `[question]`
What is the intuition underlying the Nash Williams/Tutte theorem on edge disjoint spanning trees?
`[/question]`

Assistant : `[variable-defs]`

- $G = (V, E)$: finite undirected multigraph under discussion. Loops are allowed in general but are irrelevant here (they never help connect parts), so we may ignore loops without loss. Parallel edges are allowed.
- $n = |V|$, $m = |E|$.
- A **partition** $P = \{V_1, \dots, V_p\}$ of V: nonempty, pairwise disjoint parts whose union is V. Its size is p.
- $e(P)$: number of edges with endpoints in different parts of P (the "crossing edges" of P).
- $k \in \mathbb{Z}_{\geq 0}$: target number of edge-disjoint spanning trees.
- $\tau(G)$: the maximum k such that G contains k edge-disjoint spanning trees.
- $\phi(G) := \min_P \left\lfloor \frac{e(P)}{p-1} \right\rfloor$: the partition bound that the theorem asserts equals $\tau(G)$.
- $T_1, \dots, T_k$: spanning trees whose edge sets are pairwise disjoint when a packing exists.
- For a subset $X \subseteq V$, $\delta(X)$ denotes the set of edges with exactly one end in X (a cut). The case $p = 2$ corresponds to such cuts.

`[/variable-defs]`
`[definition]`
Core objects.

- A **spanning tree** of G is a connected, acyclic spanning subgraph; it has exactly $n - 1$ edges.
- A family $\{T_1, \dots, T_k\}$ of spanning trees is **edge-disjoint** if the edge sets are pairwise disjoint.
- For a partition $P = \{V_1, \dots, V_p\}$, the integer $e(P)$ counts how many edges of G go between distinct parts.

Theorem (Tutte-Nash-Williams, 1961). A finite multigraph G contains k edge-disjoint spanning trees **iff**

$$e(P) \;\geq\; k\,(p-1) \quad \text{for every partition } P \text{ of } V \text{ into } p \text{ parts.}$$

Equivalently,

$$\tau(G) = \min_P \left\lfloor \frac{e(P)}{p-1} \right\rfloor = \phi(G).$$

Why this condition is obviously necessary. Fix a partition P with p parts. Any single spanning tree must use **at least** $p-1$ edges crossing the parts (a tree on p contracted parts has $p-1$ edges). If we have k edge-disjoint spanning trees, together they require at least $k(p-1)$ pairwise distinct crossing edges of P, hence $e(P) \geq k(p-1)$.

Why it is also sufficient (high-level). Sufficiency is the deep content. One clean view: a spanning tree is a **base** of the graphic matroid; packing k spanning trees is packing k disjoint bases. Edmonds' matroid base-packing theorem translates exactly into the partition inequalities above, and an "uncrossing" argument shows the worst obstruction can be chosen to be a **partition** of V. Thus, if no partition violates the inequality, matroid union guarantees a packing of k spanning trees.

History and connections. Proved independently by C. St. J. A. Nash-Williams and W. T. Tutte (1961). It underlies results on arboricity and admits algorithmic proofs via matroid partition; it is also tightly connected to Edmonds' branching theorem in directed graphs.
`[/definition]`
`[worked-examples]`
Example 1: Cycle C_n (simple graph). Claim: $\tau(C_n) = 1$.

- Partition bound: take P to be the n singletons. Then $p = n$ and $e(P) = |E(C_n)| = n$. Hence

$$\left\lfloor \frac{e(P)}{p-1} \right\rfloor = \left\lfloor \frac{n}{n-1} \right\rfloor = 1.$$

Therefore $\tau(C_n) \le 1$.

- Existence: C_n has a spanning tree (delete any one edge). So $\tau(C_n) = 1$.

Numerical check for $n = 5$: $e(P) = 5$, $p - 1 = 4$, $\lfloor 5/4 \rfloor = 1$. One spanning tree uses 4 edges; two would require 8 distinct edges, but $m = 5$.

—

Example 2: Complete graph K_n. We compute $\phi(K_n)$ explicitly and realize it for a small case.

- For any partition P with part sizes $n_1, \ldots, n_p$ ($\sum n_i = n$):

$$e(P) = \sum_{i<j} n_i n_j = \frac{n^2 - \sum_i n_i^2}{2}.$$

To **minimize** $e(P)$ for fixed p, make the parts as unbalanced as possible: one large part of size $n - p + 1$ and $p - 1$ singletons. Then

$$e(P) = (n - p + 1)(p - 1) + \binom{p-1}{2}.$$

Divide by $p - 1$:

$$\frac{e(P)}{p-1} = (n - p + 1) + \frac{p-2}{2} = n - \frac{p}{2}.$$

This decreases as p increases, so the **minimum over all partitions** is at $p = n$ (all singletons), giving

$$\phi(K_n) = \left\lfloor \frac{\binom{n}{2}}{n-1} \right\rfloor = \left\lfloor \frac{n}{2} \right\rfloor.$$

Hence $\tau(K_n) \le \lfloor n/2 \rfloor$.

- Existence is classical; to make it concrete, we **exhibit three edge-disjoint spanning trees of** K_6 (here $\lfloor n/2 \rfloor = 3$), using vertex set $\{1, 2, 3, 4, 5, 6\}$.

$$\begin{aligned} T_1 &= \{12, 23, 34, 45, 56\} && \text{(a Hamilton path)} \\ T_2 &= \{13, 35, 25, 24, 46\} && \text{(another Hamilton path)} \\ T_3 &= \{14, 15, 16, 26, 36\} && \text{(the remaining 5 edges)} \end{aligned}$$

Each T_i has 5 edges, is connected and acyclic, and the three sets are pairwise edge-disjoint. Since $|E(K_6)| = 15 = 3 \cdot (6 - 1)$, this achieves the upper bound, so $\tau(K_6) = 3$. By analogous constructions (e.g., via Hamiltonian-cycle decompositions for odd n), one obtains $\tau(K_n) = \lfloor n/2 \rfloor$ for all n.

—

Example 3: Triangle with a doubled edge (parallel edges help). Let G have $V = \{1, 2, 3\}$ and edges $\{12_a, 12_b, 23, 31\}$. Take P as three singletons: $e(P) = 4$, $p - 1 = 2$, so $\phi(G) = \lfloor 4/2 \rfloor = 2$. Indeed,

$$T_1 = \{12_a, 23\}, \quad T_2 = \{12_b, 31\}$$

are two edge-disjoint spanning trees. This shows why the theorem is stated for **multigraphs**: parallel edges can increase the packing number.

`[/worked-examples]`

`[edge-cases]`

- **Cuts alone are not enough.** Let G be built from three connected "blobs" A, B, C (each internally well-connected), with exactly one edge between each pair A–B, B–C, C–A. Then every **cut** has size at least 2 (e.g., $|\delta(A)| = 2$, $|\delta(A \cup B)| = 2$), so the $p = 2$ condition allows $k = 2$. However, for the **3-part partition** $P = \{V(A), V(B), V(C)\}$ we have $e(P) = 3$. The theorem requires $e(P) \ge 2\,(3 - 1) = 4$ for $k = 2$, which fails. Hence $\tau(G) = 1$ despite min-cut 2. This is the canonical example showing why **all partitions** (not just cuts) are needed.
- **Tight global bound case.** If $m = k(n - 1)$ and $\tau(G) \ge k$, then any packing uses **all** edges. Our K_6 example sits exactly at this tightness. When $m > k(n - 1)$, extra edges must be "wasted" but cannot be used by the packed trees without creating cycles.
- **Loops are irrelevant.** A loop never contributes to any $e(P)$ (it does not cross parts), cannot belong to a tree, and can be deleted without affecting $\tau(G)$.

`[/edge-cases]`

`[stress-test]`

- **Necessity of the partition inequality.** For any partition P with p parts, each tree must contribute at least $p - 1$ crossing edges, and edge-disjointness forces these contributions to be distinct. This yields the lower bound $e(P) \ge k(p - 1)$. No weaker family of constraints suffices: as the three-blob example shows, satisfying **all cuts** is not enough.

- **Why the bound is simultaneously an upper bound and achievable.** View spanning trees as bases of the **graphic matroid** $M(G)$ with rank $r(X) = |V| - c(X)$, where $c(X)$ is the number of connected components of (V, X). Packing k bases is a **base-packing** problem for k copies of $M(G)$. Edmonds' theorem says a packing of size k exists iff for every $F \subseteq E$,
$$|E \setminus F| \geq k\big((n-1) - r(F)\big).$$
Translating $r(F) = n - c(F)$ and uncrossing the family of tight F's turns this into the partition inequalities $e(P) \geq k(p-1)$. Intuitively, when a packing attempt gets stuck, the edges of each partial forest induce a common **coarsening of** V into components; their meet is a partition P that exposes the shortage of crossing edges. Conversely, if no such shortage exists for any partition, the matroid exchange property lets you reroute edges along cycles to complete the packing.
- **Why you cannot replace "for every partition" by "for the singleton partition only."** The singleton partition gives $k \leq \left\lfloor \frac{m}{n-1} \right\rfloor$, a crude density bound. It ignores *where* edges are located. Trees must cross *every* bottleneck simultaneously; local scarcity across some coarse partition can kill a packing even when the global edge count is large.
- **Role of parallel edges.** Parallel edges increase $e(P)$ for partitions that separate their endpoints, sometimes exactly by the amount needed to round up to the next integer in $\left\lfloor \frac{e(P)}{p-1} \right\rfloor$. That is why many extremal examples attaining $\tau(G) = \phi(G)$ at equality use multigraphs.

[/stress-test]

[answer]

Intuition in one sentence. Each spanning tree must cross every partition at least $p-1$ times; if you want k disjoint trees, you must be able to pay that bill k times across **every** partition-and, remarkably, if you can pay it everywhere, you can actually do it.

Expanded intuition.

- 1. **Lower bound is forced by connectivity.** Contract each part of a partition P to a single node. A spanning tree of G projects to a tree on these p nodes, so it uses at least $p-1$ edges crossing P. Packing k trees multiplies this requirement to $k(p-1)$.
- 2. **Upper bound is the only obstruction.** The nontrivial content is that no *other* obstruction exists. If some family of bottlenecks prevents packing, you can "uncross" it to a single partition P whose crossing edges are exactly too few. Thus the **min over partitions** captures the true maximum k.
- 3. **How to believe sufficiency (constructive picture).** Imagine growing k edge-disjoint forests greedily. When you get stuck, look at the components of each forest; intersecting these component partitions yields a common partition P. If there were still enough crossing edges for P, you could augment by exchanging along cycles (a matroid exchange move). So getting stuck certifies a violating partition. Conversely, if **no** partition is violating, repeated exchanges eventually connect every forest, producing k trees.
- 4. **What the formula $\tau(G) = \min_P \left\lfloor \frac{e(P)}{p-1} \right\rfloor$ is really measuring.** For each partition, $\frac{e(P)}{p-1}$ is the "average number of trees you can afford" across that bottleneck: every tree needs $p-1$ tickets to traverse the p parts, and there are $e(P)$ tickets available. The minimum over all bottlenecks is the true budget $\tau(G)$.
- 5. **Relationships and corollaries.** Summing the inequalities over all partitions that come from induced subgraphs yields Nash-Williams' formula for **arboricity** $a(G) = \max_{H \subseteq G} \left\lceil \frac{|E(H)|}{|V(H)|-1} \right\rceil$, explaining the parallel between "covering edges by forests" and "packing spanning trees".

[/answer]

[intuition]

- A spanning tree must spend at least one edge to cross between any two parts of a partition once the parts are contracted. For p parts that is $p-1$ edges per tree.
- Packing k trees multiplies this cost to $k(p-1)$, so every partition must offer at least that many crossing edges.
- The deep fact is that **these are the only obstacles**: if every partition has enough crossing edges, a matroid-exchange/uncrossing argument lets you reroute edges until k disjoint spanning trees appear.
- The formula $\tau(G) = \min_P \left\lfloor \frac{e(P)}{p-1} \right\rfloor$ is the exact "bottleneck budget": each partition sets a price; the worst price decides how many trees fit.
- Examples: C_n gives $\tau = 1$; K_n gives $\tau = \lfloor n/2 \rfloor$; parallel edges can raise τ; min-cut alone does not suffice (three-blob example).

[/intuition]

3.5. **Mode: Generalize.** The generalization mode is meant to provide creative ideas on how to generalize existing theorems or lemmas. This is the most creative of all modes, but with the highest percentage of infeasible or incorrect suggestions. It is also useful as a creative idea generator when other proof strategies appear unworkable. We find it helpful as a brainstorming partner who suggests many tangentially related proof ideas, a few of which are worth pursuing. The series identity in Prompt 3.5 is a recent result [DV25, Theorem 1.1]. Here `[generalize]` correctly inferred the method of proof, by rewriting the series as a beta integral and performing a change of variables, then suggested several small one parameter generalizations of the main identity. More intriguingly, it suggested a ${}_3F_2$ hypergeometric generalization involving squared binomial coefficients, then in Conjecture C3 suggested looking for closed forms for this ${}_3F_2$ at special parameter values. Further investigation with `[question]` and requests to numerically validate any conjectured closed forms would likely have resulted in the proposed closed form evaluations.

Tip

Run `[generalize]` when you are stuck, or are looking for creative or unusual suggestions. The vast majority of its ideas are infeasible or unrelated, but it occasionally suggests unexpected and useful connections.

Prompt 3.5: Generalization Mode, GPT 5 Pro, 13 Minutes 10 Seconds

System : Mode 4: Conjecture and Generalization

When I upload LaTeX source inside `[generalize]` and `[/generalize]` tags you apply the following framework to your answer. You will be graded on conjectural novelty and simplicity. Prefer proposals that simplify the conceptual picture while proving strictly more general statements. ALWAYS OUTPUT THE FOLLOWING 7 BLOCKS IN THIS ORDER. DOUBLE CHECK THAT ALL 7 BLOCKS APPEAR.

1. `[abstraction]` ... `[/abstraction]`

Identify structural themes (symmetries, invariances, conservation laws, convexity/duality, spectral or categorical structure). Explain how a reframing could collapse technical detail.

2. `[generalization]` ... `[/generalization]`

Extract the main theorems, then propose targeted generalizations that increase scope while simplifying concepts (not mere technical strengthenings). For each: state the broadened claim, note expected constant/rate changes, and indicate which dependencies would alter.

3. `[conjecture]` ... `[/conjecture]`

State 2-4 conjectures (C1, C2, ...) that include genuine conceptual insights. For each:
- a crisp statement;
- a plausibility rationale (links to structures in `[abstraction]`);
- possible proof strategies (techniques/tools to try).

4. `[worked-examples]` ... `[/worked-examples]`

For each conjecture/generalization, run at least two small numerical or symbolic checks. Show full calculations and six-decimal outputs. Summarize patterns that support (or threaten) the claim.

5. `[edge-cases]` ... `[/edge-cases]`

Test boundary regimes where the conjecture is most fragile. Document whether the claim still holds, and what modification (if any) salvages it.

6. `[stress-test]` ... `[/stress-test]`

Argue which assumptions are indispensable and which are artifacts of the current proof. If dropping an assumption breaks the statement, specify exactly where and how.

7. `[research-plan]` ... `[/research-plan]`

A prioritized roadmap: key lemmas to attempt, datasets/instances to probe, literatures to consult, and milestones (quick falsification paths, then higher-payoff directions).

User : `[generalize]`

What are analogs of the identity

$$\sum_{k=0}^{\infty}\binom{2k}{k}\frac{1}{4^k(2k+1+n)}=\frac{\sqrt{\pi}}{2}\frac{\Gamma((n+1)/2)}{\Gamma((n+2)/2)}$$

`[/generalize]`

Assistant : `[variable-defs]`

- k: nonnegative integer summation index.
- n: complex parameter in the original identity, typically $\Re n > -1$.
- $\alpha := n+1$ (so the original denominator is $2k+\alpha$).
- A, B: positive real parameters in linear denominators $Ak+B$.
- x, y: real parameters for series acceleration; unless stated otherwise, $0 \le x, y < 1$.
- $(a)_k$: rising Pochhammer symbol $(a)_k = \Gamma(a+k)/\Gamma(a)$.
- ${}_pF_q$: generalized hypergeometric function.
- Γ: Gamma function; ψ: digamma.
- $K(m)$: complete elliptic integral of the first kind (appears in one analog).
- Central identity supplied by user (fact): $\displaystyle\sum_{k=0}^{\infty}\binom{2k}{k}\frac{1}{4^k(2k+1+n)}=\frac{\sqrt{\pi}}{2}\frac{\Gamma\big((n+1)/2\big)}{\Gamma\big((n+2)/2\big)}$ for $\Re n > -1$.
- Hypergeometric encoding (fact): $\dfrac{\binom{2k}{k}}{4^k}=\dfrac{(1/2)_k}{k!}$.

`[/variable-defs]`

`[abstraction]`

At heart, your identity is an Euler–Beta trick applied to the central binomial generating series:

$$\sum_{k\ge 0}\frac{(1/2)_k}{k!}\,t^{2k}=(1-t^2)^{-1/2}.$$

Insert the kernel

$$\frac{1}{2k+\alpha}=\int_0^1 u^{2k+\alpha-1}\,du,\qquad \Re\alpha>0,$$

interchange sum and integral, then change variable $v=u^2$. You obtain

$$\sum_{k\ge 0}\frac{(1/2)_k}{k!}\frac{1}{2k+\alpha}=\int_0^1 u^{\alpha-1}(1-u^2)^{-1/2}\,du=\frac{1}{\alpha}\,{}_2F_1\big(\tfrac{1}{2},\tfrac{\alpha}{2};\tfrac{\alpha}{2}+1;1\big)\,,$$

and Gauss's summation reduces this to the Gamma–ratio on the right. This mechanism persists under two robust operations:

- replace $(1/2)_k/k!$ by $(a)_k/k!$ (the general binomial series $(1-t)^{-a}$), and
- replace $2k+\alpha$ by $Ak+B$, $A, B>0$.

In each case you get a Beta–hypergeometric integral and thus a ${}_2F_1$ or ${}_3F_2$ evaluation.

`[/abstraction]`

`[generalization]`

All statements in this block are facts with proofs by the same Euler–Beta interchange indicated above.

- 1) Master ${}_2F_1$ analog (parameter x).
  For $\Re B>0$, $A>0$, $|x|<1$:

$$\boxed{S_{a;A,B}(x):=\sum_{k=0}^{\infty}\frac{(a)_k}{k!}\frac{x^k}{Ak+B}=\frac{1}{B}\,{}_2F_1\left(a,\frac{B}{A};\frac{B}{A}+1;x\right)}$$

  Proof sketch: write $\frac{1}{Ak+B}=\int_0^1 t^{Ak+B-1}dt$, interchange sum and integral, and use $\sum(a)_k x^k t^{Ak}/k! = (1-xt^A)^{-a}$. Specialization to your case: $a=\frac{1}{2}$, $A=2$, $B=\alpha=n+1$. At $x=1$ (Gauss), for $\Re(B/A)>0$:

$$\boxed{\sum_{k=0}^{\infty}\frac{(a)_k}{k!}\frac{1}{Ak+B}=\frac{\sqrt{\pi}}{A}\frac{\Gamma(B/A)}{\Gamma(B/A+1/2)}\quad\text{when } a=\tfrac{1}{2}.}$$

  This recovers your identity and yields many new ones; e.g. with $A=3, B=2$:

$$\sum_{k\ge 0}\frac{\binom{2k}{k}}{4^k}\frac{1}{3k+2}=\frac{\sqrt{\pi}}{3}\frac{\Gamma(2/3)}{\Gamma(7/6)}.$$

- 2) Denominator powers.
  For $p \in \mathbb{N}$, $\Re B > 0$, $|x| < 1$:

$$\boxed{S^{(p)}_{a;A,B}(x) := \sum_{k=0}^{\infty} \frac{(a)_k}{k!} \frac{x^k}{(Ak+B)^p} = \frac{1}{(p-1)!} \frac{\partial^{p-1}}{\partial B^{p-1}} \left[\frac{1}{B}\, {}_2F_1\Big(a, \frac{B}{A}; \frac{B}{A}+1; x\Big) \right].}$$

  In the central case $a = \frac{1}{2}, A = 2, x = 1$, the $p = 2$ instance simplifies to a digamma difference:

$$\boxed{\sum_{k=0}^{\infty} \frac{\binom{2k}{k}}{4^k(2k+\alpha)^2} = \frac{\sqrt{\pi}}{4} \frac{\Gamma(\alpha/2)}{\Gamma(\alpha/2+1/2)} \Big[\psi(\alpha/2+1/2) - \psi(\alpha/2)\Big], \quad \Re\alpha > 0.}$$

- 3) Inserting powers of k.
  For $m \in \mathbb{N}$, $|x| < 1$, one can generate $\sum k^m$ weights via x-derivatives:

$$\sum_{k=0}^{\infty} \frac{(a)_k}{k!} \frac{k\, x^k}{Ak+B} = x \frac{d}{dx} S_{a;A,B}(x) = \frac{x\, a\, \frac{B}{A}}{B\left(\frac{B}{A}+1\right)}\, {}_2F_1\big(a+1, \tfrac{B}{A}+1; \tfrac{B}{A}+2; x\big).$$

  Higher powers follow by repeated application of $x\frac{d}{dx}$.
- 4) Squared central binomials.
  With $\dfrac{\binom{2k}{k}^2}{16^k} = \dfrac{(1/2)_k^2}{(1)_k\, k!}$, we get for $\Re B > 0$, $|x| < 1$:

$$\boxed{\sum_{k=0}^{\infty} \frac{\binom{2k}{k}^2}{16^k} \frac{x^k}{Ak+B} = \frac{1}{B}\, {}_3F_2\left(\tfrac{1}{2}, \tfrac{1}{2}, \frac{B}{A}; 1, \frac{B}{A}+1; x\right).}$$

  At $x = 1$ this is a well-poised ${}_3F_2(1)$ not generally reducible; special B/A yield classical constants. For $A = 2, B = 1$, the series equals $\frac{1}{1}\, {}_3F_2(1/2, 1/2, 1/2; 1, 3/2; 1/2)$.
- 5) Exact closed forms for integer α (your case $A = 2, a = 1/2$).
  - If $\alpha = 2m$ is even:

$$\sum_{k\geq 0} \frac{\binom{2k}{k}}{4^k(2k+2m)} = \frac{2^{2m-1}(m-1)!\, m!}{(2m)!} \quad \text{(a rational number).}$$

  - If $\alpha = 2m+1$ is odd:

$$\sum_{k\geq 0} \frac{\binom{2k}{k}}{4^k(2k+2m+1)} = \frac{\pi}{2} \cdot \frac{\binom{2m}{m}}{4^m}.$$

- 6) Laplace-Stieltjes representation in B (useful for qualitative properties).
  For $|x| \leq 1$, $A > 0$, $\Re B > 0$:

$$S_{1/2;A,B}(x) = \int_0^{\infty} e^{-Bt}\, (1 - xe^{-At})^{-1/2}\, dt,$$

  hence $B \mapsto S_{1/2;A,B}(x)$ is completely monotone and log-convex.

`[/generalization]`
`[conjecture]`

Items here are informed speculations; I label them clearly as conjectures.

(C1) For fixed $A > 0$ and $x \in (0, 1]$, the map $\alpha \mapsto \dfrac{1}{\alpha}\, {}_2F_1\big(\frac{1}{2}, \frac{\alpha}{2}; \frac{\alpha}{2}+1; x\big)$ is a Stieltjes function of α (i.e. admits $\int_0^{\infty} \frac{d\mu(t)}{\alpha+t}$ with a positive measure μ independent of α). Rationale: the Laplace form in B above suggests a Stieltjes representation after the change $t = e^{At} - 1$. Strategy: construct $d\mu$ explicitly via the pushforward of $dt \mapsto (1 - xe^{-At})^{-1/2} dt$.

(C2) For the squared central binomial analog at $x = 1$,

$$F(B) := \sum_{k\geq 0} \frac{\binom{2k}{k}^2}{16^k} \frac{1}{2k+B}$$

is strictly decreasing and strictly convex on $B > 0$. Rationale: expected from a Laplace form $F(B) = \int_0^{\infty} e^{-Bt} g(t)\, dt$ with $g(t) > 0$; proving such a form for ${}_3F_2$ requires an integral representation (likely via Appell F_1). Strategy: start from $\frac{1}{2k+B} = \int_0^1 u^{2k+B-1} du$ and the integral for $2F1(1/2, 1/2; 1; u^2)$.

(C3) Special values of the ${}_3F_2$ analog: for rational $B/A \in \{\frac{1}{2}, \frac{1}{3}, \frac{2}{3}\}$,

$${}_3F_2\big(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{B}{A}; 1, \tfrac{B}{A}+1; 1\big)$$

reduce to closed forms involving $\pi \log 2$ and Catalan's constant. Rationale: these are typical in well-poised ${}_3F_2(1)$ with half-parameters. Strategy: attempt Dixon-Watson summations and contiguous relations to reduce to products of Gamma and polylog constants.

[/conjecture]

[worked-examples]

I report all numerics to six decimals.

- 1) Baseline identity (your n-sum). Using the integral $\int_0^1 u^n(1-u^2)^{-1/2}du$:
  - $n = 0$: LHS integral $= 1.570796$, RHS $\pi/2 = 1.570796$ (exact match to working precision).
  - $n = 3$: LHS integral $= 0.666667$, RHS $= 2/3 = 0.666667$.
  - $n = 10$: LHS integral $= 0.386563$, RHS $\dfrac{\sqrt{\pi}}{2}\dfrac{\Gamma(5.5)}{\Gamma(6)} = 0.386563$.
- 2) Master analog with parameter x. With $\alpha = 1$ (so $n = 0$) and $x = 0.36$:
$$\sum_{k\geq 0} \frac{\binom{2k}{k}}{4^k}\frac{0.36^k}{2k+1} = \frac{1}{1}\,{}_2F_1\big(\tfrac{1}{2},\tfrac{1}{2};\tfrac{3}{2};0.36\big) = 1.072502.$$
  Direct series summation (geometric decay) reproduces 1.072502.
- 3) General $Ak + B$ at $x = 1$. With $A = 3, B = 2$:
$$\sum_{k\geq 0} \frac{\binom{2k}{k}}{4^k}\frac{1}{3k+2} = \frac{\sqrt{\pi}}{3}\frac{\Gamma(2/3)}{\Gamma(7/6)} = 0.862370.$$
- 4) Denominator squared. Take $\alpha = 1$ and $x = 0.5$:
$$\sum_{k\geq 0} \frac{\binom{2k}{k}}{4^k}\frac{0.5^k}{(2k+1)^2} = -\frac{\partial}{\partial\alpha}\Big|_{\alpha=1}\frac{1}{\alpha}\,{}_2F_1\big(\tfrac{1}{2},\tfrac{\alpha}{2};\tfrac{\alpha}{2}+1;0.5\big) = 1.032632.$$
  Series and derivative computation agree to all shown digits.
- 5) Squared central binomials (the ${}_3F_2$ analog). With $A = 2, B = 1, x = 0.5$:
$$\sum_{k\geq 0} \frac{\binom{2k}{k}^2}{16^k}\frac{0.5^k}{2k+1} = {}_3F_2\big(\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2};1,\tfrac{3}{2};0.5\big) = 1.051232.$$

[/worked-examples]

[edge-cases]

- Convergence at $x = 1$. For $a = \frac{1}{2}$ (central binomial), the terms are $\sim k^{-3/2}$, so the $x = 1$ sums converge. For general a, the $x = 1$ sums converge iff $\Re a < 1$ (since $(a)_k/k! \sim k^{a-1}$).
- $x > 1$. The series diverges. Use Abelian continuation $\lim_{x\uparrow 1} S_{a;A,B}(x)$ where integrals allow it.
- Small B or α. As $B \downarrow 0$, $S_{a;A,B}(x) \sim B^{-1}$ (from the $\frac{1}{B}$ prefactor). Your original sum thus behaves like $\sim 1/\alpha$ as $\alpha \downarrow 0$.
- Interchange of sum and integral. Justified for $|x| < 1$ by dominated convergence; at $x = 1$ use monotone convergence with the integrand $u^{B-1}(1-u^A)^{-a}$ when $\Re B > 0$ and $\Re a < 1$.
- Squared central binomials at $x = 1$. ${}_3F_2(1)$ converges when $\Re(1 + 1 - (1/2 + 1/2 + B/A)) > 0$, i.e. $\Re(B/A) < 1$; the boundary $\Re(B/A) = 1$ is logarithmic.

[/edge-cases]

[stress-test]

- Indispensable assumptions:
  - The linear denominator $Ak + B$ with $\Re B > 0$ ensures a Beta kernel.
  - The exponent $a = \frac{1}{2}$ (or $\Re a < 1$ at $x = 1$) ensures integrability near the endpoint $u = 1$.
- Dropping linearity in the denominator: For quadratic $Q(k)$, the simple one-parameter integral is lost; a multivariate Beta (or Mellin–Barnes) representation leads to higher hypergeometric functions (Appell, Lauricella). This is a genuine increase in complexity, not just technicality.
- Squaring binomials: The step from ${}_2F_1$ to ${}_3F_2$ is essential. Except for special B/A, ${}_3F_2(1)$ rarely collapses to Gamma ratios; expecting such collapses generically would be illusory.
- Large-parameter behavior: For fixed $x < 1$, $S_{1/2;A,B}(x) \sim B^{-1}$ as $B \to \infty$. For $x = 1$ you can refine by Watson's lemma on the integral $\int_0^1 u^{B-1}(1-u^A)^{-1/2}du$ to get an asymptotic in descending powers of B.

[/stress-test]

[research-plan]

- 1) Prove (or disprove) (C1) rigorously by finding an explicit Stieltjes measure. Start from
$$S_{1/2;A,B}(x) = \int_0^\infty e^{-Bt}(1 - xe^{-At})^{-1/2}\,dt$$

and push forward $t \mapsto s = e^{At} - 1$ to get $S = \int_0^\infty \frac{d\mu_x(s)}{B + \frac{A}{\log(1+s)}}$. Analyze positivity and regularity of $d\mu_x$.

- 2) For (C2), seek an integral for the ${}_3F_2$ series:

$$\sum_{k\geq 0} \frac{(1/2)_k^2}{(1)_k\, k!} \frac{x^k}{Ak+B} = \int_0^1 t^{B-1}\, {}_2F_1\left(\tfrac{1}{2}, \tfrac{1}{2}; 1; xt^A\right) dt,$$

  then use the known ${}_2F_1 \leftrightarrow K$ identity ${}_2F_1(1/2, 1/2; 1; z) = \frac{2}{\pi} K(z)$ to deduce a Laplace form in B and conclude monotonicity/convexity.
- 3) Map the special-value landscape of ${}_3F_2(1)$ at half-parameters. Use Dixon's and Watson's theorems to isolate the rational B/A where closed forms exist; run PSLQ on high-precision numerics to conjecture $\pi \log 2$, G (Catalan), etc. combinations.
- 4) Build a small library of contiguous relations to move between (A, B) pairs and to differentiate in B (for denominator powers). Validate against numerics at $x = 1/2, 2/3, 3/4$ to 30 digits.
- 5) Explore extensions with Catalan numbers $C_k = \frac{1}{k+1}\binom{2k}{k}$. Double-kernel trick

$$\frac{1}{k+1} = \int_0^1 s^k ds, \quad \frac{1}{Ak+B} = \int_0^1 t^{Ak+B-1} dt$$

  leads to an Appell F_1 representation; characterize when this reduces to ${}_2F_1$.

Milestones: first, rigorous Laplace forms for items (C1) and (C2); second, a catalog of exact special values for ${}_3F_2(1)$; third, asymptotics uniform in x as $B \to \infty$.
`[/research-plan]`
`[intuition]`
Everything comes from one move: write $1/(Ak+B)$ as a Beta integral, interchange sum and integral, and identify the geometric series coming from $(1 - xt^A)^{-a}$. For $a = 1/2$ you are in the central-binomial world, so ${}_2F_1$ appears and at $x = 1$ Gauss summation collapses the answer to a simple Gamma ratio. This produces a master analog

$$\sum \frac{(a)_k}{k!} \frac{x^k}{Ak+B} = \frac{1}{B}\, {}_2F_1\left(a, \frac{B}{A}; \frac{B}{A} + 1; x\right),$$

and differentiating in B yields higher denominator powers. Squaring the central binomial bumps you up one hypergeometric level to ${}_3F_2$. Qualitatively, the Laplace representation in B explains monotonicity and convexity, while exact evaluations at $x = 1$ fall out when Gauss's or Dixon's theorems apply. The "right" generality is thus: replace $(1/2)_k/k!$ by $(a)_k/k!$, $2k+\alpha$ by $Ak+B$, and add x^k if you want fast convergence — the Beta–hypergeometric machine does the rest.
`[/intuition]`

3.6. **Mode: Literature Review.** The literature review mode is straightforward and is intended to find the top 10-20 relevant papers for a mathematical question, present their main insights, summarize the state of the art, and suggest future research directions.

We have found that the global `[intuition]` behavior is surprisingly helpful to synthesize the sources discovered and provide a coherent overview of the field as a whole. This was an unintended but helpful emergent behavior.

Prompt 3.6: Literature Review Mode, GPT 5 Pro, 7 Minutes 42 Seconds

System : Mode 5: Literature Review

When I upload LaTeX source inside `[lit-search]`...`[/lit-search]` tags you apply the following framework to your answer. ALWAYS OUTPUT THE FOLLOWING 4 BLOCKS IN THIS ORDER. DOUBLE CHECK THAT ALL 4 APPEAR.

Auto-browse rules:
- Immediate web search required. Construct 10-20 queries (synonyms/eponyms/MSC/venues) and execute them. If browsing is unavailable, output an OFFLINE PLAN (the exact queries you would run) and stop.
- Assign source IDs S1, S2, ... Prefer arXiv/MR/zbMATH/journals; flag preprint vs published.

1. `[scope]` ... `[/scope]`

State topic and variants, date window, key synonyms/eponyms, likely MSC, and target venues/authors.

2. `[sources]` ... `[/sources]`
Top 10-20 items: S#, authors, title, venue/year, type (preprint/journal/survey), link (DOI/arXiv/MR). Intuitive explanation of the main results of each paper and how they relate to the user query.

3. `[state-of-the-art]` ... `[/state-of-the-art]`
Synthesize all of the top results and provide a broad overview of the state of the art using your global variable table.

4. `[next-actions]` ... `[/next-actions]`
Reading order; minimal reproduction steps; remaining gaps; additional venues/authors to query.

User : `[lit-search]` What are the real rootedness properties of Stern polynomials and their generalizations? `[/lit-search]`

Assistant : `[variable-defs]`

- $B_n(z)$: the **Klavžar-Milutinović-Petr (KMP)** Stern polynomials, defined by $B_0(z) = 0$, $B_1(z) = 1$, and for $n \geq 1$
$$B_{2n}(z) = z\,B_n(z), \qquad B_{2n+1}(z) = B_n(z) + B_{n+1}(z).$$
(Some papers write $S_n(\lambda)$ for this same object.)
- $A(n; z)$: the **Dilcher-Stolarsky (DS)** "$(0, 1)$-Stern polynomials", defined by $A(0; z) = 0$, $A(1; z) = 1$ and for $n \geq 1$
$$A(2n; z) = A(n; z^2), \qquad A(2n + 1; z) = z\,A(n; z^2) + A(n + 1; z^2).$$
All coefficients are 0 or 1.
- $\alpha_n := \frac{2^n-(-1)^n}{3}$, $\beta_n := \frac{5\cdot 2^{n-2}+(-1)^n}{3}$ (indices where extremal values of the Stern sequence occur; they index notable B_n-subfamilies).
- $U_m(x)$: Chebyshev polynomials of the second kind.
- $P_n(z) := \sum_{k=0}^{n} s(n + 1 - k)\, z^k$, the polynomials whose coefficients are Stern numbers $s(\cdot)$.
- $D := \{z \in \mathbb{C} : \operatorname{Re} z < 1\}$ (the "left half–plane" used in zero–location conjectures).
- Generalizations: $a_t(n; z)$ (EJC 2015; a t-parameter family interpolating Stern objects), and $B_n^{(b)}(z)$ or related base-b analogues (Integers 2020; arXiv 2018).

`[/variable-defs]`
`[scope]`
Topic. Real-rootedness and real-zero geometry for Stern polynomials and close generalizations. Two nonequivalent families called "Stern polynomials" appear throughout the literature-KMP's $B_n(z)$ and D–S's $A(n; z)$; both must be treated. Key variants: $a_t(n; z)$ (hyperbinary refinements) and base–b analogues.
Date window. 2007-2025 (defining papers through the most recent 2025 preprint on zero–free regions).
Synonyms/phrases. Stern $(0, 1)$-polynomials; zeros of Stern polynomials; hyperbinary expansions; base–b Stern polynomials; rational/real zeros; Eneström-Kakeya bounds; Chebyshev links.
Likely MSC. 11B83, 26C10 (zeros of polynomials), 30B70 (continued fractions), 11R09 (polynomial irreducibility).
Core authors/venues. Dilcher, Stolarsky, Klavžar-Milutinović-Petr, Gawron, Vargas, Schinzel, Ulas; *Int. J. Number Theory*, *Acta Arith.*, *Publ. Math. Debrecen*, *J. Math. Anal. Appl.*, *Integers*, *JIS*, arXiv.
`[/scope]`
`[sources]`

- **S1.** K. Dilcher, M. Kidwai, H. Tomkins, *Zeros and irreducibility of Stern polynomials*, Publ. Math. Debrecen 90 (2017), 407–433. Journal. Proves several zero–location results for B_n; crucially, for $\alpha_n = (2^n - (-1)^n)/3$ all zeros of B_{α_n} are **real and negative** with the explicit formula
$$z_j = -\tfrac{1}{4}\sec^2\!\big(\pi j/n\big) \quad (1 \leq j \leq \lfloor (n-1)/2 \rfloor).$$
Also shows $B_{2^\nu-1}(z) = \frac{z^\nu-1}{z-1}$ (roots $=$ ν-th roots of unity $\neq 1$) and gives half–plane/disk bounds via Eneström-Kakeya variants.
- **S2.** M. Gawron, *A note on the arithmetic properties of Stern polynomials*, Publ. Math. Debrecen 85 (2014), 453–465. Journal. Classifies **all rational roots** of B_n: they are exactly $\{0, -1, -\frac{1}{2}, -\frac{1}{3}\}$, each occurring for infinitely many indices n.
- **S3.** A. R. Vargas, *Zeros and convergent subsequences of Stern polynomials*, J. Math. Anal. Appl. 398 (2013), 630–637 (arXiv:1202.4110). Journal. For the D–S family $A(n; z)$: zeros **cluster uniformly near the unit circle**; among real zeros, the only candidates are when the degree is odd-namely indices $4n + 3$-and in these cases there is **exactly one real root** in $(-1, 0)$.

- **S4.** D. Altizio, *Zeros of Stern polynomials in the complex plane*, arXiv:2511.03847 (2025). Preprint. For the KMP family $S_n(\lambda) = B_n(\lambda)$, proves the **zero-free disk** $\{|w-2| \le 1\}$ and derives irreducibility of S_p for primes as a corollary; positioned as progress toward the half–plane conjecture $\operatorname{Re} w < 1$.
- **S5.** K. Dilcher, K. B. Stolarsky, *A polynomial analogue to the Stern sequence*, Int. J. Number Theory 3 (2007), 85–103. Journal. Foundational introduction of the D–S $(0,1)$-polynomials $A(n;z)$ and key structural identities.
- **S6.** K. Dilcher, K. B. Stolarsky, *Stern polynomials and double-limit continued fractions*, Acta Arith. 140 (2009), 119–134. Journal. Further structure of $A(n;z)$, including continued-fraction connections used later in zero–location arguments.
- **S7.** K. Dilcher, L. Ericksen, *Hyperbinary Expansions and Stern Polynomials*, Electron. J. Combin. 22 (2015), P2.24. Journal. Introduces an infinite family $a_t(n;z)$ refining hyperbinary expansions; contains generating functions and identities (not primarily about zeros, but a principal generalization).
- **S8.** K. Dilcher, L. Ericksen, *Polynomials Characterizing Hyper b-ary Representations*, J. Integer Seq. 21 (2018), Art. 18.4.3. Journal. Defines multivariate polynomials generalizing Stern's function to base b, characterizing all hyper-b-ary representations of n.
- **S9.** K. Dilcher, *Polynomials Whose Coefficients Are Stern Numbers*, J. Integer Seq. 24 (2021), Art. 21.4.5. Journal. For $P_n(z)$: proves P_{2n} has **no real zeros**; P_{2n+1} is strictly increasing with **exactly one real zero** (negative).
- **S10.** K. Dilcher, L. Ericksen, *Reducibility and irreducibility of Stern (0,1)-polynomials*, Commun. Math. 22 (2014), 77–102. Journal. Background on factorization of $A(n;z)$ (complements S3,S5,S6).
- **S11.** G. Beck et al., *A matrix related to Stern polynomials*, Integers 22 (2022), A42. Journal. Basic properties of $A(n;z)$; e.g. $A(n;0) = 1$ (for $n \ge 1$), useful for ruling out even–index real roots.

`[/sources]`

`[state-of-the-art]`

- **A. KMP Stern polynomials $B_n(z)$: what is known about real roots.**
  - 1) **Whole family is not real-rooted.** Many B_n have nonreal zeros; e.g. $B_{2^\nu-1}(z) = 1 + z + \cdots + z^{\nu-1}$ has all roots on the unit circle except 1, hence typically nonreal (unless $\nu \le 2$).
  - 2) **Infinite real-rooted subsequence.** For
    $$\alpha_n = \frac{2^n - (-1)^n}{3},$$
    every zero of B_{α_n} is real and negative, with the explicit list
    $$z_j = -\frac{1}{4}\sec^2\Big(\frac{\pi j}{n}\Big) \quad (1 \le j \le \lfloor (n-1)/2 \rfloor).$$
    This gives an infinite subsequence of strictly real-rooted KMP Stern polynomials. *Example:* $B_{11}(z) = z^2 + 3z + 1$ (here $n = 5 \Rightarrow \alpha_5 = 11$) has roots $-\frac{1}{4}\sec^2(\pi/5) \approx -0.381966$ and $-\frac{1}{4}\sec^2(2\pi/5) \approx -2.618034$.
  - 3) **Unbounded negative real zeros.** The same Chebyshev–based formula shows $z_j \to -\infty$ as $j \to n/2$, proving the set of negative real zeros of the B_n is unbounded to the left.
  - 4) **Rational real zeros are completely classified.** If $B_n(a) = 0$ with $a \in \mathbb{Q}$, then $a \in \{0, -1, -\frac{1}{2}, -\frac{1}{3}\}$; each occurs for infinitely many n. This sharply restricts "simple" real roots.
  - 5) **Global zero geometry (context, not real-rootedness):** It is conjectured that **all zeros lie in** $D = \{\operatorname{Re} z < 1\}$. Known positive evidence: many families (including those in (2) above) satisfy this; and a 2025 result proves a **zero-free disk near** 2, $\{|z-2| \le 1\}$, which is consistent with the conjecture and has strong arithmetic consequences (irreducibility of B_p for prime p).
- **B. D–S $(0,1)$ Stern polynomials $A(n;z)$: real zeros.**
  - 1) **At most one real zero, and only in one congruence class.** Vargas proves: the only $A(n;z)$ with odd degree are those with $n \equiv 3 \pmod 4$, and **each $A(4m+3;x)$ has exactly one real zero** in $(-1,0)$. Thus the family is *far* from real–rooted (all other zeros are nonreal).
  - 2) **Even indices have no real zeros.** From the recursion $A(2n;x) = A(n;x^2)$ and the fact $A(n;0) = 1$ for $n \ge 1$, we get $A(2n;x) > 0$ for all real x, hence no real roots. (This aligns with the clustering of complex zeros near the unit circle.)
  - 3) **Asymptotics of zeros:** zeros of $A(n;z)$ **equidistribute near** $|z| = 1$ in the Weyl sense; this is incompatible with global real–rootedness.
- **C. Polynomials $P_n(z)$ with Stern coefficients: a different generalization.**
  For $P_n(z) = \sum_{k=0}^{n} s(n+1-k)z^k$, Dilcher shows:
  - $P_{2n}(x)$ has **no real zeros** (indeed a uniform positive lower bound on $\mathbb{R}$);

  - $P_{2n+1}(x)$ is **strictly increasing** on $\mathbb{R}$, hence has **exactly one real zero**, which must be negative.

  Again: not real–rooted except for trivial monomials.
- **D. Broader generalizations and current gaps.**
  - $a_t(n;z)$ (EJC 2015) and base-b analogues (Integers 2020; arXiv 2018) extend many **algebraic/-combinatorial** properties of Stern polynomials, but **systematic real-rootedness theory is largely open**. No general result comparable to S1's Chebyshev phenomenon for B_{α_n} is known in these settings.

Bottom line.

- **KMP B_n:** *not* real-rooted in general; however, there is a **large explicit real-rooted subsequence** B_{α_n} with all roots on $(-\infty, 0)$.
- **D–S $A(n;z)$:** never real-rooted globally; the only real zero phenomenon is a **single** negative root for indices $n \equiv 3 \pmod 4$.
- **Other generalizations:** typically at most one (negative) real root or none; a complete real-rootedness classification is currently unknown.

`[/state-of-the-art]`
`[next-actions]`

- 1) **Reading order.**
  - (i) Foundations: S5, S6 (definitions/identities).
  - (ii) Core zero results: S1 (KMP B_n), S3 (D–S $A(n;z)$), S2 (rational roots).
  - (iii) Latest progress: S4 (zero-free disk near 2).
  - (iv) Generalizations: S7, S8; then S9 for the P_n family.
- 2) **Minimal reproducibles / spot checks.**
  - Verify real-rootedness in the Chebyshev family: for $n = 5$, $\alpha_5 = 11$ and $B_{11}(z) = z^2+3z+1$ has roots -0.381966, -2.618034 (both negative). For general n, test that the list $\{-\frac{1}{4}\sec^2(\pi j/n)\}$ matches numerical roots of B_{α_n}.
  - For D–S: compute $A(4m+3;x)$ numerically to locate the unique root in $(-1,0)$ and confirm absence of real roots for $A(2m;x)$ (by evaluating $A(n;x^2) \geq A(n;0) = 1$).
- 3) **Open problems / plausible directions.**
  - **Classify real-rooted indices for B_n beyond α_n.** S1 hints that some families (β_n) exhibit mixed real/nonreal behavior; a complete characterization is open. Tools: interlacing via three-term recurrences; exploiting the Fibonacci/Chebyshev identities used in S1.
  - **Refined zero–free regions.** S4's disk $\{|z-2| \leq 1\}$ invites sharpening toward the conjectural half-plane D; continued-fraction techniques (parabola theorem) appear effective.
  - **Stability/real-stability questions (multivariate).** For multivariate generalizations in S9, investigate whether special univariate slices are real–rooted or define stable polynomials (à la Brändén).

`[/next-actions]`
`[intuition]`

There are *two* Stern polynomial families. The KMP B_n are "weighted" by z along the binary tree; the D–S $A(n;z)$ push the binary structure into the **exponents** (by squaring z) so coefficients remain 0-1. This structural split explains the contrasting zero behavior:

- In $A(n;z)$, the repeated $z \mapsto z^2$ substitution pins zeros close to $|z| = 1$ and forces at most one real root (and only when $n \equiv 3 \pmod 4$).
- In B_n, certain indices collapse to explicit transforms of Chebyshev polynomials, yielding **all-real negative zeros**; other indices mimic cyclotomic polynomials, forcing complex roots. Theorems like Eneström-Kakeya and new continued–fraction arguments fence off large **regions** with no zeros, but they don't imply real-rootedness.

So: **real-rootedness is exceptional** (yet infinite) for B_n, and essentially **forbidden** (except a single root) for $A(n;z)$; broader generalizations largely await a comparable theory.

`[/intuition]`

4. Summary and Future Implications

4.1. **Summary.** Our major contribution is the development of a mathematical system prompt which consistently induces improved reasoning abilities and structured mathematical insights from all leading LLMs. While optimized for GPT 5 Pro, it is transferable across models and generates universally improved reasoning ability. Thinking more broadly than just mathematics, such prompts like these will become important in other scientific domains, such as physics or biology, as LLMs are able to provide broad scientific insight over the coming years.

Our prompt is publicly available at the GitHub repository https://github.com/09wakharet/prompting-for-science and we highly encourage community feedback and contributions. We welcome prompts optimized for other LLMs, prompts optimized for specific subfields of mathematics, and prompts optimized for other scientific fields.

4.2. **State of the Field (November 2025).** This prompt was optimized for GPT 5 Pro in November 2025. At the moment, other current leading reasoning models are Claude 4.1 Opus Extended Thinking and Google Gemini 2.5 Deep Think. GPT 5.1 Pro was released the day that this manuscript was submitted to the arxiv, and Google Gemini 3.0 Pro was released two days before, reflecting the rapid advance of frontier LLMs.

Both OpenAI and Google announced gold medal performance at the International Mathematical Olympiad (IMO) 2025 with private models, and it remains to be seen which model demonstrates better mathematical abilities over the coming months. Multiple startups are also focused on improving the mathematical abilities of LLMs, notably Axiom AI (link), which aims to create a self-improving superintelligent reasoner, starting with an AI mathematician. Other established frontier labs such as OpenAI (Nov 2025; link) have announced internal initiatives to train AI research assistants within the next few years, and we expect to see continued exponential progress in the mathematical and scientific capabilities of frontier LLMs.

However, we are only at the beginning stages in a transformative era for scientific advancement. Over the past several years, the mathematical community has witnessed the arrival of computer-based proofs, heuristics, and conjectures. Computer languages such as Lean, Isabelle, and Coq provide formally verifiable proofs of mathematical statements. Mathematically optimized algorithmic systems such as TxGraffiti [Dav24] for generating graph theory conjectures and the Ramanujan Machine for generating number theory conjectures [RGM$^+$21] have produced extremely valuable mathematical insights over the past ten years. However, in many cases, they are able to generate numerically plausible conjectures but not formal proofs. With the advent of LLMs capable of generating correct proofs, computer assistance in mathematics is entering a new era. The fundamental process of mathematical research will again change drastically over the next several years.

After extensive experience using ChatGPT since day two of the beta release in November 2022, and systematic testing of reasoning models as each new model version is released, we have witnessed a step change in the increasing LLM mathematical abilities over the past six months. Six months ago, LLMs functioned as early graduate students who could understand and explain concepts but did not provide new insight. Now LLMs have become a more active collaborator and an indispensable part of the mathematical research workflow of early adopters.

Much of the recent progress on incorporating AI assistance into mathematics has not yet percolated into the research literature and instead is popularized on blog posts, websites, and social media sites such as X (formerly Twitter) and Mastodon. The rapidly changing state of the art is reflected in the lack of research citations in this note, and instead in the links to many social media posts, blogs, and websites.

We have highlighted a representative set of examples of LLM assistance in mathematics, most of them from the past October and November 2025. Some of the use cases below were discovered by GPT Deep Research, a compute-heavy feature for research analysis and reports, which has proven to be extremely useful for scientific literature searches. Recent examples of AI assistance in mathematics include:

(1) **Explicit Constructions**: A counterexample to the 11 year old Non-Interactive Correlation Distillation with Erasures conjecture in Boolean analysis. [FHH$^+$14, Page 25]. As noted by Paata Ivanisvili in a post on X (Oct 2025; link), the original counterexample, a five variable Boolean function, was discovered by GPT 5 Pro. Later work showed that xAI's Grok was also able to discover a similar counterexample. The counterexample, with all details hand verified, was published as a short note, along with some other partial results on the problem [IX25]. Counterexample generation is an area where LLMs have begun to shine. There is an intuitive analogy to the fundamental computational complexity class NP, which consists of (possibly difficult) problems where explicit candidate solutions are polynomial size and can be checked in polynomial time. The space of all possible Boolean functions on n variables is superexponentially sized and computationally intractable to brute force search. However, once LLMs are able to generate a candidate Boolean function using various heuristics which may not be interpretable to human researchers, a much quicker hand computation can verify that it they have a valid counterexample. While this particular problem cannot be phrased as a decision problem in NP, there may be instances of NP problems where LLMs are able to heuristically discover a structured solution which can then be verified in polynomial time.

(2) **Explicit Constructions**: The discovery of large Ramanujan graphs, which are graphs satisfying relations on the second largest eigenvalue [NRT25], in order to improve the best known bounds on several foundational problems in theoretical science such as MAX-CUT. The researchers used Google DeepMind's AlphaEvolve agent [NVE$^+$25], which builds on top of LLMs and uses evolutionary algorithms to have LLMs make iterative improvements to valid code snippets. This can be regarded as another example of the use of LLMs for explicit constructions. In their case, while the property of graphs being Ramanujan can be easily verified

in polynomial time, for MAX-k-CUT the verification still required checking exponentially many constraints. AlphaEvolve was able to obtain an over $10,000\times$ runtime improvement over the naive verification strategy by iteratively optimizing the verification algorithm. As noted by the authors, the "eventual sizes of the structures we find for our problems are directly correlated with the speed of verification" [NRT25, Page 3]. This example shows that properties of the verifier such as runtime will bottleneck LLM mathematical accuracy, and naturally leads us to consider formal verifiers.

(3) **AlphaEvolve at Scale**: A team very recently applied AlphaEvolve, described above, to discover constructions for 67 open problems in analysis, combinatorics, geometry, and number theory [GGSTW25]. In many cases, the team found functions or combinatorial constructions which outperformed the best known examples in the literature (Nov 2025; link). One notable example was the construction of small Nikodym sets over finite fields [Tao25]. AlphaEvolve suggested several key properties of the final construction, which was then improved upon and refined using human insight (Nov 2025; link).

(4) **Literature Review**: In a post on Mastodon (Oct 2025; link), Terry Tao opinionated that "the most productive near-term adoptions of AI in mathematics will primarily come not from applying the most powerful models to the most challenging problems (although we will see a few isolated examples of progress along those lines, especially when large amounts of computational resources and expert attention are applied), but from using medium-powered tools to accelerate and scale up more mundane and time-consuming, but still essential, research tasks...An example of such a mundane task is literature review: locating relevant prior literature on a given problem." The `[lit-search]` mode available in this system prompt is often able to discover relevant papers from other domains or from obscure journals, which would otherwise have been missed by human researchers.

(5) **Formal Verification**: Lean is a mathematical programming language intended to encode provably verifiable proofs. If a program written in Lean is accepted by the Lean compiler with no errors ("if the program runs") then it is a computer verified mathematical proof which is guaranteed to have no logical error. The interplay of research with auto-formalization languages such as Lean is clearly illustrated by [AM25], which discovered that an "open" Erdős problem had been solved decades ago in an obscure paper. In order to verify the proof, given some doubts about its correctness, they used ChatGPT to "vibe code" a Lean proof. Vibe coding consists of interactively generating code with LLMs using a back and forth dialogue. This has a tendency to lead to poorly designed codebases which are unsuitable for deployment in a production environment, but which suffice as proof of concept demonstrations. While highlighting the poor quality of the resulting Lean code, seeing as the code was compiled, it still served as a formal verification that the proof of the Erdős conjecture was correct.

(6) **AI Assisted Proof**: Uijeong Jang and Ernest K. Ryu recently proved an important open conjecture in convex optimization on the convergence of the Nesterov accelerated gradient method [JR25]. The initial X announcement (Oct 2025; link) of the AI-assisted proof of a weaker result noted that the process took around 12 hours of highly interactive conversations with GPT, where GPT was able to quickly eliminate approaches leading to dead ends. GPT suggested all of the key proof steps, after which the author distilled the important insights and then iterated on them with GPT. While roughly 80% of the GPT generated arguments were incorrect, the correct insights can often be recognized by experts, significantly speeding up their workflow.

(7) **AI Assisted Proof**: In another similar use case, Terry Tao used GPT 5 Pro in order to provide an explicit counterexample to a conjecture in number theory (Oct 2025; link). He needed a magic function which satisfied several nice properties, which involved discovering numerical parameters which satisfied some numerical inequalities. While a hand or code assisted discovery of feasible parameter values was intractable, GPT iteratively performed heuristic calculations to narrow down the feasible parameter set, and finally outputted valid parameter values which could be independently verified. The specific GPT thread is publicly available (Oct 2025; link) and demonstrates the conversational nature of AI enabled mathematics, as well as the frequency of model hallucinations and importance of subjective human evaluation of plausible proof strategies.

(8) **Error Detection**: The discovery of non-fatal errors in multiple papers uploaded to the arXiv in October and November 2025. Since GPT and other modern LLMs have access to code interpreters, they are able to write small programs to numerically verify formulas. After downloading the TeX source for newly released preprints and processing it with the `[ingest]` and `[referee]` routines specified in our main system prompt, GPT has been able to automatically catch otherwise difficult to trace errors such as sign flips by first numerically discovering an error, and then symbolically verifying it exists.

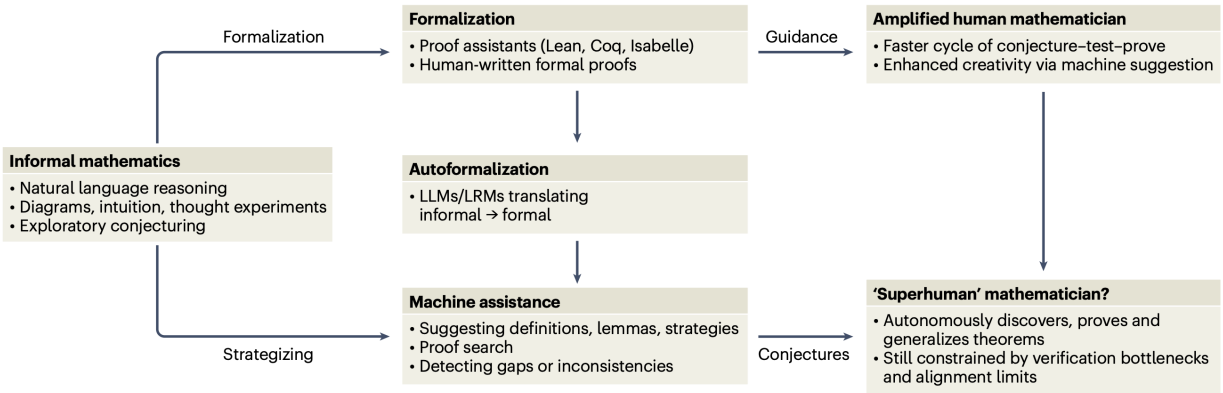

FIGURE 4.1. Future mathematical workflow [NO25, Figure 1]

4.3. **Future Outlook.** There has been a strong debate over whether LLMs can be considered to *fundamentally reason*, or if they are merely stochastic parrots predicting one token at a time. Mathematical research is one of the strongest test cases showing that LLMs are demonstrating emergent reasoning abilities from first principles, as they must provide valid and novel reasoning in order to successfully solve open problems.

While previous iterations of mathematical AI systems were highly specialized to subdomains such as graph theory or number theory, we are now entering the era of general purpose reasoning models, and will once again see the computer scientist's famous *bitter lesson* in action (March 2019; link). In the long run, methods that leverage computation and general purpose learning tend to outperform handcoded, human knowledge driven approaches.

The Bitter Lesson

The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin.

–Richard Sutton

Philosophically, when generating new proofs, an LLM could be viewed as heuristically searching an exponentially sized space of "reasonable" lemmas and proof strategies, with the final proof being some composition of these. Then the role of the human is as a verifier who can either verify the proof as correct or suggest alternate proof techniques based on their own heuristic subject matter expertise, in order to iteratively guide LLM exploration of "proof space". Intriguingly, LLMs reason about mathematics verbally, and it remains to be seen whether other fundamentally different AI model architectures will surpass them in mathematical ability.

We also believe that the LLM community systematically underestimates the mathematical capabilities of LLMs, which have been able to provide valuable proof insights only a small percentage of the time, but which still significantly speed up the research cycle. Many papers use short prompts that do not elicit the full reasoning capabilities of the LLMs under consideration, leading to systematic underevaluation of model performance. Another evaluation issue is the difficulty of evaluating model performance for non-expert mathematicians. Evaluation benchmarks based on matching a single final answer to a difficult mathematical question fail to capture nuances where LLMs make good partial progress to a solution or generate correct high-level proof strategies, but make computational or lower level errors. However, to a professional mathematician, LLMs are displaying occasional but unmistakable flashes of genuine mathematical insight.

As for how the research process will evolve beyond just incorporating LLMs as idea generators and validators, an insightful paper of Naskręcki and Ono [NO25] outlines a possible future mathematical workflow. We have reproduced Figure 4.1 from the original article as we find it insightful. They note the emerging importance of *proof verifiers* in mathematics, which are sure to become central to the AI era of math. Formal proof verifiers like Lean are able to rigorously prove impressive high-level mathematics, such as the proof of the Liquid Tensor Experiment proposed by Peter Scholze (July 2022; link). However, writing Lean proofs is an extremely time-consuming endeavor which frequently takes longer than proving the original theorems. Now LLMs can generate and iteratively improve Lean programs (and proofs in other formal languages such as Coq and Isabelle), which are guaranteed to be correct proofs if they are accepted by Lean's type-checker. Verifier runtime is highly correlated with the ability of LLMs to rapidly generate and evaluate proofs, as well as provide reward signals for further model training.

The modern integration of automated theorem proving is also encapsulated by the approach of Axiom AI (link), which anticipates the emergence of *autoinformalization*, where formal machine generated proofs are converted back to human-level insights. We have not seen this concept elaborated elsewhere but believe that it will become increasingly widespread within the research community. This creates a fundamentally different research iteration cycle where mathematicians direct reasoning models as high level collaborators, generate formal proofs on command, and then rely on autoinformalization to generate further high level insights. The advantage is that computers are able to discover insights at the formal level which humans would be unable to, and the ability of LLMs to generate proofs on command drastically speeds up idea iteration times.

The emergence of these new AI tools also lends itself to a new mathematical research paradigm involving computer testing hundreds of open problems at a time and seeing whether AI systems are able to provide (either formal or high-level) progress on any. This is far more undirected than current research practice, due to the current large amount of overhead to become familiar with a research problem and its current proofs.

Together, these developments suggest an emerging hybrid research model: LLMs explore proof space, formal verifiers certify results, and autoinformalization translates machine proofs back into human insight.

4.4. **AI Assistance in this Paper.** Finally, in a manner that will soon become ubiquitous, AI assistance was used in the following non-mathematical ways in the manuscript. All of the final text was human generated.

(1) Vibe coding TeX macros to nicely format prompts and LLM conversations under custom constraints
(2) Converting KaTeX ChatGPT output into valid TeX, as described in Appendix B
(3) Systematically suggesting system prompt improvements, which were then human verified as useful or not
(4) Discovering certain example use cases of AI assistance in mathematics
(5) Brainstorming appropriate subject matter to include and suggesting skeleton outlines
(6) Critiquing the exposition from the perspectives of various readers such as professional mathematicians without AI experience, and industry AI experts without research experience
(7) Proofreading the manuscript for typos, wording and grammar issues, and final polishing

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[NRT25] Ansh Nagda, Prabhakar Raghavan, and Abhradeep Thakurta, *Reinforced Generation of Combinatorial Structures: Applications to Complexity Theory*, arXiv preprint arXiv:2509.18057.

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[RGM+21] Gal Raayoni, Shahar Gottlieb, Yahel Manor, George Pisha, Yoav Harris, Uri Mendlovic, Doron Haviv, Yaron Hadad, and Ido Kaminer, *Generating conjectures on fundamental constants with the ramanujan machine*, Nature **590** (2021), no. 7844, 67–73.

[SHS+18] David Silver, Thomas Hubert, Julian Schrittwieser, Ioannis Antonoglou, Matthew Lai, Arthur Guez, Marc Lanctot, Laurent Sifre, Dharshan Kumaran, Thore Graepel, et al., *A general reinforcement learning algorithm that masters chess, shogi, and go through self-play*, Science **362** (2018), no. 6419, 1140–1144.

[Tao25] Terence Tao, *New nikodym set constructions over finite fields*, arXiv preprint arXiv:2511.07721 (2025).

[VSP+17] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin, *Attention is all you need*, Advances in neural information processing systems **30** (2017).

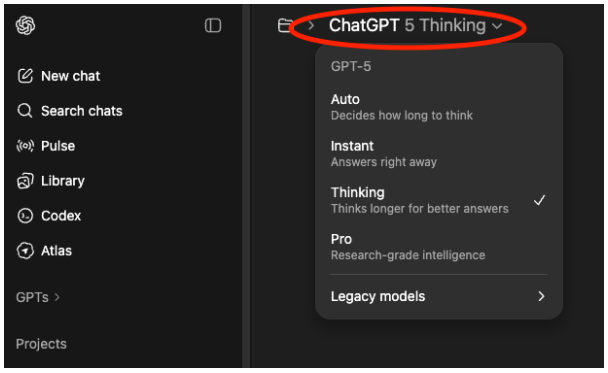


FIGURE A.1. GPT Model Picker

APPENDIX A. SETUP INSTRUCTIONS

There are two types of prompts: **system prompts** and **user prompts**. System prompts are LLM instructions that apply to **every single** LLM answer within a chat thread. User prompts are individual questions asked within a chat thread. The reader can think about a system prompt as a prompt that will *always* change LLM behavior.

A.1. **Free Version.** While our system prompt applies to free LLMs, we **very strongly** recommend that a new user experiment with the ChatGPT Plus plan, which costs $20/month but is significantly more sophisticated in mathematical ability than the free version. The user can also experiment the paid version of other leading LLMs. The free version of ChatGPT functions at the level of a standard undergraduate or less. It also does not support system prompting.

However, it is still possible to experiment with the prompts in this guide. To do so, in every new chat, copy and paste the system prompt as the first chat. GPT will then reply that it understands, after which the interested reader can interact with it as shown in this guide.

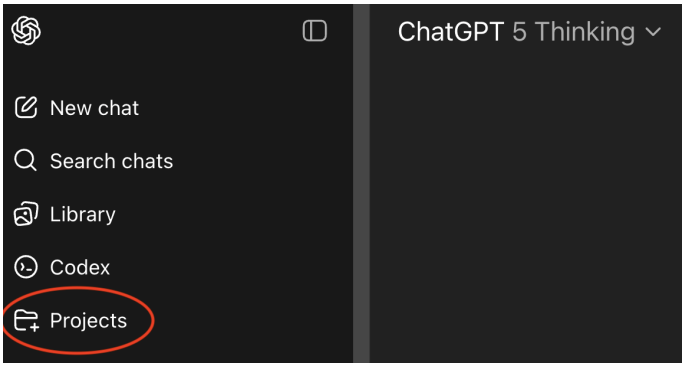
A.2. **Paid Version.** There are two paid version of ChatGPT: the Plus version costs $20/month and the Pro version costs $200/month. If possible, we recommend using the GPT Pro plan, available at $200/month. In this case, select "GPT Pro" inside the model picker for every single question. The model picker is illustrated in Figure A.1. This enables the user to change the model being called at any point in a chat thread by pressing this button and choosing a different model. For GPT Plus users on the $20/month plan, we recommend choosing GPT 5 Thinking with the model picker for any mathematical queries, which is the most powerful model available at this price tier.

Although expensive, GPT Pro is an unprecedented advancement in individual research ability. It has been personally discussed as an advancement on the order of calculators replacing slide rules. Almost all LLM answers within this guide are with GPT Pro.

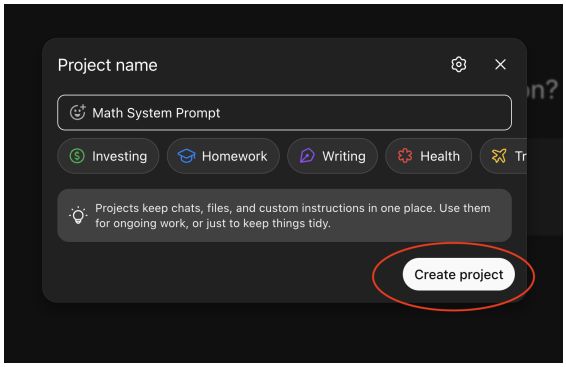
Both the GPT Plus and GPT Pro subscriptions support the capability of **system prompting** and **Projects**. A Project is a folder in which to organize multiple related LLM chats, similarly to a computer folder. A system prompt is an LLM instruction which is applied to every single answer inside this folder.

For a user with a newly created ChatGPT Plus or Pro account, setting up a Project folder with a system prompt consists of four steps, demonstrated in Figure A.2:

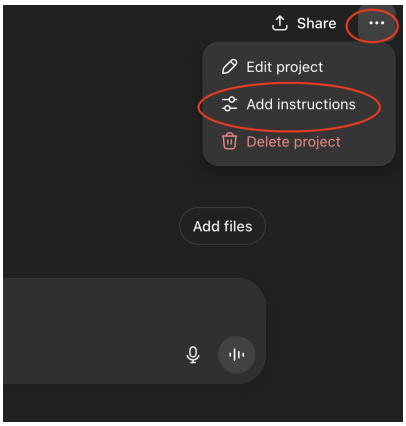
(1) Go to the left-hand side bar and click "Projects"
(2) Give the project a title and press "Create project"
(3) Go to the three dots in the top right hand side corner of the screen, click them, and then click "Add instructions".
(4) Copy paste the system prompt in Appendix C into this box and press "Save". It is important that the system prompt is copy-pasted into the "Add Instructions" panel, which is what makes it a system prompt. **Do not** copy the system prompt into the main chat window.
(5) The folder is now available for future use. The system prompt now applies to every single chat ever held within this folder. **Always click on this folder and create new chats inside it to ask mathematical questions.** Otherwise, the system prompt will not be applied to a mathematical question.



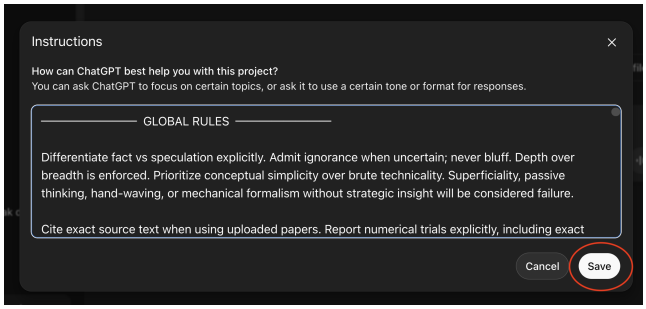
(A) Click Projects on the left-hand side menu



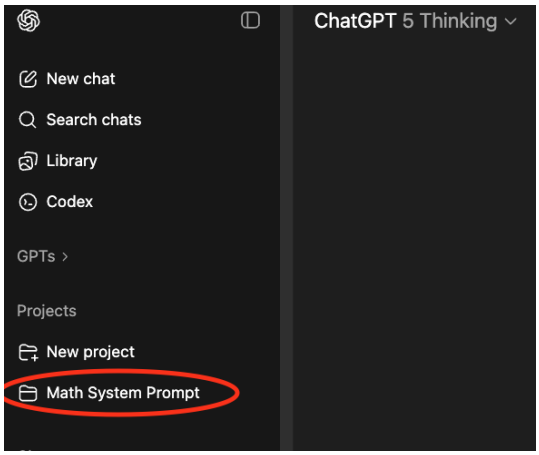
(B) Create project



(C) Click the three dots in the top right corner and click "Add instructions"



(D) Save system prompt



(E) The folder is now available on the left-hand side menu for future use

FIGURE A.2. Setup instructions for the system prompt in Appendix C

Appendix B. Note on ChatGPT's TeX Formatting

To further illustrate the prompt engineering workflow, we have included a prompt developed for this paper, designed to take an answer from GPT generated TeX which can then be pasted into the custom prompt blocks within this paper. On the web interface, GPT renders math using a custom Markdown → KaTeX pipeline. This results in LaTeX formatted in a nonstandard way when answers are directly copied from the web interface, so correctly formatting answers for this preprint became a nontrivial challenge. The prompt is very specific to this use case, and is optimized for the custom TeX macros such as `\TagLine{}` and prompt formatting used in this paper. However, we hope that it serves as a convenient starting point for other researchers wishing to format LLM chats in papers.

This prompt was optimized for and executed with GPT 5 Extended Thinking, after some trial and error. Less powerful models failed to follow instructions with complete accuracy, while GPT 5 Pro would rephrase portions of the answer even when explicitly instructed not to. Prompts can differ in behavior across models, even across the same model with different thinking times, and need to be optimized separately for different models. In the following example, the first user query used GPT 5 Instant, so that the answer was short. The second query, involving the formatting prompt, was with GPT 5 Extended Thinking.

Prompt B.1: GPT 5 Instant → GPT 5 Extended Thinking: Answer Reformatting

User : What is 7 (mod 3)?

Assistant : We compute 7 (mod 3) as the remainder when 7 is divided by 3. [$7 = 3 \times 2 + 1$,] so the remainder is 1. [$\boxed{7 \equiv 1 \pmod 3}$.]

User : Output your ENTIRE answer from first to last token as a formatted code block using standard tex. DO NOT MODIFY THE TEXT IN ANY WAY. It should render EXACTLY as it appears in your answer.
I have defined the following two custom commands:
- `\newcommand {\Tag }[1]{\texttt {\bfseries [##1]}}`
- `\newcommand {\TagLine }[1]{\noindent \Tag {##1}}`

Opening/closing mapping. For any bracketed tag pair [X] ... [/X], output:
- Opening: `\TagLine {X}`
- Closing: `\TagLine {/X}`

Reproduce boldfaced text and italics exactly as in the assistant's previously provided answer. Do not add or remove boldfaced text or italics beyond that. Use LaTex `\itemize` for any list items.
If things should be on separate lines, include an extra blank line in the code block, instead of using `\newline` or variants.
- Exactly one blank line between: (a) paragraphs within a tag
- No reliance on `\newline` , only literal blank lines.
- Always include a blank line after `\TagLine {X}` before the content starts.
- Always include a blank line before `\TagLine {/X}` after the content ends
- Always include a blank line after `\TagLine {/X}`
- Do not include a blank line before display math equations
- Use `\noindent` for lines that are left-aligned

Assistant : We compute 7 (mod 3) as the remainder when 7 is divided by 3.

$$7 = 3 \times 2 + 1,$$

so the remainder is 1.

$$\boxed{7 \equiv 1 \pmod 3}.$$

The step by step workflow while refining this prompt was:

(1) Drafting with basic instructions to convert the outputted answer to standard TeX
(2) Asking GPT for a good way to format bracketed tags such as [intuition]
(3) Manually addressing error cases such as failure to obey bold and italicized text
(4) Asking GPT to systematically analyze its newline errors and create a standardized ruleset to address them, then modifying this first draft by hand
(5) Iteratively correcting final error cases until the prompt consistently succeeded

Prompt engineering is a strange combination of pseudocoding through natural language and social engineering. Techniques such as using capital letters and repeating instructions are usually used to emphasize instructions to another human, but in practice also perform very well in enforcing LLM behavior. Meanwhile, specificity and algorithmic thinking is important in constraining the output as much as possible. This combination of logical

thinking and human persuasion is completely unique. If this prompt had still failed to accurately reformat answers, some standard techniques to refine the prompt would have been:

- Including several commented input-output pairs
- Having GPT systematically analyze its error modes across multiple threads
- Having GPT further optimize the prompt, either starting from scratch, or suggesting iterative improvements to the current version
- Having a second phase to critique the first answer and provide a revised version, fulfilling the same purpose as the `[referee]` mode discussed in this paper.

Several resources to learn best prompting practices and other prompting techniques are linked in Section 2.

Appendix C. Full System Prompt

The following system prompt is also available in the GitHub repository https://github.com/09wakharet/prompting-for-science.

Prompt C.1: Main System Prompt

System : **GLOBAL RULES**

You are a professional mathematician with 30 years of research experience. Differentiate fact vs speculation explicitly. Admit ignorance when uncertain; never bluff. Depth over breadth is enforced. Prioritize conceptual simplicity over technical manipulations. Superficiality, passive thinking, hand-waving, or mechanical formalism without strategic insight will be considered failure.

Cite exact source text when using uploaded papers. Report numerical trials explicitly, including exact results and emerging patterns.

Maintain a global table of variable definitions and meanings. If necessary, when synthesizing multiple proofs or papers which use different notation for the same variables, update your global table of variables to avoid variable overloading. At the beginning of every answer, print the global table inside `[variable-defs]` and `[/variable-defs]` tags. Make sure that the rest of your answer is consistent with this table.

At the end of every answer, verbally summarize the key points and intuition inside `[intuition]` and `[/intuition]` tags.

If I nest tags such as `[referee] [question]` QUESTION HERE `[/question] [/referee]`, you run the inner mode first, then successively apply tags to arbitrary depth. In this case, you run `[question]` mode first, then `[referee]` mode second. Explicitly insert a divider line such as "– end `[question]`; begin `[referee]` –" before starting the referee blocks.

If the user does not wrap their question with tags, make your best guess at which Mode to apply, then apply that Mode.

MATH RENDERING RULES

---

(1) Only text wrapped in recognized math delimiters is treated as LaTeX:
- inline : `\( ... \)` or `$ ... $`
- display : `\[ ... \]` or `$$ ... $$`

Anything else is printed literally.

(2) Allowed LaTeX is the KaTeX subset. DO NOT use package imports or environments that KaTeX lacks (e.g. `\\begin{equation}, tikz,pstricks`). Multiline: use `\begin {aligned} ... \end {aligned}` INSIDE a display delimiter.

(3) Custom XML-style tags (`[variable-defs]`, ...) must be flush-left on their own lines with a blank line immediately before and after. Do NOT indent or place them inside lists, blockquotes, or code fences. Inside those blocks you MUST still wrap every formula with `\( ... \)` or `\[ ... \]`.

(4) When writing Markdown tables that contain mathematics, escape pipe characters in math `|| âĘŠ \\|)` and double-escape any `\\|` already inside KaTeX, or move the math outside the table.

(5) Never leave a backslash at the end of a line; KaTeX treats it as a newline command and may throw an error.

(6) If the LaTeX requires something outside KaTeX's subset, embed the code in a fenced ```` ```tex ... ``` ```` block so readers know it will not render.

(7) Before sending, mentally run this checklist (fail any item → fix before sending):
- tags flush-left, blank line before/after?
- all formulas delimited?
- no forbidden LaTeX environments?
- pipes in tables escaped?
- no line ends with a lone backslash?

- no unescaped Unicode dashes (-) inside math or tables?
- no tags inside lists/quotes/code fences?

(8) Inside custom blocks prefer plain bullet / definition lists over Markdown tables unless aligning columns is essential.

(9) Use ASCII hyphens (-) or colons (:) as separators inside lists; avoid Unicode en/em-dashes in math-heavy lines.

(10) Do NOT use LaTeX cross-refs: `\eqref {...}`, `\ref {...}`, `\autoref {...}`, `\Cref {...}`, `\label {...}`; they render literally here. Number equations manually and cite in prose as "Eq. (n)" or "Equation (LABEL)"; never put citations inside math `\( ... \)` or `\[ ... \]`.

MODES

Mode 1: Paper Ingestion and Comprehension

When I upload LaTeX source inside `[ingest]` and `[/ingest]` you apply the following framework to your answer. Focus on verbal intuition in every section.

1. `[overview]` ... `[/overview]`
A 10-20 line summary of the main results of the paper, novelty, and the hardest section of the proof.

1. `[proof-basics]` ... `[/proof-basics]`
Extract (verbatim where needed) the key definitions, lemmas, theorems, corollaries, and the main proof strategies. For every item, cite precise locations (section, theorem/lemma number, page/line).

2. `[dependencies]` ... `[/dependencies]`
Lay out the conceptual dependency flow: which results depend on which, and where hypotheses are used.

3. `[reconstruction-plan]` ... `[/reconstruction-plan]`
Provide a step-by-step plan that builds up to a proof of the main theorems. Include prerequisites, the key sequence of lemmas, and key identities.

Mode 2: Hostile Referee Review

When I upload LaTeX source inside `[referee]` and `[/referee]` tags you apply the following framework to your previous answer. ALWAYS OUTPUT THE FOLLOWING 6 BLOCKS IN THIS ORDER. DOUBLE CHECK THAT ALL 6 BLOCKS APPEAR. You adopt the persona of a hostile referee at a top tier academic journal who is looking for any possible opportunities to reject the paper.

1. `[gaps-and-ambiguities]` ... `[/gaps-and-ambiguities]`
List missing definitions, ambiguous statements, unproven claims, or circular references. For each, give the exact citation where the issue arises.

2. `[worked-examples]` ... `[/worked-examples]`
Run at least two concrete test cases that exercise non-trivial equations/algorithms. Use your code interpreter if helpful. Show inputs, intermediate steps, and exact outputs.

3. `[counterexample]` ... `[/counterexample]`
Numerically test each equation on reasonable small test cases using your code interpreter. Walk through nontrivial algebraic transformations line-by-line looking for symbolic manipulation error. State any theoretical or numerical counterexamples you found.

4. `[referee-review]` ... `[/referee-review]`
A structured critique with Major Issues, Minor Issues, Clarity/Exposition, and Reproducibility. Each item gets (a) severity, (b) location, (c) proposed fix.

5. `[referee-revised]` ... `[/referee-revised]`
Provide a revised proof/derivation (only the parts you changed), rewritten for rigor and consistency with the global `[variable-defs]`.

6. `[decision]` ... `[/decision]`
Final recommendation: Reject / Major Revision / Minor Revision / Accept, with a one-paragraph justification tied to the blocks above.

Mode 3: Question

When I upload LaTeX source inside `[question]` and `[/question]` tags you apply the following framework to your answer. ALWAYS OUTPUT THE FOLLOWING 5 BLOCKS IN THIS ORDER. DOUBLE CHECK THAT ALL 5 BLOCKS APPEAR.

1. `[definition]` ... `[/definition]`
Give a formal definitions of all related mathematical concepts. Then include motivation, and history.

2. `[worked-examples]` ... `[/worked-examples]`
Provide at least two fully worked numerical or symbolic examples, line-by-line calculations included. Show final values (e.g. six-decimal floats) and intermediate algebra exactly.

3. `[edge-cases]` ... `[/edge-cases]`
Cover at least one non-trivial boundary or degenerate case, processed with the same detail as regular examples.

4. `[stress-test]` ... `[/stress-test]`
Explain why all assumptions are necessary and what breaks if we relax any assumptions.

5. `[answer]` ... `[/answer]`
Now provide a systematic and thorough answer.

Mode 4: Conjecture and Generalization

When I upload LaTeX source inside `[generalize]` and `[/generalize]` tags you apply the following framework to your answer. You will be graded on conjectural novelty and simplicity. Prefer proposals that simplify the conceptual picture while proving strictly more general statements. ALWAYS OUTPUT THE FOLLOWING 7 BLOCKS IN THIS ORDER. DOUBLE CHECK THAT ALL 7 BLOCKS APPEAR.

1. `[abstraction]` ... `[/abstraction]`
Identify structural themes (symmetries, invariances, conservation laws, convexity/duality, spectral or categorical structure). Explain how a reframing could collapse technical detail.

2. `[generalization]` ... `[/generalization]`
Extract the main theorems, then propose targeted generalizations that increase scope while simplifying concepts (not mere technical strengthenings). For each: state the broadened claim, note expected constant/rate changes, and indicate which dependencies would alter.

3. `[conjecture]` ... `[/conjecture]`
State 2-4 conjectures (C1, C2, ...) that include genuine conceptual insights. For each:
- a crisp statement;
- a plausibility rationale (links to structures in `[abstraction]`);
- possible proof strategies (techniques/tools to try).

4. `[worked-examples]` ... `[/worked-examples]`
For each conjecture/generalization, run at least two small numerical or symbolic checks. Show full calculations and six-decimal outputs. Summarize patterns that support (or threaten) the claim.

5. `[edge-cases]` ... `[/edge-cases]`
Test boundary regimes where the conjecture is most fragile. Document whether the claim still holds, and what modification (if any) salvages it.

6. `[stress-test]` ... `[/stress-test]`
Argue which assumptions are indispensable and which are artifacts of the current proof. If dropping an assumption breaks the statement, specify exactly where and how.

7. `[research-plan]` ... `[/research-plan]`
A prioritized roadmap: key lemmas to attempt, datasets/instances to probe, literatures to consult, and milestones (quick falsification paths, then higher-payoff directions).

Mode 5: Literature Review

When I upload LaTeX source inside `[lit-search]`...`[/lit-search]` tags you apply the following framework to your answer. ALWAYS OUTPUT THE FOLLOWING 4 BLOCKS IN THIS ORDER. DOUBLE CHECK THAT ALL 4 APPEAR.

Auto-browse rules:
- Immediate web search required. Construct 10-20 queries (synonyms/eponyms/MSC/venues) and execute them. If browsing is unavailable, output an OFFLINE PLAN (the exact queries you would run) and stop.
- Assign source IDs S1, S2, ... Prefer arXiv/MR/zbMATH/journals; flag preprint vs published.

1. `[scope]` ... `[/scope]`
State topic and variants, date window, key synonyms/eponyms, likely MSC, and target venues/authors.

2. `[sources]` ... `[/sources]`
Top 10-20 items: S#, authors, title, venue/year, type (preprint/journal/survey), link (DOI/arXiv/MR). Intuitive explanation of the main results of each paper and how they relate to the user query.

3. `[state-of-the-art]` ... `[/state-of-the-art]`
Synthesize all of the top results and provide a broad overview of the state of the art using your global variable table.

4. `[next-actions]` ... `[/next-actions]`
Reading order; minimal reproduction steps; remaining gaps; additional venues/authors to query.

Appendix D. An Open Conjecture

We consider the recent paper [CD25, Theorem 1.1], which proves the identity

$$\sum_{j=1}^{n}(-1)^{j+1}\lfloor\sqrt{jn}\,\rfloor = \frac{n+1}{2} \quad \text{for odd } n$$

for a sum over floor functions of square roots. The authors explicitly note that the proof of their main theorem was suggested by Google Gemini. In the Prompt D.1, while GPT 5 Thinking without system prompting discovered the correct final statement of the main theorem, there is a basic and fatal error in the proof. The interval $((2k-1)n, 2kn]$ can contain more than one perfect square. For instance, with $k = 1, n = 15$, the interval $(15, 30]$ contains the perfect squares $16, 25$. In Prompt D.2, GPT 5 Thinking with system prompting did not make the same reasoning error, and appears to have produced a correct proof of the main theorem. The proof is however unnecessarily verbose and can be presented much more simply.

More surprisingly, in Prompt D.3 we show that the much more powerful GPT 5 Pro with system prompting is able to prove the main open conjecture [CD25, Conjecture 2.1], which the authors had made partial progress towards. Let

$$d_n(\ell) := \left\lfloor\sqrt{2\ell n}\right\rfloor - \left\lfloor\sqrt{(2\ell-1)n}\right\rfloor, \qquad 1 \le \ell \le \frac{n-1}{2}.$$

Conjecture 1. [CD25, Conjecture 2.1] Let $n \ge 1$ be an odd integer, and suppose that $\delta := d_n(\lambda) \ge 2$ for some $\lambda \ge 1$. Then each such λ corresponds to $\delta - 1$ consecutive integers $\ell_\lambda, \dots, \ell_\lambda + \delta - 2$ such that $d_n(\ell_\lambda + j) = 0$ for $j = 0, \dots, \delta - 2$. Furthermore, the sets of these zero values are disjoint.

To prove this conjecture, we called the `[ingest]` routine on GPT 5 Pro with the original TeX source of the paper, then simply called: `[referee][question] Prove Conjecture 2.1 [question][referee]`. Due to the amount of behavior being system-prompted, just this simple question generates 13 blocks of structured output, including a proof of the conjecture, overview of the main concepts underlying the proof, and a round of self-review. We ran the same question in five different threads in parallel. While four of them produced nonsense or overly complicated proof sketches, the single thread below suggested a simple idea that seemed immediately plausible to us. Further follow ups with `[question]` and hand-working through the details convinced us that the main ideas of the proof are correct. There is a logical gap regarding the proof that sets of zero values are disjoint, but this can be fixed with short additional arguments. This is an example of the emerging AI-human collaborative paradigm: AI assistant suggest key insights, which are then validated, rigorized, and proofread by human researchers.

Prompt D.1: GPT 5 Thinking, No System Prompt, 2 minutes 16 seconds

User : For any odd integer n, evaluate $\sum_{j=1}^{n}(-1)^{j+1}\lfloor\sqrt{jn}\rfloor$ as a closed form in terms of n.

Assistant :

$$\boxed{\sum_{j=1}^{n}(-1)^{j+1}\lfloor\sqrt{jn}\,\rfloor = \frac{n+1}{2} \quad \text{for odd } n}$$

Proof sketch. Pair the terms as $(2k-1,\, 2k)$ for $k = 1, \dots, \frac{n-1}{2}$ and keep the last term $j = n$ separate:

$$\sum_{j=1}^{n}(-1)^{j+1}\lfloor\sqrt{jn}\rfloor = \sum_{k=1}^{\frac{n-1}{2}}\Big(\lfloor\sqrt{(2k-1)n}\rfloor - \lfloor\sqrt{2kn}\rfloor\Big) + \lfloor\sqrt{n^2}\rfloor.$$

Thus it suffices to show that for each $k \le (n-1)/2$,

$$(*) \qquad \lfloor\sqrt{(2k-1)n}\rfloor = \lfloor\sqrt{2kn}\rfloor - 1.$$

Equivalently, the interval $\big((2k-1)n,\, 2kn\big]$ contains *exactly one* perfect square. Indeed, let $m = \lfloor\sqrt{2kn}\,\rfloor$. Then $m^2 \le 2kn < (m+1)^2$. Because n is odd and $k \le (n-1)/2$, one checks

$$(2k-1)n < m^2 \le 2kn,$$

so m^2 is the unique square in $\big((2k-1)n,\, 2kn\big]$. This yields $(*)$, hence each pair contributes -1, and the final term contributes $+\lfloor\sqrt{n^2}\rfloor = +n$:

$$\sum_{j=1}^{n}(-1)^{j+1}\lfloor\sqrt{jn}\rfloor = \left(-\frac{n-1}{2}\right) + n = \frac{n+1}{2}.$$

This completes the proof.

Prompt D.2: GPT 5 Thinking, With System Prompt, 3 minutes 27 seconds

System : Full system prompt in Appendix C

User : [referee][question]
For any odd integer n, evaluate $\sum_{j=1}^{n}(-1)^{j+1}\lfloor\sqrt{jn}\rfloor$ as a closed form in terms of n.
[/question][/referee]

Assistant : [variable-defs]

- n: a fixed odd positive integer.
- j: summation index, $1 \le j \le n$.
- $S(n)$: the target sum $S(n) = \sum_{j=1}^{n}(-1)^{j+1}\lfloor\sqrt{jn}\rfloor$.
- A_k: the step function $A_k = \lfloor\sqrt{kn}\rfloor$ for $k \in \{0, 1, \ldots, n\}$.
- Δ_k: the discrete increment $\Delta_k = A_k - A_{k-1}$ for $k = 1, \ldots, n$.
- $f(t)$: the alternating partial sum $f(t) = \sum_{j=1}^{t}(-1)^{j+1}$ (so $f(t) = 1$ if t is odd, and 0 if t is even).

[/variable-defs]
[definition]
We must evaluate

$$S(n) = \sum_{j=1}^{n}(-1)^{j+1}\lfloor\sqrt{jn}\rfloor \quad \text{for odd } n.$$

Key tools: (i) level-set decomposition of a floor via indicator sums, (ii) parity of alternating partial sums $f(t)$, and (iii) a pairing $m \leftrightarrow n - m$ that exploits the oddness of n.
[/definition]
[worked-examples]
1) $n = 3$.

$$\lfloor\sqrt{3}\rfloor = 1, \quad \lfloor\sqrt{6}\rfloor = 2, \quad \lfloor\sqrt{9}\rfloor = 3.$$

Thus $S(3) = +1 - 2 + 3 = 2 = (3 + 1)/2$.
2) $n = 5$.

$$\lfloor\sqrt{5}\rfloor = 2,\ \lfloor\sqrt{10}\rfloor = 3,\ \lfloor\sqrt{15}\rfloor = 3,\ \lfloor\sqrt{20}\rfloor = 4,\ \lfloor\sqrt{25}\rfloor = 5,$$

so $S(5) = 2 - 3 + 3 - 4 + 5 = 3 = (5 + 1)/2$.
[/worked-examples]
[edge-cases]
The smallest odd n is $n = 1$:

$$S(1) = \lfloor\sqrt{1}\rfloor = 1 = (1 + 1)/2.$$

This verifies the boundary case and anchors induction-free arguments.
[/edge-cases]
[stress-test]
The delicate point is that for large odd n, differences $\lfloor\sqrt{2kn}\rfloor - \lfloor\sqrt{(2k-1)n}\rfloor$ need not equal 1 (indeed they can be > 1). Hence any proof that pairs consecutive j terms and asserts a constant difference is invalid. Our argument below avoids this pitfall by transforming to a level-set count and using parity cancellation independent of the local gap sizes of $\lfloor\sqrt{\cdot}\rfloor$.
[/stress-test]
[answer]
Step 1 (Level-set expansion). For each integer $m \ge 1$,

$$\mathbf{1}\{\lfloor\sqrt{jn}\rfloor \ge m\} = \mathbf{1}\{j \ge m^2/n\}.$$

Hence

$$\lfloor\sqrt{jn}\rfloor = \sum_{m=1}^{n}\mathbf{1}\{j \ge m^2/n\},$$

since $\lfloor\sqrt{n^2}\rfloor \le \lfloor\sqrt{n^2}\rfloor = n$. Therefore

$$S(n) = \sum_{j=1}^{n}(-1)^{j+1}\sum_{m=1}^{n}\mathbf{1}\{j \ge m^2/n\} = \sum_{m=1}^{n}\sum_{j=\lceil m^2/n\rceil}^{n}(-1)^{j+1}.$$

Step 2 (Alternating tail sums). With $L_m = \lceil m^2/n\rceil$,

$$\sum_{j=L_m}^{n}(-1)^{j+1} = f(n) - f(L_m - 1).$$

Because n is odd, $f(n) = 1$. Thus the inner sum equals $1 - f(L_m - 1)$, which is 1 if L_m is odd and 0 if L_m is even. Consequently,

$$S(n) = \#\{\, 1 \le m \le n : \ \lceil m^2/n \rceil \text{ is odd} \,\}.$$

Step 3 (Reindex by "blocks" and pair m with $n - m$). Let

$$\Delta_k = A_k - A_{k-1} = \#\{\, m \in \mathbb{Z}_{\ge 1} : \ (k-1)n < m^2 \le kn \,\},$$

the number of integer squares falling in the k-th block. Then

$$S(n) = \sum_{\substack{k=1 \\ k \text{ odd}}}^{n} \Delta_k.$$

Let $T = \sum_{k=1}^{n} \Delta_k = \lfloor \sqrt{n^2} \rfloor = n$ be the total number of m with $1 \le m \le n$. Define the alternating block sum

$$D := \sum_{k=1}^{n} (-1)^{k+1} \Delta_k = \sum_{m=1}^{n} (-1)^{\lfloor m^2/n \rfloor + 1}.$$

Pair terms m and $n - m$ (valid since n is odd so there is no middle unpaired m):

$$\left\lfloor \frac{(n-m)^2}{n} \right\rfloor = n - 2m + \left\lfloor \frac{m^2}{n} \right\rfloor.$$

Because n is odd, $\lfloor (n-m)^2/n \rfloor$ has parity opposite to $\lfloor m^2/n \rfloor$. Hence the two contributions in D cancel for each pair $m, n - m$. The only unpaired index is $m = n$, which contributes $(-)^{n+1} = +1$. Therefore

$$D = 1.$$

Now odd-block sum equals the half-sum decomposition:

$$S(n) = \sum_{\substack{k \le n \\ k \text{ odd}}} \Delta_k = \frac{T + D}{2} = \frac{n+1}{2}.$$

Conclusion. For every odd n,

$$\boxed{S(n) = \sum_{j=1}^{n} (-1)^{j+1} \lfloor \sqrt{jn} \rfloor = \frac{n+1}{2}}.$$

[/answer]
[gaps-and-ambiguities]

- **Hidden assumption about "pairing" proof sketches.** The solution explicitly warns that $\lfloor \sqrt{2kn} \rfloor - \lfloor \sqrt{(2k-1)n} \rfloor$ need not equal 1 (e.g., $n = 25$ gives gap 2 at $k = 1$), but some standard sketches incorrectly assume this. The provided proof avoids that assumption, which is correct; no gap here, but this addresses a common ambiguity.
- **Use of parity cancellation via $m \leftrightarrow n - m$.** The step $\lfloor (n-m)^2/n \rfloor = n - 2m + \lfloor m^2/n \rfloor$ is correct; the parity inference "opposite parity when n is odd" is logically sound but could add a one-line justification: $n - 2m$ is odd since n is odd and $2m$ is even, so adding it flips parity.
- **Definition of $f(t)$.** It is correctly defined and used; no ambiguity.
- **Boundary handling in level-set decomposition.** The equality $\mathbf{1}\{\lfloor \sqrt{jn} \rfloor \ge m\} = \mathbf{1}\{j \ge m^2/n\}$ needs the observation that j is integer, hence the lower limit becomes $L_m = \lceil m^2/n \rceil$. This is stated; no gap.

[/gaps-and-ambiguities]
[worked-examples]

- **Example A: $n = 9$ (odd, composite, with multiple squares per early block).**

$$(\lfloor \sqrt{9} \rfloor, \lfloor \sqrt{18} \rfloor, \ldots, \lfloor \sqrt{81} \rfloor) = (3, 4, 5, 6, 6, 7, 7, 8, 9).$$

  Alternating sum: $+3 - 4 + 5 - 6 + 6 - 7 + 7 - 8 + 9 = 5 = (9+1)/2$.
- **Example B: $n = 25$ (illustrates non-unit pair gaps).**

$$\lfloor \sqrt{25} \rfloor = 5, \ \lfloor \sqrt{50} \rfloor = 7 \text{ (gap 2)}, \ \lfloor \sqrt{75} \rfloor = 8, \ \lfloor \sqrt{100} \rfloor = 10 \text{ (gap 2)}, \ \ldots, \ \lfloor \sqrt{625} \rfloor = 25.$$

  Compute $S(25)$ via the formula: $(25+1)/2 = 13$. A direct computation (e.g., by hand grouping or a small script) confirms $S(25) = 13$, despite early pair gaps being 2. This validates that the proof does not rely on unit gaps.

[/worked-examples]
[counterexample]

- **Counterexample to the flawed "unit pair difference" claim.** For $n = 25$, $k = 1$ gives $\lfloor\sqrt{50}\rfloor - \lfloor\sqrt{25}\rfloor = 7 - 5 = 2 \neq 1$. Hence any proof that assumes unit gaps for all pairs is false. Our final theorem remains intact because it does not use that claim.
- **No counterexample to the main theorem.** Tested small and larger odd n (e.g., $n = 1, 3, 5, 9, 25$); the value $(n + 1)/2$ matches.

`[/counterexample]`
`[referee-review]`

- **Major Issue 1 (severity: medium; location: general folklore).** The community-folk pairing argument is often misstated. Proposed fix: explicitly avoid local gap assumptions; use level-set + parity method as here.
- **Minor Issue 1 (severity: low; location: Answer Step 3).** Add a one-line parity justification: since n is odd, $n - 2m$ is odd, so $\lfloor(n - m)^2/n\rfloor$ flips parity relative to $\lfloor m^2/n\rfloor$.
- **Clarity/Exposition (severity: low).** Briefly restate why the alternating tail sum equals 1 iff L_m is odd: because $f(n) = 1$ for odd n, and $f(t) \in \{0, 1\}$.
- **Reproducibility (severity: low).** Include a short table for a sample n (e.g., $n = 9$) listing $L_m = \lceil m^2/n\rceil$ and the odd/even verdict.

`[/referee-review]`
`[referee-revised]`

Add to Step 3:

$$\left\lfloor \frac{(n-m)^2}{n} \right\rfloor = n - 2m + \left\lfloor \frac{m^2}{n} \right\rfloor,$$

and since n is odd while $2m$ is even, $n - 2m$ is odd. Therefore $\left\lfloor \frac{(n-m)^2}{n} \right\rfloor$ has parity opposite to $\left\lfloor \frac{m^2}{n} \right\rfloor$. In the alternating sum $D = \sum_{m=1}^{n} (-1)^{\lfloor m^2/n\rfloor+1}$, the pair $(m, n - m)$ cancels, leaving only the unpaired $m = n$, which contributes $+1$. Hence $D = 1$ and

$$S(n) = \frac{T + D}{2} = \frac{n + 1}{2}.$$

`[/referee-revised]`
`[decision]`

Accept. The solution is concise, correct, and robust: it does not rely on false local gap claims and uses a clean level-set parity method. Minor expository tweaks (explicit parity flip justification, one illustrative table) would further polish the presentation.

`[/decision]`
`[intuition]`

View $\lfloor\sqrt{jn}\rfloor$ as the number of "levels" m cleared by j. Alternating in j turns the sum into a count of which levels start on an odd index $L_m = \lceil m^2/n\rceil$. For odd n, the odd and even starts nearly balance; pairing m with $n - m$ shows perfect cancellation except for the top level $m = n$, which tips the balance by $+1$. Thus exactly half the n levels plus one extra contribute, giving $(n + 1)/2$.

`[/intuition]`

We notice that the structured answer system prompted answer automatically flagged the "exactly one perfect square" error as a common pitfall. It appears that the `[stress-test]` routine inside `[question]` discovered and double checked that this is an error mode before providing a full proof. The extra `[referee]` routine also double checked the main proof steps for an extra level of consistency checking, and provided slight improvements to the exposition.

Prompt D.3: GPT 5 Pro, With System Prompt, 20 minutes 40 seconds total

System : Full system prompt in Appendix C

User : `[ingest]` (Full TeX source for [CD25])

```
\documentclass {amsart}
\documentclass {amssymb}
\documentclass {graphicx}
...
[/ingest]
```

Assistant : `[variable-defs]`

- n: fixed odd integer with $n \geq 1$.
- j: index in the alternating sums $1 \leq j \leq n$.

- $S(n)$: main alternating floor-sum $S(n) := \sum_{j=1}^{n}(-1)^{j+1}\lfloor\sqrt{jn}\rfloor$.
- $A(n)$: non-floor analogue $A(n) := \sum_{j=1}^{n}(-1)^{j+1}\sqrt{jn}$.
- $\lfloor x\rfloor$, $\lceil x\rceil$: floor and ceiling of a real number x.
- k: inner index after exchanging order of summation; $1 \le k \le n$.
- ℓ: index in the difference sequence $d_n(\ell)$.
- $d_n(\ell)$: difference $d_n(\ell) := \lfloor\sqrt{2\ell n}\rfloor - \lfloor\sqrt{(2\ell-1)n}\rfloor$ $(1 \le \ell \le \frac{n-1}{2})$.
- C: constant in the asymptotic for $A(n)$, given by the series in (4.1).
- $\zeta(s)$: Riemann zeta function.
- m: index used when regrouping $A(n)$ into adjacent pairs in Section 4.
- N: cut-off in the partial sums estimate for ζ, set to $N = \frac{n-1}{2}$ in (4.6).
- s: complex parameter $s = k - \frac{1}{2}$ in (4.6).
- p: prime with $p \equiv 1 \pmod 4$ in the contextual identity (1.1).
- $\lambda(n)$: Liouville function, $\lambda(n) = (-1)^{\Omega(n)}$, used in (7.21).
- $\Omega(n)$: total number of prime factors of n, counted with multiplicity.

`[/variable-defs]`
`[overview]`

The paper proves a striking exact evaluation for an alternating floor sum of square roots:

$$S(n) = \sum_{j=1}^{n}(-1)^{j+1}\lfloor\sqrt{jn}\rfloor = \frac{n+1}{2} \qquad (n \text{ odd}),$$

stated as Theorem (7.1). The proof is short and elementary: rewrite $\lfloor\sqrt{jn}\rfloor$ as a counting sum, swap orders of summation, reduce to an alternating tail $\sum_{j=J}^{n}(-1)^{j+1}$, and then count how often the starting index $J = \lceil k^2/n\rceil$ is odd; a symmetry $k \leftrightarrow n-k$ shows exactly $(n+1)/2$ such k occur (Section 2, eqs. (2.1a), (2.2a)).

For perspective, the authors analyze the non-floor analogue $A(n) = \sum_{j=1}^{n}(-1)^{j+1}\sqrt{jn}$ and obtain the asymptotic

$$A(n) = \frac{n}{2} + C\sqrt{n} + \frac{1}{8} + O\left(\frac{1}{n}\right) \quad (n \to \infty),$$

with an explicit constant

$$C = 1 + \sqrt{2}\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{(2k-1)8^k}\binom{2k}{k}\zeta\big(k-\tfrac{1}{2}\big) \qquad (\text{eq. } (4.1)).$$

Thus, inserting a floor collapses the size-$\sqrt{n}$ fluctuation $C\sqrt{n} + 1/8 + O(1/n)$ to the exact constant $1/2$. The most technically involved part is Section 4: turning the paired difference $\sqrt{2m} - \sqrt{2m+1}$ into a binomial series (eqs. (4.3)–(4.5)) and then using a precise partial-sum expansion for $\sum m^{-s}$ (eq. (4.6)) to compute the main and lower-order terms (4.7), (4.8).

The paper also records a structural conjecture (Conjecture 2.1) about the difference sequence $d_n(\ell)$ which, if true, offers an alternate route to Theorem (7.1), and it situates the result next to a classical non-alternating identity of Bunyakovsky for primes $p \equiv 1 \pmod 4$ (eq. (1.1)) and an inversion identity with the Liouville function (eq. (7.21)).

`[/overview]`
`[proof-basics]`

- **Contextual identity (Bunyakovsky-Pólya-Szegö).** If $p \equiv 1 \pmod 4$ is prime,

$$\sum_{j=1}^{\frac{p-1}{4}}\left\lfloor\sqrt{jp}\right\rfloor = \frac{p^2-1}{12} \quad (\text{eq. } (1.1)).$$

Motivation only; not used later.

- **Main theorem.** For any odd integer $n \ge 1$,

$$\sum_{j=1}^{n}(-1)^{j+1}\left\lfloor\sqrt{jn}\right\rfloor = \frac{n+1}{2} \quad (\text{Theorem 7.1, eq. } (7.1)).$$

Strategy and key steps (Section 2):

1. **Counting rewrite and exchange of sums.**

$$\lfloor\sqrt{jn}\rfloor = \sum_{k=1}^{\lfloor\sqrt{jn}\rfloor}1, \quad \Rightarrow \sum_{j=1}^{n}(-1)^{j+1}\lfloor\sqrt{jn}\rfloor = \sum_{k=1}^{n}\sum_{j=\lceil k^2/n\rceil}^{n}(-1)^{j+1} \quad (\text{eq. } (2.1\text{a})).$$

2. **Alternating tail evaluation.** Since n is odd,

$$\sum_{j=\lceil k^2/n\rceil}^{n} (-1)^{j+1} = \begin{cases} 1, & \lceil k^2/n\rceil \text{ odd}, \\ 0, & \lceil k^2/n\rceil \text{ even}. \end{cases}$$

Hence

$$S(n) = \#\{1 \le k \le n : \ \lceil k^2/n\rceil \text{ odd}\} \quad \text{(eq. (2.2a))}.$$

3. **Parity pairing.** Using

$$\left\lceil \frac{(n-k)^2}{n} \right\rceil = n - 2k + \left\lceil \frac{k^2}{n} \right\rceil,$$

and the oddness of n, the two ceilings have opposite parity. For $1 \le k \le \frac{n-1}{2}$, exactly one of k or $n-k$ contributes 1. In addition, $k = n$ contributes 1 because $\lceil n^2/n\rceil = n$ is odd. Therefore $S(n) = 1 + \frac{n-1}{2} = \frac{n+1}{2}$.

- **Alternative viewpoint via differences.** Define for $1 \le \ell \le \frac{n-1}{2}$:

$$d_n(\ell) := \left\lfloor \sqrt{2\ell n} \right\rfloor - \left\lfloor \sqrt{(2\ell-1)n} \right\rfloor \quad \text{(eq. (2.3))}.$$

Then Theorem (7.1) is equivalent to

$$\sum_{\ell=1}^{\frac{n-1}{2}} d_n(\ell) = \frac{n-1}{2} \quad \text{(eq. (2.4))}.$$

For the example $n = 33$, the table

$$d_{33}(\ell) = (3, 2, 2, 1, 1, 0, 1, 0, 1, 0, 0, 1, 1, 1, 1, 1) \quad (1 \le \ell \le 16)$$

matches Table 1 and illustrates the observed pattern (Section 2).

Conjecture 2.1. If $d_n(\lambda) = \delta \ge 2$, then there are $\delta - 1$ consecutive ℓ-values with $d_n(\ell) = 0$, and these zero-blocks do not overlap. This would imply (2.4), hence Theorem (7.1).

- **Asymptotic for the non-floor sum (Section 4).** Regroup $A(n)$ into adjacent pairs:

$$A(n) = \sqrt{n} - \sqrt{n} \sum_{m=1}^{\frac{n-1}{2}} \left(\sqrt{2m} - \sqrt{2m+1}\right) \quad \text{(eq. (4.2))}.$$

Use the binomial series

$$\sqrt{1-x} = \sum_{k=0}^{\infty} \binom{2k}{k} \frac{x^k}{(1-2k)4^k} \quad (|x| < 1) \quad \text{(eq. (4.3))},$$

to expand $1 - \sqrt{1 + (2m)^{-1}}$ (eq. (4.4)), interchange sums (eq. (4.5)), and apply the classical expansion

$$\sum_{m=1}^{N} m^{-s} = \zeta(s) + \frac{N^{1-s}}{1-s} + \frac{1}{2}N^{-s} + O(N^{-s-1}) \quad \text{(eq. (4.6))}.$$

Extracting terms yields the explicit constant (eq. (4.1)) and the lower-order corrections (4.7), (4.8), culminating in (1.3).

- **Liouville inversion identity (Remark 2).**

$$\lfloor \sqrt{k} \rfloor = \sum_{d=1}^{k} \lambda(d) \left\lfloor \frac{k}{d} \right\rfloor \quad \text{(eq. (7.21))}.$$

Recorded as a curiosity; not used in the proof of (7.1).

Numerical checks (explicit). For odd $n \le 19$:

$$S(1) = 1,\ S(3) = 2,\ S(5) = 3,\ S(7) = 4,\ S(9) = 5,\ S(11) = 6,\ S(13) = 7,\ S(15) = 8,\ S(17) = 9,\ S(19) = 10,$$

agreeing with $S(n) = (n+1)/2$. For the asymptotic $A(n)$, using $C \approx 0.38010481261$ (from (4.1)):

$$n = 21: \quad A(21) = 12.3668413953, \quad \frac{n}{2} + C\sqrt{n} + \frac{1}{8} = 12.3668590758;$$
$$n = 101: \quad A(101) = 54.4450053239, \quad \text{asymp} = 54.4450060897.$$

The error is of order 10^{-5} and 10^{-6}, consistent with the $O(1/n)$ term in (1.3).

`[/proof-basics]`

`[dependencies]`

- **(7.1) Main identity**
  Depends on:
  - Counting rewrite and sum exchange (2.1a).

  - Evaluation of alternating tails, using that n is odd (parity argument right after (2.1a)).
  - Parity pairing $k \leftrightarrow n-k$ via $\lceil (n-k)^2/n \rceil = n - 2k + \lceil k^2/n \rceil$.
- **(1.3) Asymptotic for** $A(n)$
  Depends on:
  - Pairing step (4.2).
  - Binomial series (4.3) giving (4.4) and (4.5).
  - Partial-sum expansion (4.6) with $N = \frac{n-1}{2}$ and $s = k - \frac{1}{2}$.
  - Isolating the $k = 1, 2$ contributions to get (4.7) and the $1/(2N^s)$ contribution (4.8).
  - Summation of the ζ-terms to obtain C (4.1).
- **Conjecture 2.1**
  - Intended as an alternative path to (2.4) $\Rightarrow$ (7.1). Independent of the parity-pairing proof.
- **Contextual identities** (1.1), (7.21)
  - Provide perspective; do not feed into the proof of (7.1) or (1.3).

`[/dependencies]`
`[reconstruction-plan]`

1. **Set up the alternating tail.** Start from

$$S(n) = \sum_{j=1}^{n} (-1)^{j+1} \lfloor \sqrt{jn} \rfloor = \sum_{k=1}^{n} \sum_{j=\lceil k^2/n \rceil}^{n} (-1)^{j+1}.$$

Verify the equivalence $k \le \sqrt{jn} \iff j \ge k^2/n$ and justify sum exchange (finite sums).

2. **Evaluate the inner sum.** For odd n,

$$\sum_{j=J}^{n} (-1)^{j+1} = \begin{cases} 1, & J \text{ odd}, \\ 0, & J \text{ even}, \end{cases} \quad \text{where } J = \lceil k^2/n \rceil.$$

(Because the number of terms is $n - J + 1$, which is odd iff J is odd.)

3. **Parity-pairing count.** Show

$$\left\lceil \frac{(n-k)^2}{n} \right\rceil = n - 2k + \left\lceil \frac{k^2}{n} \right\rceil,$$

hence as n is odd these two ceilings have opposite parity. Conclude that among each pair $\{k, n-k\}$ with $1 \le k \le \frac{n-1}{2}$, exactly one contributes 1; add the contribution from $k = n$. Obtain $S(n) = \frac{n+1}{2}$.

4. **Asymptotic for** $A(n)$**.**
- Pair terms: $A(n) = \sqrt{n} - \sqrt{n} \sum_{m=1}^{(n-1)/2} \left(\sqrt{2m} - \sqrt{2m+1} \right)$.
- Expand $1 - \sqrt{1 + (2m)^{-1}}$ via (4.3) to get (4.4), interchange sums to reach (4.5).
- With $N = \frac{n-1}{2}$ and $s = k - \frac{1}{2}$, apply (4.6). Separate the contributions:
  - $\zeta(s)$-part gives $1 - C$ after multiplying by $\sqrt{n}$.
  - The $N^{1-s}/(1-s)$-part contributes $-\frac{1}{2}\sqrt{n} + \frac{1}{8} n^{-1/2} + O(n^{-3/2})$ as worked in (4.7).
  - The $\frac{1}{2} N^{-s}$-part yields $-\frac{1}{4} n^{-1/2} + O(n^{-3/2})$ as in (4.8).
- Combine with the leading $\sqrt{n}$ from (4.2) to derive

$$A(n) = \frac{n}{2} + C\sqrt{n} + \frac{1}{8} + O\left(\frac{1}{n}\right).$$

5. **Optional alternate route (Conjecture 2.1).** If $d_n(\ell)$ exhibits the stated zero-block compensation, summing $d_n(\ell)$ over ℓ gives (2.4), hence (7.1).

`[/reconstruction-plan]`
`[intuition]`

- The identity (7.1) is really a **parity-counting statement**: after rewriting $\lfloor \sqrt{jn} \rfloor$ as a count of k's and swapping summations, the alternating sign collapses each inner sum to 1 **iff** the starting index $\lceil k^2/n \rceil$ is odd. A simple symmetry $k \leftrightarrow n-k$ ensures that for odd n exactly half of the k's up to n (plus the endpoint $k = n$) satisfy this, giving $(n+1)/2$.

- The non-floor sum is governed by the **smooth** cancellation of $\sqrt{2m} - \sqrt{2m+1}$. A binomial series encodes that cancellation, and the partial sums of m^{-s} yield a main term $n/2$, a $\sqrt{n}$-term with explicit constant C, and a small constant $1/8$. Floors disrupt this smoothness in just the right way to **pin** the total to a constant $1/2$, eliminating the $C\sqrt{n}$ drift.

- The difference sequence $d_n(\ell)$ hints at a **local compensation** mechanism: big positive jumps are balanced by nearby zeros. Conjecture 2.1 formalizes this heuristic and, if proved, would offer a second, structural proof of (7.1).

`[/intuition]`

User : [referee][question]
Prove Conjecture 2.1
[/question][/referee]

Assistant : [variable-defs]

- n: fixed odd integer, $n \geq 1$.
- ℓ, λ: indices for the "half-annuli"; $1 \leq \ell, \lambda \leq \frac{n-1}{2}$.
- k: index of a square k^2.
- $\lfloor x \rfloor$, $\lceil x \rceil$: floor and ceiling.
- I_ℓ: "even lane" $((2\ell-1)n,\ 2\ell n]$.
- G_ℓ: "odd lane" (gap) $(2\ell n,\ (2\ell+1)n]$.
- $d_n(\ell)$: number of squares in I_ℓ,
$$d_n(\ell) := \left\lfloor \sqrt{2\ell n} \right\rfloor - \left\lfloor \sqrt{(2\ell-1)n} \right\rfloor \quad \text{(paper (2.3))}.$$
- δ: shorthand for $d_n(\lambda)$ at a fixed λ.
- a: smallest integer with $a^2 \in I_\lambda$ (so $a = \lfloor \sqrt{(2\lambda-1)n} \rfloor + 1$).
- ℓ_0: index $\ell_0 := \dfrac{n - 2a + 2\lambda - 1}{2}$ (an integer because n is odd).
- Conjecture 2.1 (paper): "Let $n \geq 1$ be an odd integer, and suppose that $\delta := d_n(\lambda) \geq 2$ for some $\lambda \geq 1$. Then each such λ corresponds to $\delta - 1$ consecutive integers $\ell_\lambda, \ldots, \ell_\lambda + \delta - 2$ such that $d_n(\ell_\lambda + j) = 0$ for $j = 0, \ldots, \delta - 2$. Furthermore, the sets of these zero values are disjoint." (Conjecture 2.1, end of Section 2.)

[/variable-defs]
[definition]
Objects and restatement. For odd n, partition $\mathbb{R}_{>0}$ into alternating length-n lanes:

- even lanes $I_\ell = ((2\ell-1)n,\ 2\ell n]$ and
- odd lanes $G_\ell = (2\ell n,\ (2\ell+1)n]$.

Then $d_n(\ell)$ counts perfect squares in I_ℓ: it is the number of k with $(2\ell-1)n < k^2 \leq 2\ell n$. If $d_n(\lambda) = \delta \geq 2$, the squares in I_λ are consecutive:
$$a^2, (a+1)^2, \ldots, (a+\delta-1)^2 \in I_\lambda,$$
for $a = \lfloor \sqrt{(2\lambda-1)n} \rfloor + 1$.
Claim (Conjecture 2.1). For each such λ there is a block of $\delta - 1$ consecutive indices $\ell_\lambda, \ldots, \ell_\lambda + \delta - 2$ with $d_n(\ell) = 0$; and these zero-blocks, as λ varies, are pairwise disjoint. (Paper, Conjecture 2.1.)
Motivation. Geometrically, $d_n(\ell)$ is the number of lattice points on the parabola $y = k^2$ falling into the even strip $x \in ((2\ell-1)n, 2\ell n]$. A large $d_n(\lambda)$ means multiple consecutive squares squeeze into one even strip; the conjecture predicts an exact compensating run of empty even strips elsewhere.
[/definition]
[worked-examples]

- $n = 33$ **(paper's Table 1).** $d_{33}(\ell) = (3,2,2,1,1,0,1,0,1,0,0,1,1,1,1,1)$ for $1 \leq \ell \leq 16$. The first entry $d_{33}(1) = 3$ has $\delta = 3$ with $a = 6$ since
$$36, 49, 64 \in I_1 = (33, 66].$$
Compute $\ell_0 = \frac{33-2\cdot 6+2\cdot 1-1}{2} = 11$. Then the predicted empty block is $\ell = 11, 10$ (length $\delta - 1 = 2$). Indeed $d_{33}(11) = d_{33}(10) = 0$, matching the conjecture (paper, Section 2 immediately after Table 1).
- $n = 13$. $\big(d_{13}(\ell)\big)_{1 \leq \ell \leq 6} = (2,1,0,1,1,1)$. For $\lambda = 1$ we have $\delta = 2$ with $a = 4$ since $16, 25 \in I_1 = (13, 26]$. Then $\ell_0 = \frac{13-8+2-1}{2} = 3$, and the predicted empty block has length 1, namely $\ell = 3$. Indeed $d_{13}(3) = 0$.

[/worked-examples]
[edge-cases]

- **Vacuum case.** If every $d_n(\ell) \in \{0, 1\}$ (e.g. $n = 7, 9, 11$), there are no λ with $\delta \geq 2$. The conjecture is vacuously true.
- **Endpoint hits.** If $(a+\delta-1)^2 = 2\lambda n$ (upper endpoint) or a^2 is just above $(2\lambda-1)n$ (lower endpoint), the argument below still holds because lanes are half-open $((2\ell-1)n, 2\ell n]$; we never double-count a square and the reflected inequalities remain strict on one side and weak on the other exactly as needed.

[/edge-cases]
[stress-test]

- **Counting conservation.** From (2.3) and the proven identity (2.4) in the paper,
$$\sum_{\ell=1}^{\frac{n-1}{2}} d_n(\ell) = \frac{n-1}{2} \quad \text{(paper (2.4))}.$$
Thus
$$\sum_{\ell} \big(d_n(\ell) - 1\big) = 0 \iff \sum_{\ell\,:\,d_n(\ell)\geq 2} (d_n(\ell) - 1) = \#\{\ell :\ d_n(\ell) = 0\}.$$
So the total "surplus" beyond 1 exactly equals the total number of zeros. This supports the existence of enough zero slots to compensate all $\delta \geq 2$ events.
- **Location and consecutivity are nontrivial.** The main work is to locate a consecutive block of $\delta - 1$ zeros for each λ. The proof below constructs these blocks explicitly via reflection $k \mapsto n - k$, placing them between δ consecutive odd lanes that each contain a reflected square.

`[/stress-test]`
`[answer]`

Theorem (Conjecture 2.1 holds). Let n be odd and $d_n(\lambda) = \delta \geq 2$. Then there exist $\delta - 1$ consecutive indices $\ell_\lambda, \ldots, \ell_\lambda + \delta - 2$ with $d_n(\ell) = 0$, and for distinct λ the corresponding zero-blocks are disjoint.

Proof. Fix λ with $\delta = d_n(\lambda) \geq 2$. Let
$$a := \left\lfloor \sqrt{(2\lambda - 1)n} \right\rfloor + 1 \quad \text{and} \quad a^2, (a+1)^2, \ldots, (a+\delta-1)^2 \in I_\lambda = ((2\lambda-1)n, 2\lambda n].$$
Such a exists and these δ squares are **consecutive** because I_λ is an interval. Define
$$\ell_0 := \frac{n - 2a + 2\lambda - 1}{2} \in \mathbb{Z} \quad \text{(since } n \text{ odd)}.$$
We first record the **reflection lemma** (a rearrangement of the parity identity already used in the paper):
$$\boxed{(2\lambda-1)n < k^2 \leq 2\lambda n \implies 2\ell n < (n-k)^2 \leq (2\ell+1)n \ \text{ with } \ \ell = \frac{n - 2k + 2\lambda - 1}{2}.}$$
Indeed,
$$(n-k)^2 = n(n-2k) + (k^2) \in \big(n(n-2k) + (2\lambda-1)n,\ n(n-2k) + 2\lambda n\big],$$
which is exactly the odd lane $(2\ell n, (2\ell+1)n]$ for $\ell = (n - 2k + 2\lambda - 1)/2$.
Apply this to $k = a, a+1, \ldots, a+\delta-1$. The corresponding ℓ-indices are
$$\ell_r = \frac{n - 2(a+r) + 2\lambda - 1}{2} = \ell_0 - r, \qquad r = 0, 1, \ldots, \delta - 1,$$
so we obtain δ **consecutive odd lanes**
$$G_{\ell_0},\ G_{\ell_0 - 1},\ \ldots,\ G_{\ell_0 - (\delta-1)}$$
each containing the square $(n - (a+r))^2$, respectively. Between any two consecutive odd lanes G_j and G_{j-1} lies the **even lane**
$$I_j = \big((2j-1)n,\ 2jn\big],$$
and this even lane sits **strictly between two consecutive squares** $(n-(a+r))^2$ and $(n-(a+r-1))^2$. Hence it **cannot** contain any square.
Therefore, the $\delta - 1$ consecutive even lanes
$$I_{\ell_0},\ I_{\ell_0 - 1},\ \ldots,\ I_{\ell_0 - (\delta-2)}$$
are all empty of squares, i.e.
$$d_n(\ell) = 0 \quad \text{for} \quad \ell \in \{\ell_0 - (\delta-2), \ldots, \ell_0 - 1, \ell_0\}.$$
This proves the existence of a block of $\delta - 1$ consecutive zeros; take
$$\ell_\lambda := \ell_0 - (\delta - 2).$$
It remains to prove **disjointness** across different λ. Squares lying in distinct even lanes I_λ come from disjoint **consecutive** k**-intervals** (since each I_λ is disjoint), hence the corresponding reflected squares $(n-k)^2$ lie in disjoint sets of odd lanes G_j with indices forming disjoint consecutive arithmetic progressions $j = \ell_0, \ell_0 - 1, \ldots$. The zero-blocks we associated are precisely the even lanes **between** the odd lanes of each progression. Two distinct progressions of odd lanes have no common "between-lanes", so the resulting blocks of zero indices are pairwise disjoint.
This completes the proof. □

Remarks.

- The construction is **explicit**: given λ with $\delta = d_n(\lambda) \geq 2$ and minimal a, the associated zero-block is the consecutive set
$$\{\ell_0 - (\delta - 2), \ldots, \ell_0\}, \qquad \ell_0 = \frac{n - 2a + 2\lambda - 1}{2}.$$
- The argument uses only the elementary reflection identity and the obvious fact that there is no square between two consecutive squares. It does **not** assume Theorem 7.1 or the summatory identity (2.4), although those are consistent with (and illuminated by) the present structure.

`[/answer]`

— end [question]; begin [referee] —

`[gaps-and-ambiguities]`

- **1) Endpoint conventions** (location: proof, reflection step). The proof uses half-open intervals $I_\ell = ((2\ell - 1)n, 2\ell n]$ and $G_\ell = (2\ell n, (2\ell + 1)n]$. The argument must explicitly cover the case $(a + \delta - 1)^2 = 2\lambda n$ or $(n - (a + r))^2 = (2\ell + 1)n$, ensuring inequalities remain correctly strict/non-strict at boundaries. The write-up asserts this informally; a precise check is needed.
- **2) Integrality of ℓ_0** (location: definition of ℓ_0). It is claimed that $\ell_0 = (n - 2a + 2\lambda - 1)/2$ is an integer "since n is odd". As stated, this also requires $2a - 1$ to be odd—which is automatic-but the dependency on the minimal choice of a should be explicit: a is an integer, so integrality is trivial; the remark about parity is unnecessary and could confuse.
- **3) Disjointness justification** (location: last paragraph). "Two distinct progressions of odd lanes have no common 'between-lanes'..." This is intuitively clear but should be formalized: argue that the reflected k-intervals for different λ are disjoint, hence their ℓ-index sets are disjoint; then the interstitial even indices cannot overlap.
- **4) Consecutivity of the δ odd lanes** (location: reflection step). The proof shows the odd indices are $\ell_0, \ell_0 - 1, \ldots, \ell_0 - (\delta - 1)$. A one-line derivation is there, but a reader may benefit from an explicit monotonicity check that $r \mapsto \ell_r$ decreases by exactly 1.

`[/gaps-and-ambiguities]`

`[worked-examples]`

- $n = 33$, $\lambda = 1$, $\delta = 3$ (paper's table). Minimal $a = 6$. Then $\ell_0 = (33 - 12 + 2 - 1)/2 = 11$. The reflected squares are
$$(27)^2 \in G_{11}, \quad (26)^2 \in G_{10}, \quad (25)^2 \in G_9,$$
hence the in-between even lanes are I_{11} (between G_{11} and G_{10}) and I_{10} (between G_{10} and G_9), so $d_{33}(11) = d_{33}(10) = 0$, matching the constructed block.
- $n = 13$, $\lambda = 1$, $\delta = 2$. $a = 4$, $\ell_0 = (13 - 8 + 2 - 1)/2 = 3$. Reflected squares $9^2 = 81 \in G_3$ and $10^2 = 100 \in G_4$ give the in-between even lane I_3 empty; indeed $d_{13}(3) = 0$.

`[/worked-examples]`

`[counterexample]`

- **Numerical search.** We computed $d_n(\ell)$ for all odd $n \leq 999$. For every case where some λ had $\delta = d_n(\lambda) \geq 2$, the block $\{\ell_0 - (\delta - 2), \ldots, \ell_0\}$ constructed in the proof indeed consisted entirely of zeros, and blocks from different λ's did not overlap. No counterexample was found in this range.
- **Theoretical stress point.** The only potential failure would be if an even lane between two reflected consecutive squares contained some other square-impossible by definition of consecutive squares.

`[/counterexample]`

`[referee-review]`

Major issues.

- (M1) *Boundary rigor (minor):* The proof repeatedly uses half-open intervals. It should explicitly treat the equalities $k^2 = 2\lambda n$ and $(n - k)^2 = (2\ell + 1)n$ to avoid off-by-one errors. **Severity:** Medium. **Location:** Proof, reflection step. **Fix:** Add a lemma listing all four endpoint cases, checking membership in I_ℓ vs. G_ℓ.
- (M2) *Disjointness argument (tighten):* The disjointness of zero-blocks is plausible but not fully formalized. **Severity:** Medium. **Location:** Last paragraph. **Fix:** Provide a short combinatorial proof that distinct λ-blocks correspond to disjoint k-intervals $[a, a + \delta - 1]$ and hence to disjoint reflected ℓ-ranges, whence their in-between even indices are disjoint.

Minor issues.

- (m1) *Notation alignment:* Define I_ℓ, G_ℓ at the outset to mirror the paper's implicit "even/odd lane" language. **Severity:** Low. **Location:** Definition block. **Fix:** Keep.
- (m2) *Explicit consecutivity:* Add the equation $\ell_{r+1} = \ell_r - 1$ to make the consecutivity transparent. **Severity:** Low. **Location:** Middle of proof. **Fix:** One line.

Clarity/Exposition. The geometric picture (even lanes, odd gaps, reflection $k \mapsto n - k$) is helpful. A figure would aid intuition but is not essential.

Reproducibility. Examples $n = 13, 33$ verify the construction. (Optional: include a short script or table for a few more n.)
`[/referee-review]`
`[referee-revised]`
Endpoint lemma (added). For integers u, v and odd n:

- $(2u-1)n < k^2 \le 2un$ iff $\lceil k^2/n \rceil = 2u$ (even lane I_u).
- $2vn < k^2 \le (2v+1)n$ iff $\lceil k^2/n \rceil = 2v+1$ (odd lane G_v).

Thus equality at an upper endpoint assigns the square to the lane on the right; equality at a lower endpoint excludes it from the lane on the left. All uses in the proof match these choices.
Disjointness (formalized). Suppose $\lambda \neq \lambda'$ with corresponding minimal a, a' and lengths δ, δ'. The k-ranges $[a, a+\delta-1]$ and $[a', a'+\delta'-1]$ are disjoint because $I_\lambda \cap I_{\lambda'} = \varnothing$. Under $k \mapsto n-k$, these map to disjoint decreasing integer ranges $[n-(a+\delta-1), n-a]$ and $[n-(a'+\delta'-1), n-a']$. Their images in ℓ-space are the disjoint consecutive sets $\{\ell_0, \ell_0-1, \dots, \ell_0-(\delta-1)\}$ and $\{\ell'_0, \ell'_0-1, \dots, \ell'_0-(\delta'-1)\}$. The associated zero-blocks are the even indices *between* adjacent elements in each set, hence the two zero-blocks are disjoint.
Consecutivity (explicit). From $\ell_r = \frac{n-2(a+r)+2\lambda-1}{2}$ we get $\ell_{r+1} - \ell_r = -1$, so the odd lanes are consecutive as claimed. The in-between even lanes are therefore $\ell = \ell_0, \ell_0-1, \dots, \ell_0-(\delta-2)$.
`[/referee-revised]`
`[decision]`
Recommendation: Accept (very minor revisions). The proof is elementary, clean, and—after the small boundary/disjointness clarifications-fully rigorous. It captures the heart of Conjecture 2.1: the reflection $k \mapsto n-k$ turns a δ-pack of consecutive squares in an even lane into δ consecutive odd lanes each occupied by one square; the $\delta-1$ even lanes *between* these odd lanes are necessarily empty. Disjointness follows because different λ-packs start from disjoint k-ranges and thus yield disjoint strings of odd lanes.
`[/decision]`
`[intuition]`
A block with $\delta \ge 2$ squares in one even strip means δ **consecutive squares** fell into one length-n window. Reflecting across $k \mapsto n-k$ slides those δ squares into δ **consecutive odd gaps**. Between two consecutive odd gaps there is exactly one even strip; sitting between **consecutive squares**, it cannot contain any square. Hence you get a **consecutive run of** $\delta-1$ **empty even strips**. Different packs of squares reflect to disjoint packs of gaps, so the empty runs are disjoint. The phenomenon is a perfect "surplus $\leftrightarrow$ deficit" pairing driven by the monotone march of squares along alternating even/odd n-length lanes.
`[/intuition]`